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WORK PLAN
CADILLAC AREA
GROUNDWATER CONTAMINATION

REMEDIAL INVESTIGATION/ FEASIBILITY STUDY

CONTRACT NO. 1525

E.C. JORDAN PROJECT NO. 4656.51

PREPARED FOR

MICHIGAN DEPARTMENT OF NATURAL RESOURCES

MAY 1986

E.C. JORDAN CO.

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1.0 WORK PLAN SUMMARY

1.1 PROBLEM STATEMENT

The City of Cadillac Industrial Park is located on the north edge of the City. Its northern and western borders are delineated by 13th Street and Leeson Street, respectively (see Figure 1-1). There is a residential subdivision, North Park, located due north of the Industrial Park and population density thins out to scattered homes and businesses as one progresses west along 13th Street. Residences are also present to the south of the Industrial Park, the closest being a mobile home park near Sixth and Leeson Streets. All of the homes north of 13th Street are in Haring Township and are served by their own individual water supply wells. Cadillac has a municipal water supply system which is provided by the City's well field located in the northeast corner of the industrial park.

Since 1978, a number of water supply wells have been found to be contaminated with various organic industrial chemicals (principally solvents). The municipal wells have not been affected to this date. Several industries in the park have been determined to have contaminated groundwater underlying the industrial park as a result of their manufacturing processes and waste handling practices. Several of these industries have performed limited hydrogeologic investigations to evaluate the impact or potential impact of these discharges. None of these studies has yet completely described the extent of contamination attributable

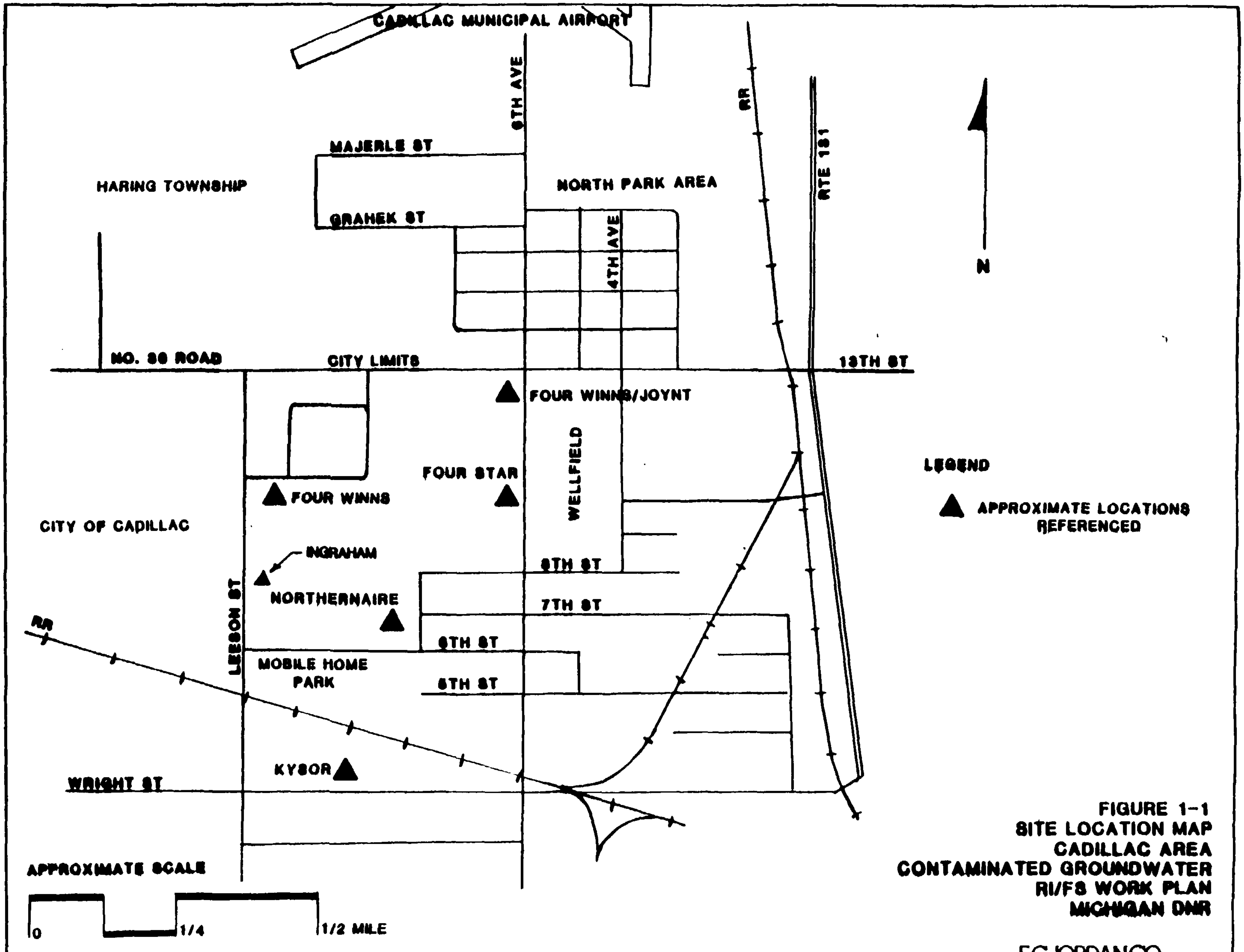

FIGURE 1-1
SITE LOCATION MAP
CADILLAC AREA
CONTAMINATED GROUNDWATER
RI/FS WORK PLAN
MICHIGAN DNR

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to releases from the particular industry being investigated.

Further, it appears that none of the studies, except for the Remedial Investigation (RI) now being conducted for the Northernaire site, a Superfund site, will be finished in a timely manner.

Most of the contaminated water supply wells are located north of 13th Street in the North Park area. The contaminants detected in wells there consist mainly of chlorinated hydrocarbons, such as trichloroethylene and perchloroethylene, with fewer occurrences of aromatic hydrocarbons such as xylene, toluene and benzene. Concentrations of volatile organics found in contaminated groundwater have ranged from a few parts per billion to about 70 parts per million. Concentrations of chromium above drinking water standards have also been found in monitoring wells in the industrial park in conjunction with the RI/FS study of the Northernaire site under CERCLA (see Figure 1-2).

During August, 1985, the Cadillac District Office of the Michigan Department of Natural Resources (MDNR) conducted a survey and site inspection of 37 industrial sites in the area to aid in determining probable sources of existing groundwater contamination. The survey established low, medium and high levels of risk associated with each of the selected sites based on chemicals used (past and present), volumes, processes and handling and storage practices. Based on this survey and existing data, MDNR identified several industries and sites of concern. The industries or sites which are strongly suspected or, in some cases, known to have released volatile organic contaminants to the environment which subsequently affected groundwater quality are listed below and shown on Figure 1-1:

- o Kysor of Cadillac, Inc.;
- o Four Star Corporation (former);
- o 4-Winns Boats; and
- o Ingraham Residence

In addition, hexavalent chromium was discharged to the groundwater at the Northernaire site. Limited hydrogeologic studies have been performed for the Kysor and Four Star sites and a RI/FS for the Northernaire site is ongoing. Cadillac Molded Rubber had an accidental spill of trichloroethylene in the fall of 1984 and has performed a limited hydrogeologic study in connection with its ongoing clean-up activities. The contamination at the 4-Winns facility was only recently discovered and the company has agreed to perform investigations to define the extent of the problem caused by their discharges. A report based on preliminary findings at the 4-Winns site has not yet been made available to the MDNR. In addition, it is believed that drums of solvents have been stored on the Ingraham property on Lesson Street and that investigation of this area is warranted.

CMI-Cadillac Inc. is not currently suspected in the known contamination of wells, but has recently performed limited hydrogeologic work in support of its application for a wastewater discharge permit.

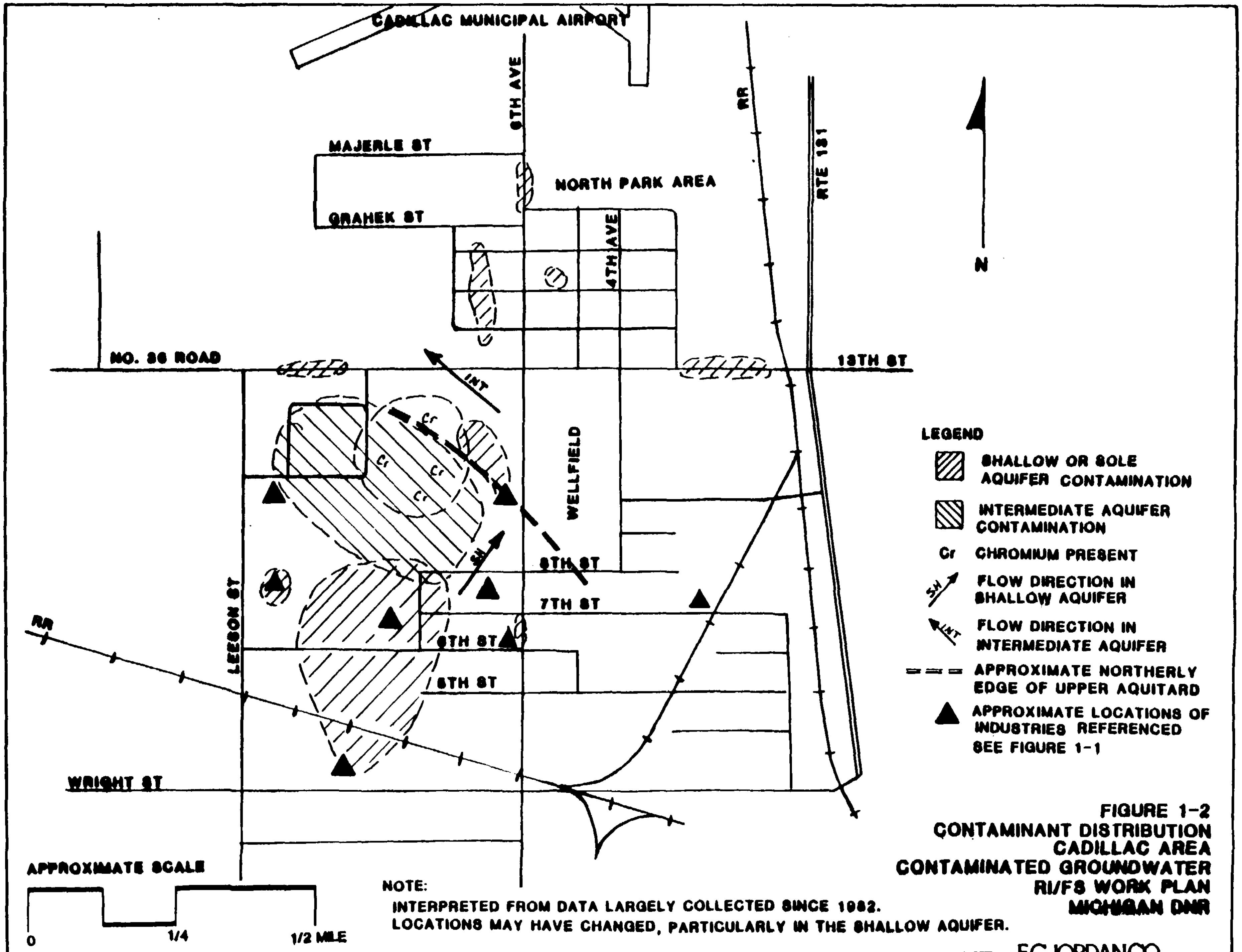
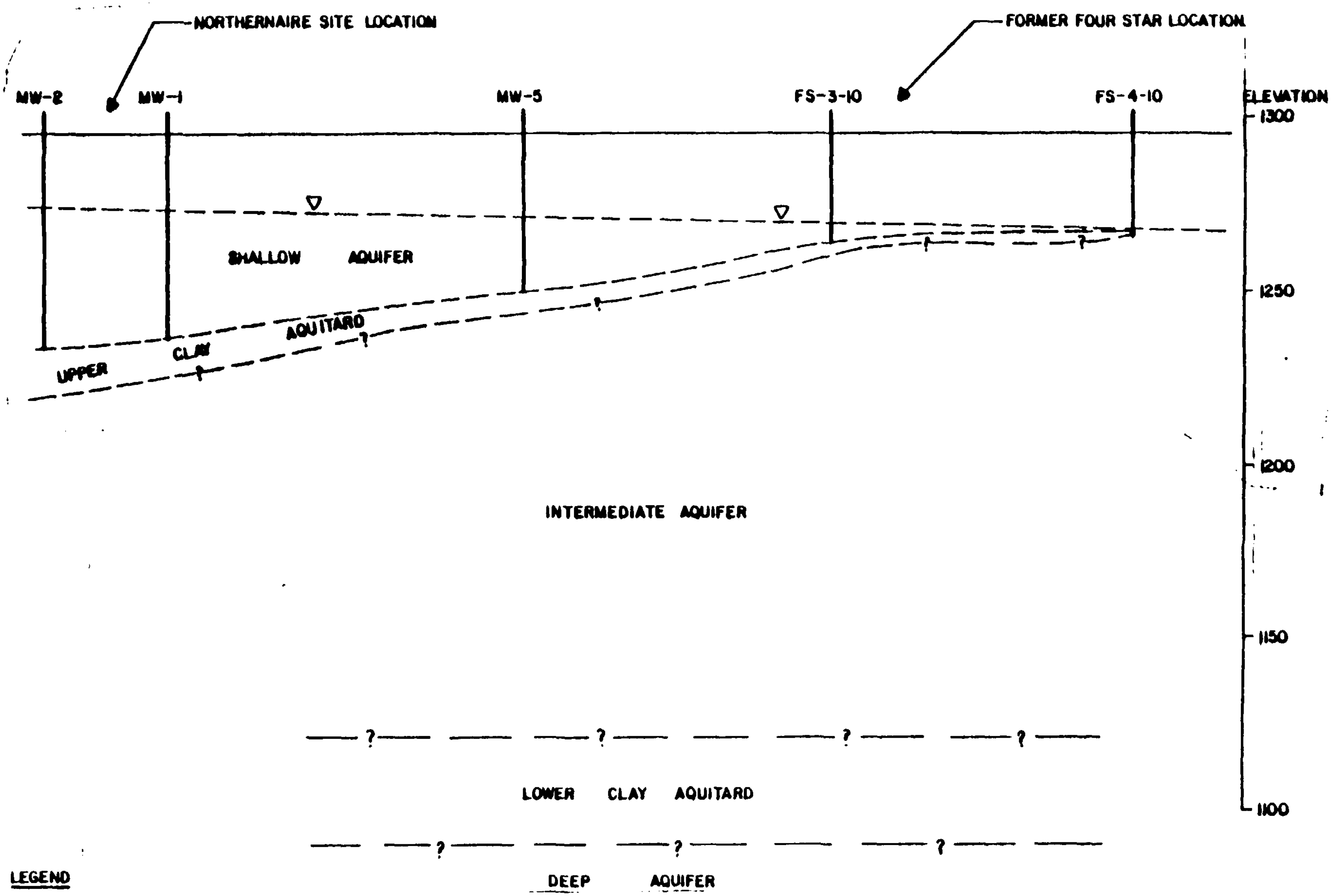


FIGURE 1-2
CONTAMINANT DISTRIBUTION
CADILLAC AREA
CONTAMINATED GROUNDWATER
RI/FS WORK PLAN
MICHIGAN DNR

A review of the data available thus far has indicated that the general hydrogeologic setting in the area consists of three aquifers separated by two clay aquitards which limit the hydraulic connections between adjacent water-bearing units (see Figure 1-3). The shallow and intermediate aquifers merge where the upper clay aquitard pinches out along a NW-SE line in the vicinity of the former Four Star facility. Based on data collected during the RI for the Northernaire site and on interpretations of flow directions for both the Kysor and Four Star hydrogeologic studies, the shallow aquifer flow is to the north-east at a rate of about 200 feet per year, and the intermediate aquifer flow is to the northwest at a rate of about 25 feet per year (see Figure 1-2).

The hydrogeologic studies and other information thus far have provided individual portions of a larger picture of area groundwater contamination in the upper and intermediate aquifers. These studies have as yet defined neither the lateral boundaries of the contamination nor the hydrogeology sufficiently enough to be able to evaluate the risk associated with the contamination, its ultimate fate or appropriate measures for remediation.

The areas of contamination indicated in the various studies or water monitoring programs are shown in Figure 1-2. The data provided from analyses of private water supply wells along 13th Street and in North Park in particular, seem to show many individual areas of contamination. This may be due to the varying depths of the private wells and the lack of data reported for one particular point in time, or may be due to intermittent occurrences of contaminant release. The chemical distribution data represent a time span of about six years, but have been collected largely since 1982.



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∇ WATER TABLE ELEVATION

—?— ESTIMATED SURFACE

NOTES

1. PROFILE FROM NORTHERNAIRE TO FOUR STAR
2. AQUITARD SURFACE ESTIMATED FROM AVAILABLE BORING LOGS
3. ELEVATIONS IN FEET ABOVE MSL

FIGURE 1-3
HYDROGEOLOGIC SETTING
NORTHERNAIRE SUPPLEMENTAL RI/FS
CADILLAC, MICHIGAN

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Based on the data available, MDNR has expressed concerns for the general water quality in the Cadillac area and the potential for impact on human health and the environment. These concerns are:

- contaminated individual residential wells and the wells in their proximity which are not yet contaminated;

- the threat which may be posed to the City of Cadillac well field; and

- the potential degradation of a major usable aquifer.

1.2 OBJECTIVES

The overall objective of the remedial investigation/feasibility study of the Cadillac industrial park and environs is to undertake studies which will determine the nature, extent and probable origin, of the environmental contamination in the area, the public health and environmental hazard posed by the contamination in the aquifers and to identify a cost-effective, environmentally sound and socially acceptable remedial program for the site.

Specific objectives outlined by the MDNR in its request for a work plan for the Cadillac Area Contaminated Groundwater study include:

- compile and correlate existing data generated by Federal, State and privately funded groundwater studies performed in the area to serve as a basis for initiating this study;

- define the horizontal and vertical extent of the contaminant plume(s);

- define more completely individual contaminant species (Hazardous Substance List);

- determine the probable sources of groundwater contamination;

- determine the fate of the groundwater contamination plumes (including potential for impact on the deepest aquifer and the municipal water supply);

- perform an evaluation and analysis of remedial action alternatives ranging from complete purging and treatment of the contaminated groundwater and removal of contaminated soils to the consequences of taking no action to restore or rehabilitate the aquifer;

- chain-of-custody and sampling procedures as outlined in EPA SW-846 and analytical procedures in accordance with 40 CFR 136.3 or equivalent; and

- provide additional data for input to a groundwater flow and transport model to be prepared by MDNR. The model will be used to explore the site hydrogeology, estimate flow and contaminant transport rates and to evaluate remedial action alternatives.

1.3 SCOPE OF WORK

1.3.1 General

The Work Plan for the Cadillac Area Groundwater Contamination study has been developed based on information made available by the MDNR. This information includes studies of the Northernaire site conducted by Jordan and others, studies of the Kysor, Four Star, and Four Winns/Joynts sites, water quality data of area private wells and other pertinent information. This additional information consists of the project summary included in MDNR's request for a work plan for the proposed groundwater contamination study and information disclosed in meetings with the MDNR.

The RI/FS for this project has been divided into three general phases. The first consists of initial activities leading to the initiation of field investigations. The second consists of work leading to and including the completion of a draft RI/FS report for MDNR review and selection of a remedial action from among those favorable alternatives remaining after screening. The third phase includes the preparation of a conceptual design of the selected remedial action and the completion and delivery of a final report. The three general phases have been divided into 20 detailed tasks which are presented and discussed in Section 2.0.

Several assumptions have been made in the preparation of this work plan. They are:

- MDNR will provide or obtain rights of entry or access to specific areas of exploration.

- Level D personnel protection equipment will suffice for all field activities. Respiratory protection will be available if high levels of volatile organics are encountered as indicated by monitoring with a photoionization meter. Use of higher levels of protection will require adjustments to costs and possibly schedules.

- Investigations for this project by MDNR/Jordan will be based on coordination with studies conducted by participating Potentially Responsible Parties (PRPs) before field activities commence. This may also require amendments to this proposed Work Plan.

- Existing wells will be available for sampling, if necessary, for this project.

- Laboratory costs have been estimated assuming analyses of samples by a laboratory in EPA's Contract Laboratory Program (CLP) and according to CLP protocal.

- Data collected to assess the distribution of volatile organic analytes (VOA) in MW-7 and MW-8 for the Northernaire RI and in the deep wells installed for the Northernaire Amendment No. 1 RI can be incorporated into the data base for this study.

- If Jordan assumes primary lead for the formulation of a groundwater and contaminant transport model, it will use a model provided by MDNR unless Jordan already has the specified model in-house. If Jordan is to acquire a model under MDNR's specification, these costs would be additional.

1.3.2 Site Specific Work

MDNR is affording the opportunity to probable and known PRPs to perform parts of the hydrogeologic and contamination study of the overall remedial investigation. For those PRPs electing to perform new or complete on-going studies, MDNR and Jordan's function will be as program coordination, program standardization, scheduling, report review and ultimately assembling individual reports into a final RI document for the study. For those PRPs electing not to participate in the hydrogeologic or contaminant studies, MDNR and Jordan will perform those site related tasks for that portion of the study. MDNR and Jordan will also perform those investigations in areas not clearly the province of any individual PRP and overall tasks, such as survey, which are clearly more cost-effectively done by MDNR, and which require a consistency of field operations for their successful accomplishment.

In the Work Plan that follows, each task that may be performed by a PRP will also contain or reference objectives, guidelines, procedures, methodologies and minimum requirements that the individual site specific study must meet. This will permit the development of an accurate and easily coordinated final report. The more important areas of standardization include:

- o performance of a soil gas survey;
- o installation of monitoring wells and piezometers;
- o horizontal location and elevation survey of wells/piezometers;
- o use of common base maps;
- o protocol for sampling and analysis;
- o data collection, reduction and presentation; and
- o format for the site specific study report.

MDNR/Jordan will accept recommendations by the PRPs for corrective action , but will perform the feasibility study for the area; MDNR will select the most acceptable remedial measure(s) subject to public review. The selected remedial alternative will be chosen based on a number of criteria as explained in Section 2.2.

In this Work Plan, areas in which the PRPs may participate effectively and for which guidance is given are denoted by asterisks (**). At the end of each of these identified tasks describing Jordan's approach to the task assuming no PRP participation, is a section which deals with the specific requirements or guidelines for conduct of the study. If a PRP elects to participate, there must be a complete commitment of resources sufficient to accomplish the study in a timely manner and a firm schedule must be submitted to MDNR.

1.4 BUDGET

The budget for this Work Plan has been prepared based on the assumption of no participation by the PRPs. Should a number of PRPs decide to participate, then costs associated with Jordan and its subcontractor's efforts will be decreased. However, a minimum number of hours will still be required to coordinate field activities, for review of documents and for whatever other consulting support MDNR may require even if the PRPs perform all field work. This minimum number of manhours has not been estimated. However, funding for the project should be requested at the maximum figure because the level of PRP participation is unknown and likely to be limited, or non-existent.

The level of effort (manhours) for each phase of this RI/FS for the Cadillac Area Groundwater Contamination study is as follows:

Phase I - Initial Activities	1,044 manhours
Phase II - Draft RI/FS	9,288 manhours
Phase III - Conceptual Design and Final Report	800 manhours

The actual manpower requirements and duration of field activities will depend on the data obtained in the initial phases of the field explorations. It may be necessary to place additional wells in excess of the estimated number if the outermost wells are found to be contaminated or if it is necessary to place additional explorations to acquire sufficient data to identify a source. The budget and schedule include an estimate for a number of these additional explorations, but it may prove that this number is not sufficient and require changes in scope, schedule and costs.

A total of 11,132 manhours will be required for the RI/FS. This does not include manhours needed to perform any laboratory or field studies that may be required under Task 16 (optional). The manpower estimates for these studies will be determined during the preparation of the laboratory study work plan, if one is required. The manhour and cost estimates for the RI/FS are presented in Section 4.0.

1.5 SCHEDULE

It is estimated that the RI/FS will take from 13 to 15 months depending on the need to perform additional field explorations to attain the program objectives of plume definition and source identification. The schedule also assumes a 30-day turnaround for laboratory analytical results. In addition, MDNR review time for draft and final reports is estimated at one to three weeks as specified in the schedule. The detailed schedule is presented in Section 4.0.

The above estimated project duration has been based on no participation by PRPs. If some PRPs participate in the program, it is anticipated that the project may be extended from 1 to 2 months. The additional time is expected to be required to coordinate activities among the parties, review of multiple documents and consolidation of the individual documents into a final remedial investigation report.

2.0 PLAN OF WORK

2.1 INITIAL ACTIVITIES (PHASE I)

A total of 7 tasks have been identified for Phase I.

Participating PRPs will be required to submit a work plan and a health and safety plan (HASP) for MDNR review and approval prior to commencement of field activities. In addition, and as part of the work plan, the PRPs should identify the location of a field office with facilities suited to the particular scope of work to be carried out at the specific site location. PRPs need address only Tasks 1, 3, 4 and 6 of the initial activities.

Task 1: Prepare Work Plan

A Work Plan will be prepared for the Cadillac Area Groundwater Contamination Remedial Investigation/Feasibility Study (RI/FS). The Work Plan will describe:

- the initial activities to be undertaken prior to the RI/FS;

- the tasks to be completed during the RI to establish the nature and extent of the contamination;

- tasks to provide sufficient hydrogeologic information to predict the fate of the contaminants in the groundwater and the potential for impact on the municipal well field and on other private wells in the probable path of the contaminant plumes;

- o tasks to determine the source(s) of the contamination determined to be present in the aquifer, if possible;

- o tasks to be completed during the FS which will lead to the identification of acceptable alternatives for remediation; and following MDNR's selection of the most appropriate remedial action,

- o tasks to develop a conceptual design of the selected remedial action and the completion of final RI/FS report.

<u>**PRP Work Plan**</u>

The work plan submitted by a participating PRP to MDNR should follow the same format as this work plan, but should be tailored to the site specific conditions. Prior work completed at a site should be summarized as indicative of partial or complete fulfillment of a task item. The work plan submitted by the PRPs need not address task items beyond Task 13 and need not include Task 11 (Aquifer Testing) which will be done by MDNR/Jordan. Under the description of the contents of the work plan listed above, the PRPs should address only the first four bulleted items, leaving the FS and the conceptual design activities to MDNR and Jordan.

The work plan submitted by the participating PRP should identify project management structure and key individuals with project responsibility who will act as liaison and contact with MDNR/Jordan. The schedule should address the timeframe set by MDNR for completion of studies. For complex sites or if unavoidable delays in work completion are encountered, MDNR will consider reasonable requests for extension.

MDNR is not necessarily interested in being apprised of costs for conduct of the investigation. However, the work plan should contain a statement of commitment and an acknowledgment of sufficient financial capacity to carry the study to completion. If this is not the case, then the PRP should acknowledge this to MDNR and waive its right to perform these studies.

Task 2: Describe Current Situation

The objective of Task 2 is to prepare an up-to-date description of the conditions currently existing at the site. This description will be accomplished by a thorough review of existing files of information regarding the problem of contaminants, primarily volatile organic solvents, in the aquifer and related studies or sources of information. These data will be gathered in conjunction with a site visit to Cadillac to perform a detailed reconnaissance of the proposed investigation locations and of the potential sources in the industrial park area. Meetings will be held to coordinate and/or identify other planned studies with respect to the determination of area groundwater quality or activities relevant to the performance of this RI/FS. One key element in assessing the current situation has already been provided through the MDNR District Office survey and site assessment visits to the industrial park.

During reconnaissance, level D protection will be implemented by the reconnaissance team in the field. No situations are foreseen in which a higher degree of protection would be required. However, as is customary, the team will, in areas where there is a higher potential for the presence of volatile organic chemicals, monitor the area with an HNU photoionization meter, and be equipped with escape respirators.

There is a body of information available for the area which was developed by Jordan for the Northernaire site RI/FS and presented in a Draft RI report dated May 1985. This information will be augmented by the other data collected during this task. These data will be incorporated into a task report describing the current situation. The report will make recommendations for any apparently required modifications to the proposed investigation locations as presented in this RI/FS Work Plan. The content of the report, issued as a technical memorandum, will include observations on the site geology, hydrogeology, contaminant distribution and identification of all possible source or receptor locations as determined during the data collection activities.

Task 2 will also include the preparation of a base map for use during the project by MDNR/Jordan and participating PRPs and their contractors. The base map will be on a scale of 1 inch equals 200 feet. Registered surveyors will be contracted to update and expand an existing map of the study area with the surveyed location of existing monitoring wells and new buildings. Local bench marks will be established in areas likely to require elevation surveys for monitoring well installations. Bench marks will be relative to USGS datum.

Task 3: Prepare Health and Safety Plan

Based on the experiences of Jordan at the Northernaire site and review of data from Kysor and Four Star hydrogeologic studies, the potential for adverse health or safety impacts from identified contamination is rated low. Only relatively low levels of volatile organics have been determined to be present in the groundwater and soils. The highest value reported has been about 79 mg/l in a sample of groundwater from a Kysor well (K-4) in 1983. Levels reported at other locations have been much less. The precautions taken during drilling, however, or for the collection of soil samples, will be done in a manner prescribed by acceptable protocol as outlined in the health and safety plan prepared for the Northernaire Superfund site investigation. Monitoring of wells and soils will be performed with an HNU photoionization meter and, if prescribed level of protection criteria are exceeded, then respiratory protection, maintained at the site, will be utilized through the remainder of the exploration. Specific conditions encountered which may trigger the need for local or general evacuation will be detailed in the HASP as well as the procedures to be followed in case of such an emergency.

The HASP will be written to Jordan's specifications which are in accord with the requirements of the MDNR. Much of the HASP will be directly taken from the EPA approved Northernaire HASP and modified as necessary for the particular tasks of the Cadillac study.

** PRP HASP**

The HASP provided by the PRP or its agent for a specific site should follow the format of the HASP produced by MDNR for the Northernaire site. This HASP is provided as Appendix A for reference. Specific sections of the Northernaire HASP may not be applicable for a specific site; however, other concerns or contact names may need to be added.

*Task 4: Prepare a Quality Assurance Project Plan

A Quality Assurance Project Plan (QAPP) will be prepared for the Cadillac Area Contaminated Groundwater Study. The QAPP will describe the procedures to be used in the collection of samples, generation of data and assessment of data during the project to assure that valid, appropriate methods are consistently applied toward achieving project objectives. The QAPP will also present laboratory QA/QC procedures employed by CLP to ensure conformance with EPA protocol. The QAPP will employ many of the same procedures and methods as the Northernaire program. Economies in the preparation of the QAPP will be attained through use of appropriate sections from the Northernaire QAPP.

** PRP QAPP **

PRPs electing to participate in the study will be provided with copies of the Northernaire Quality Assurance Project Plan (QAPP) to provide guidance for site specific investigation methodology. The work plan should provide references to the appropriate sections of the QAPP to assure data quality. Analytical laboratories used by PRPs for sample analyses must provide QA/QC documentation and be acceptable to MDNR on the basis of capability, capacity, instrumentation, methodology, documentation and other areas of quality control.

Task 5: Preinvestigative Evaluation

Based on information developed during Task 2 - Description of Existing Conditions, a list of potential remedial technologies and remedial alternatives will be prepared for review by MDNR. By identifying potential remedial alternatives at this time, it will be possible to review the RI tasks to be certain that adequate information will be developed to allow the FS to be properly completed in accordance with the project schedule.

** PRP Participation**

Participating PRPs may suggest potential remedial alternatives, but the final list for evaluation in the Feasibility Study will be selected by MDNR.

*Task 6: Site Office

A site office will be established for the duration of the on-site investigations. The site office will be divided into two sections: 1) office space; and 2) staging, equipment preparation, sample handling and storage space. Level D safety procedures are anticipated for all project activities; therefore, an isolated decontamination facility and pressurized water supply will not be provided with the site office. Decontamination and sanitary facilities will be located outside, adjacent to the site office. The decontamination area will be described in the HASP developed in Task 3. The site office will provide space for storing safety equipment and monitoring and sampling gear to be used during the site investigation and will also provide a clean work space for field staff.

** PRP Site Office**

PRPs participating in the study should identify a site office in the HASP and also indicate its acceptability on the basis of providing necessary facilities for conducting field activities.

Task 7: Community Relations

The community relations program for the Cadillac Area Groundwater Contamination study will be conducted by the MDNR with support from Jordan to provide support in the following areas, as directed by MDNR:

- o meet with local officials to determine their concerns and community relations needs;
- o periodically brief local officials regarding current activities during the on-site activities;
- o provide information for press releases during the RI/FS; and

- prepare information to be presented at public meetings.

Community relations activities will be ongoing during the entire project.

2.2 PREPARATION OF DRAFT RI/FS (PHASE II)

A total of 11 tasks have been identified for Phase II.

*Task 8: Geophysical and Geochemical Investigation

Based on:

- experiences with the lack of definitive information from earth resistivity investigations conducted at Northernaire;
- the unfavorable geology for investigation by other techniques;
- the necessity for installation of the borings and wells for collection of chemical and hydrogeologic data as indicated in Tasks 9 and 10;
- the nonionic nature of the hydrocarbon contamination;
- and conversations with MDNR personnel regarding current use of geophysical methods in the vicinity of the former Four Star facility;

it is believed that geophysical methods in general are not likely to be cost effective in generating primary or complementary data for site assessment. As such they are not recommended in this work plan. The results of geophysics conducted at Northernaire and at the former Four Star facility will be reviewed and referenced in the RI/FS report as appropriate.

One sensing method which may offer potential for use at the site consists of probing the shallow unsaturated zone for indications of volatile organics in the soil gas space. First, a hollow tube is driven into the soil. A sample of the soil gas is then withdrawn, e.g., with a syringe, and analyzed with a portable organic vapor analyzer (OVA). Jordan has used a similar technique with some success at other hazardous waste sites. The presence of volatile organics as the primary contaminant of concern at the site and the general presence of coarse grained soils which would promote upward migration of gases may make this method useful. This type of investigation is newly developed and must be applied with care because some geologic or hydrogeologic conditions may not yield good results. The presence of clay layers above contaminated ground-water may inhibit upward migration of volatiles. Also, although the coarse-grained materials would promote high rates of migration, these soils would also permit the greatest amount of infiltration which would, again, suppress volatile organic contaminant migration.

MDNR anticipates conducting a pilot study to indicate the value and sensitivity of the soil gas test. If successful, the method will provide an inexpensive and rapid screening of sites for further study and may eliminate the need for some of the more expensive drilling methods.

Should the pilot study prove favorable, the technique will be used as a screening tool to indicate which suspect sites warrant further investigation as part of the Cadillac Area Study. Readings will be taken at site property boundaries up and down gradient of the indicated groundwater flow direction. Positive indications of the presence of volatile organic contaminants in groundwater will require additional tasks for assessment. Readings will be taken along the property lines at 25 foot intervals. Due to the potential influence of meteorologic conditions on the relative readings, all readings (both up and down gradient) will be performed at a site on the same day.

Analysis would be done using a Photovac organic vapor analyzer (OVA) model 10A10 or similar instrument, probably in the general (non-GC) mode. Detection limits by headspace measurement for commonly encountered solvents should be equal to or less than 10 ug/l in water. An HNU, Photovac TIP or a Foxboro Century OVA is not deemed sensitive enough for purposes of the soil gas testing. If qualitative indications of chemical species present were requested, the instrument could be operated in the GC mode.

Papers dealing with the theory and mechanisms of volatile organic chemical migration through the vadose zone have been published by Swallow and Gschwend (1983) and case histories of the use of the soil gas technique by Lapala and Thompson (1984). MDNR has also had experience in applying the method at other sites. Appendix B provides a description of the soil gas testing as it would be applied in the Cadillac study. Should this prove to be a useful method of contaminant assessment, Keck Consulting Services will be subcontracted for this phase of work.

** PRP Participation**

While a PRP may elect to perform the soil gas testing at a specific site, MDNR would urge that MDNR/Jordan coordinate this task to maintain consistency in methodology, instrumentation (which a PRP or its agent may not have or not wish to acquire), interpretation of data, and to exercise its knowledge gained through experience in applying the method.

Task 9: Contamination Investigation

Soils

The available geologic data provided in the Kysor study, the Four Star study and the Northernaire RI/FS cover a significant portion of the probable extent of the regional groundwater contamination. It will be the purpose of this task to undertake explorations to provide additional soils and geologic data over a larger area than has been explored thus far and to provide data for soils parameters which have not yet been measured. For example, no data have been provided in previous studies regarding the grain size distributions or hydraulic conductivities of aquifer and aquitard materials. This task will include such testing.

Detailed logs of all proposed boring/well installations shown in Figure 2-1 will be developed in the field with special attention given to the depth and thickness of strata encountered. Geologic profiles will be determined from the

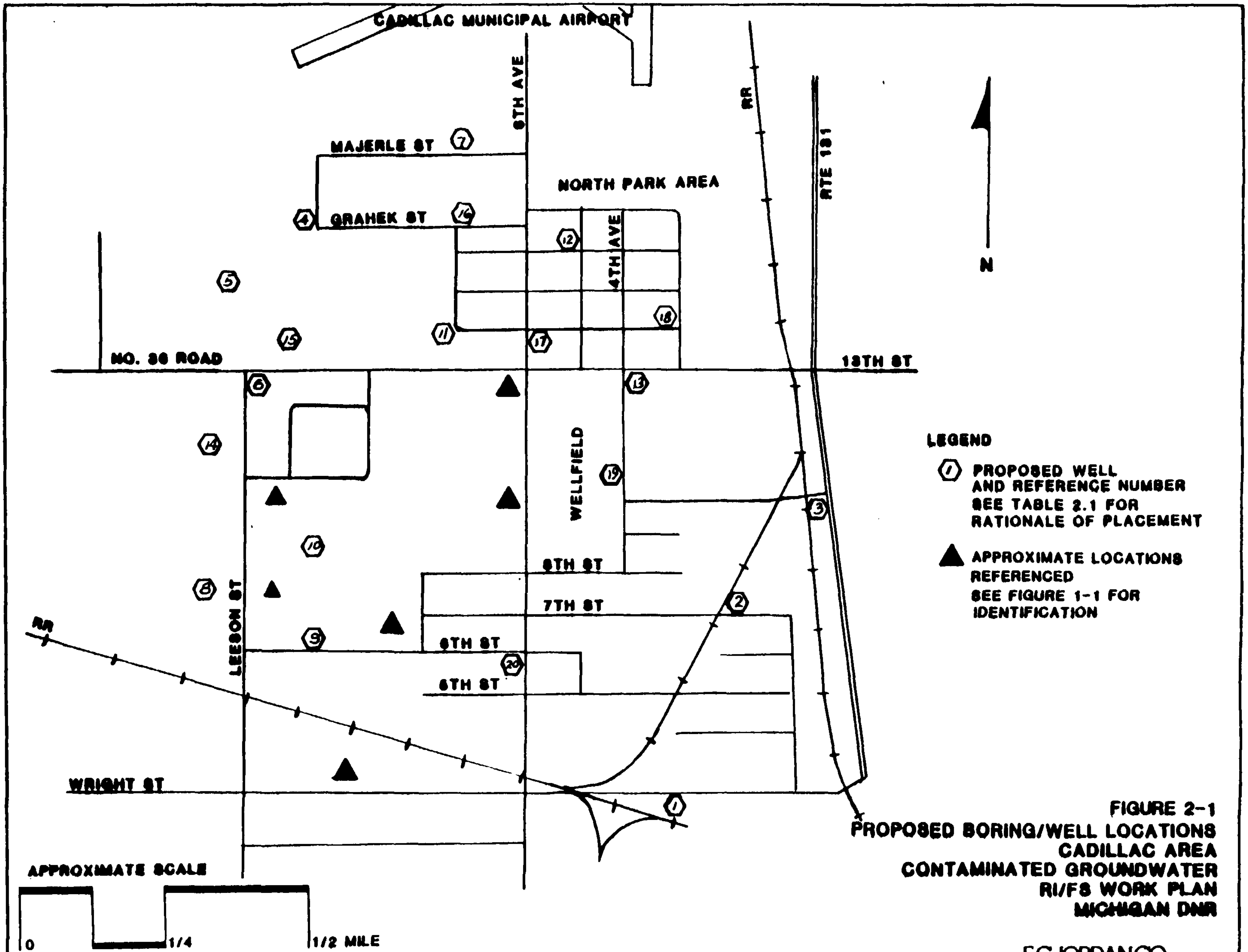

FIGURE 2-1
PROPOSED BORING/WELL LOCATIONS
CADILLAC AREA
CONTAMINATED GROUNDWATER
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boring logs. Samples of both aquitard and aquifer soil materials will be collected for grain size distribution analysis. A soil sample for grain size distribution analysis will be taken at the screen interval of each new well. In addition, four samples each of the upper and lower aquitard materials will be tested for grain size distribution and to determine the saturated hydraulic conductivity of these materials. These data will be necessary in the evaluation of the potentials for vertical gradients and the impact of contaminant migration on the municipal well field as discussed above and in Task 11 (Aquifer Testing).

Soil samples at 5 foot intervals will be taken in each of the 60 proposed source-location holes. The unsaturated zone is approximately 30 feet deep, therefore as many as 360 samples will be taken in the 60 borings. The subsurface conditions encountered may warrant taking surficial or near surface soil samples. It is anticipated that about 200 of the collected soil samples (i.e., the 360 subsurface and 60 surface samples) will be submitted for analysis. EPA protocol for sampling, handling, shipment, chain-of-custody procedures and analysis will be mandatory for soil and groundwater sampling.

<u>** PRP Soil Investigation **</u>

Based on the results of the soil gas testing, visual inspection or historical information, soil samples at a site may be required to define levels of solvent contaminants remaining in the soils. Because much of the natural soils in the area are sands and gravels with little tendency to retain solvents, the expectation of the presence of the solvents in soils would be low. However, if fills, sludges or organic materials are present at the surface or in the subsurface, or if clay lenses or layers inhibit percolation, there may be a potential for such retention. Recent spills may also increase the potential for the presence of contamination in unsaturated soils. Unusually high soil gas readings may also indicate concentrations of solvents in the soils. The PRP should address soil sampling in the work plan, either as an adjunct to the soil gas program or the boring program. If the PRP declines participation in the soil gas program, but accepts the remainder, then MDNR will indicate "hot areas" where soil samples are to be taken by PRPs in addition to the boring program defined in the PRPs work plan.

Soil samples for volatile organic analysis should be collected in 40 ml VOA vials with a minimum of headspace above the sample. The remaining free space in the vial will be filled with deionized water, the amount poured into each VOA vial being recorded in the field book and on the bottle label. Although hold times for soil samples are not clearly specified in EPA protocol, these samples should be iced and sent to the analytical laboratory as promptly as possible.

Chain-of-custody forms and sample well log forms appear in Appendix C and Appendix D, respectively. Copies of these will be required to be used by participating PRPs for recording and transferring information and project documentation.

Groundwater

MDNR/Jordan will conduct a screening survey of the study area using the soil gas testing technique (assuming pilot testing of the method is confirmed) to aid in determining areas requiring further investigation (Task 8). As soil gas data are evaluated, those individual sites evidencing contamination by volatiles will be further explored using borings, field screening and monitoring wells.

The general approach at sites with potential for solvent releases will be to install a series of borings with monitoring wells in the direction of groundwater flow to determine the longitudinal extent of the contaminant plume. Screened auger water samples taken with depth (10 foot increments) will provide vertical profiles of contaminant distribution and indicate the depth at which the screen will be positioned. A second line of wells will be positioned at right angles to the first at the point of highest concentration to determine both lateral extent and vertical distribution. Additional wells will be placed to provide complete definition of the plume.

At sites with known contamination, borings/wells will be positioned so as to complete the description of the contaminant plume based on currently available data. The decision to utilize data generated in the past will partially depend on the groundwater flow velocity in the aquifer and the date of last sampling. In the shallow aquifer, where the groundwater flow velocity has been estimated at 200 feet per year, data of greater than one year old are probably of little use in describing the current extent of the plume, but may be valuable in assessing movement and fate.

The number of wells to be installed during this phase of investigation (source and plume definition) is indeterminant pending the results of the soil gas exploration. However, for purposes of budgeting for this remedial investigation, a total of 60 wells each to a depth of 90 feet has been assumed. This has been based on the assumption of shallow contamination in groundwater associated with contaminants close to their sources. If a plume is more extensive than believed, deeper wells (particularly in the intermediate aquifer) may be required for plume definition.

A typical well installation would consist of:

- o 2-inch ID galvanized steel riser completed approximately 2 to 2.5 feet above ground surface;
- o 5-foot section of stainless steel screen (.010-inch slotted or wire wound) situated at the level of greatest contamination as determined by field screening;
- o backfill of clean sand (natural soils may be acceptable in this setting) to about 5 feet above the screen;
- o bentonite or bentonite/cement seal at any aquiclude encountered or penetrated;
- o bentonite or bentonite/cement slurry seal to the surface;
- o PVC adapter with threaded plug to cap the riser;
- o steel protective casing grouted at least 24-inches into the ground surface over the riser, lockable with padlock;
- o identifying metal tag (ID number, depth and installation date) inside casing.

The well location and well riser elevation will be surveyed relative to the USGS datum by a local registered surveyor under subcontract to Jordan.

This investigation of sources will provide a great deal of information relative to the contaminants present in the study area aquifer systems. However, the soil gas survey is not considered to be infallible and in some areas, plumes may have migrated far enough off-site to ellude detection by this approach. To avoid missing such contaminants and to investigate the potential for deeper contaminants as has been seen in Kysor well K-16 (deep) and MW-7 and MW-8 installed for the Northernaire RI/FS, an array of deep and shallow wells will be installed. The exact locations of these wells will be influenced by the findings of the source and plume extent identification wells, but, based on current knowledge, 26 wells (6 shallow and 20 deep) have been tentatively located on Figure 2-1 and listed on Table 2.1. These wells will provide geologic and hydrogeologic data including vertical gradients at some locations as well as chemical data on groundwater quality.

These wells will be constructed similarly to the shallow wells in the source identification program, but are anticipated to range from about 150 to 185 feet deep based on available information from the Northernaire RI/FS and other studies which have located the upper surface of the lower aquitard. Explorations to this depth have been successfully completed using hollow stem augers.

Each well will be developed by hydraulic surging using the air lift technique. The air compressor used will be screw type equipped with a Deltech Engineering Inc. Model 130 hydrocarbon filter or equivalent. The filter is designed to

TABLE 2.1

WELL INSTALLATION SUMMARY[1]

PHASE	WELL ID	APPROX. DEPTH	PURPOSE
General geology, hydrogeology and contaminant distribution determination	1S, 1D	60 & 180	Background
	2	180	Background
	3	180	Background
	4	180	Downgradient, flow direction
	5	180	Downgradient, flow direction
	6S, 6D	60 & 180	Downgradient, flow direction
	7	180	Downgradient, flow direction
	8	180	Lateral limit
	9S, 9D	80 & 180	Matrix data, flow direction
	10S 10D	80 & 180	Matrix data, flow direction
	11	180	Matrix data, flow direction
	12	180	Matrix data, flow direction
	13	180	Matrix data, flow direction
	14S, 14D	60 & 180	Downgradient limit
	15	180	Matrix data, flow direction
	16	180	Matrix data, flow direction
	17	180	Matrix data, flow direction
	18	180	Matrix data, flow direction
	19	180	Matrix data, flow direction
	20S, 20D	60 & 180	Matrix data, flow direction
Source and plume extent determination	21-80	90	Source location

(1) Refers to numbered locations on Figure 2-1

change color with use and will be replaced with sufficient frequency to ensure that no oil enters the well. The filter will be inspected prior to and during each well development episode. Upon completion, each monitoring well will be provided with a locking protective steel casing securely grouted in place.

** PRP Groundwater Investigation **

PRPs participation in the groundwater contaminant investigation will be limited to the explorations to identify the presence, extent and distribution of site related contaminant plumes. Numbers of wells required for this task will obviously vary from site to site, but must be sufficient to achieve the goals of the investigation. MDNR/Jordan will provide participating PRPs with well log forms, well installation details and a preassigned block of identification numbers to maintain uniformity in designations and avoid confusion of possible duplication in well identification. In addition, MDNR will require that the boring/well installations will be monitored and logged by a qualified geologist. Drilling contractors who have experience with installation of monitoring wells (as opposed to water supply wells only) should be selected.

The drilling monitor/supervisor will use the Unified Soil Classification System of soil descriptions to maintain uniformity in recording and interpreting boring logs. The guidelines of this classification system are provided in Appendix D. Soil horizons encountered and features such as clay lenses or layers should be indicated as closely as possible. The presence of clay layers will be a determining factor in positioning the screened zone of the well and setting seals. In no instance should the lower clay aquitard be penetrated (which will be encountered approximately 150-180 feet deep). If the upper aquitard is penetrated for an installation, a bentonite seal will be placed to completely reseal the aquitard zone around a well.

As in the program description above, groundwater will be sampled with depth with field screening of headspaces above water samples with an organic vapor analyzer to determine the zone of greatest contamination. The well screen will be placed in this zone. Data obtained from the screenings will be included in the report of findings. The instrument used for the screenings must be specified in the work plan and provide sufficient sensitivity for detection by headspace measurement of volatile organics in water at concentrations less than or equal to 10 ug/l.

The general sequence of installations after the determination of local ground-water flow direction will include placement of monitoring wells from the source in the direction of flow at 200' intervals (suggested) to determine the longitudinal extent and vertical distribution of the contaminant plume. At the point of highest contamination, a second set of wells, perpendicular to the first, will be installed to assess the lateral spread of the plume and vertical distribution. Sampling at 10-foot intervals with depth will be conducted to identify the zone of highest contamination for proper screen placement.

Detailed well installation specifications and guidelines appear in Appendix D.

*Task 10: Groundwater Sampling

The purpose of this task is to collect groundwater quality data to assess current and future impacts of contamination which is present in the study area aquifers and which may be present in soils in areas near sources.

All new wells and available wells that are pertinent to this investigation will be sampled during this task. Key to the detailed representation of the vertical distribution of contaminants will be the collection of samples of groundwater at ten foot intervals as the borings are advanced. Samples from the completed wells will be taken at levels at which the initial samples have been determined by screening to be the most contaminated. Existing wells that will be of particular interest are those defining shallow contamination in the vicinity of Kysor and Four Star. The groundwater velocity in the shallow aquifer is estimated to be 200 feet per year, thus tending to move plumes away from sources rapidly. If contamination is detected in the near vicinity of these facilities, it may be an indication that some solvents remain in soils to provide a continuing source.

As part of the sampling methodology, the well will first be checked for proper identification and location. The height of protective casing will be measured and recorded. The photoionization meter will be used to measure the ambient air levels at each well site location. After removing the well caps, the ambient and well-mouth vapor levels will be measured and recorded. If the ambient air quality at breathing level reaches 5 ppm, as described in the

Health and Safety Plan, sampling personnel will utilize the appropriate safety equipment.

The electronic water level meter will be used to measure and record the static water level in the well (to the nearest 0.01 ft) and the depth to the well bottom (in wells <100 ft deep). Upon removing the water level measuring line, it will be rinsed with methanol and distilled water. Water levels are to be obtained from wells before any purging and/or sampling.

Following these measurements, sampling will commence in the sequence below:

1. Lower the submersible pump into the well to the top of the water surface. Begin flushing the well lowering the pump as the water level drops. Remove a minimum of three well volumes of groundwater from the well.

2. Obtain a groundwater sample by means of the submersible pump or bailer.

3. Record the in situ parameters, i.e., pH, temperature and specific conductance, immediately after sampling.

4. Remove the pump from the well and decontaminate the pump and tubing by flushing two gallons of methanol/water mix through the pump and tubing followed by two gallons of distilled water. Using paper towels and methanol, hand-wipe the outside of the teflon tubing and pump. Rinse with distilled water. Sampler blanks will also be taken as part of the QA/QC

for the project with a minimum of two sampler blanks (before and after sample event) or 10 percent if more than 20 samples.

5. All sample bottles will have the appropriate labels attached beforehand. The chain-of-custody forms will be completed after each well is sampled.

6. All sample containers will be filled directly from the discharge tubing of the pump.

7. Secure the well cap and lock.

The monitoring well samples will be collected by Jordan and shipped to a laboratory in the EPA CLP. Sample collection and handling will be in accordance with EPA guidelines as described in the QA/QC plan. The analytical program is described in Task 12.

** PRP Participation **

PRPs participating in the program and taking samples will use standard EPA protocol for sample collection. These procedures are described in several references, notably EPA SW-846 entitled "Test Methods for Evaluating Solid Waste". This document is listed as stock number 055-002-81001-2 and may be obtained through the U.S. Government Printing Office in Washington, DC. PRPs may defer sampling to MDNR/Jordan but have the samples sent to the laboratory of their choice (approved by MDNR) for analysis. This option should be clearly stated in the work plan for scheduling purposes. The primary considerations in addition to proper containers and sampling technique are the provision of chain of custody documentation and meeting required hold times specified in the protocols for various parameters.

Task 11: Aquifer Testing

Current information concerning the hydrogeology of the study area indicates the need for aquifer testing. The purposes of the proposed aquifer testing are to provide: 1) data which may be utilized in determining the potential for contamination reaching the municipal well field, 2) more data for estimating groundwater velocities in the intermediate aquifer and, 3) data to aid in remedial design evaluation. The potential for impact on the well field depends on the hydraulic conductivity and thickness of the aquitard material over the well field aquifer and the gradient which develops across the aquitard as a result of the difference between the heads in the upper aquifer (which would remain relatively constant) and those in the municipal well aquifer which would decrease with increased pumpage (increasing the differential head across the aquitard).

It is possible, however, that the hydraulic conductivity and thickness of the lower aquitard may be sufficient to prevent any measurable change in the heads in the upper aquifer due to pumping from a select municipal well or wells. Therefore, it is recommended that a careful analysis be performed on the probability of such a test producing usable data. The proposed 72-hour pump test on the well field could be simulated with a numerical model using available data and data resulting from determinations of the grain size and hydraulic conductivity testing in Task 8 to provide more guidance for this decision.

The decision on the exact course of action should await data obtained during the earlier phases of this RI/FS. If data indicate no significant potential for impact on the lowest aquifer, the need for further testing of the municipal well field under this task item may be negated. The larger of the estimated labor and other costs have been included as budget items for the task alternatives in the budget summary. An estimate to perform some limited and simple modeling of the proposed pump test has also been included.

No determinations of aquifer hydraulic conductivity (K) have yet been made based on actual field data. K values have been estimated based on literature values for the observed soil types in the area. The selected value for K (0.01 cm/sec) is somewhat confirmed by the approximate travel times for known sources and plumes. However, the estimates may still be only correct within an order of magnitude. To provide more accurate data for the purposes of the groundwater model which will be used to assess remedial actions, in situ permeability tests will be performed on selected wells. Because the soil permeability is so high, and changes in the water levels are expected to change very rapidly during the tests, electronic apparatus with pressure transducers and recording capabilities will be used to measure and record the water levels with time. Rising head tests are preferred to eliminate the introduction of non-formation water into the well which may affect water quality determinations.

If the soil matrix is so permeable and slug tests for measuring hydraulic conductivity in wells cannot be accurately measured, a second pump test is recommended. This test would gather data on the intermediate aquifer where most of the contaminants are present and for which only estimates of hydraulic conductivity are currently available. Available data will be reviewed to assess the need and scope of this test. For purposes of budgeting this task, a 24-hour pump test is planned using the existing extraction pumps and with the installation of 5 3/4-inch PVC observation wells to an average depth of 60 feet to measure resultant drawdowns.

Also included in this task will be the measurement of water levels (to the nearest 0.01 ft) in all new wells and in selected existing wells. Although water levels will have been obtained during the collection of water samples for analysis, those data will have been collected over time. The water level data collected for this task will minimize the potential for time variations and also include data from wells not sampled for chemical analysis. The water level data will be used to describe groundwater flow in the shallow and intermediate aquifers over a larger area than has yet been studied. Also, because the thickness and quality of the lower aquitard is not known over the study area, maximum potentials for transfer from the intermediate aquifer to the lowest may actually occur away from the well field. Of particular concern are conditions downgradient of known contaminants.

Task 12: Analytical Program

The proposed analytical program to be conducted is summarized in Table 2.2. Analyses will be performed by a laboratory approved for EPA's Contract Laboratory Program (CLP). All sample collection, handling, shipment and analytical procedures will be in accordance with approved EPA CLP protocol and will be described in detail in the QAPP (Task 4) to be prepared for the project.

Bids will be solicited from at least 3 CLP qualifying laboratories for cost savings purposes. Costs may range considerably from lab to lab depending on current laboratory capacities and work loads. Typical costs for CLP analyses for metals and VOA have been budgeted to estimate laboratory costs.

** PRP Participation **

PRPs participating in the study may select their own laboratory. A list of certified CLP laboratories appears in Appendix E. In this event, the PRP will be required to submit quality assurance/quality control information in general for the lab performing the analyses and for the particular methodologies to be used for sample analyses. MDNR will also require 10 percent split samples to be sent to the EPA CLP as an additional QA/QC check. Costs for the analyses of these split samples will be assumed by the PRP.

PRPs should identify the selected laboratory as early as possible to allow MDNR to review qualifications (to be supplied by the PRP/lab) and to approve the

TABLE 2.2
PROPOSED SAMPLES FOR ANALYSIS

Media/Description	Locations	Samples(1) VOA	Metals(2)	CN
Groundwater				
New wells for chemical distribution				
During advancement	20	360*	0	0
After completion (2 rounds)	26	60	24	24
Existing wells (2 rounds)	14	50	4	4
New wells for source identification				
During advancement	60	360*	0	0
After completion (2 rounds)	60	144	0	0
Existing wells (2 rounds)	25	60	0	0
Soil				
During source identification	60	220	0	0

(1) Includes 10% blanks plus 10% duplicates.
(2) Metals analysis for total and hexavalent chromium, cadmium, lead and copper.
* Field screening only; headspace by OVA
Second sampling after assessment of first round data.

selection. PRPs should select a lab based on certification, experience, QA/QC procedures, turnaround time and capacity as well as cost.

Task 13: Data Interpretation/Contaminant Assessment

The objective of this task is to compile all data available from the area into a complete hydrogeologic evaluation of the study area. All existing information and data collected during Tasks 2, 5 and 8 through 12 will be used for this portion of the work.

The interpretation and assessment for this project will include:

- Determination of the horizontal and vertical distribution of contaminants in the groundwater.
- Determination of the directions of groundwater flow in the shallow and intermediate aquifers over the study area.
- Evaluation of the probable fate of the contaminant plume(s).
- Evaluation of the hydraulic connection between the individual aquifers.
- Identification of sources of groundwater contaminants, if possible.
- Evaluation of the potential for contamination from the upper aquifers to impact water quality in the aquifer used for the municipal water supply.

- Evaluation of the potential for impact on other private water supply wells in the study area.

A contaminant exposure assessment will be conducted to assess the potential risks posed by the groundwater contamination and to assist in the establishment of project response objectives. A key element of this task is the development of an appropriate predictive analog model of the site. The development of a comprehensive and well-calibrated model is a valuable tool in achieving an understanding of present and future transport pathways for contaminants in groundwater. This model is an integral feature of the work program, inasmuch as it provides the basis for predicting behavior and dispersion of contaminants under a "no-action" alternative and is used to estimate the effectiveness of selected remedial action alternatives. MDNR has expressed a desire to assume the lead in the development of a model for the area. Meetings will be held with MDNR to discuss the modeling effort and the level at which Jordan will provide support. The schedule for model development is critical in the schedule for the completion of the feasibility study.

Jordan is prepared to provide complete development of the model in cooperation with and under the guidance of MDNR's principal modeler, Mr. David Hamilton, if the schedule and MDNR's work load so require. Budget estimates for the modeling have been established assuming the higher level of Jordan involvement.

** PRP Participation **

PRPs will provide a presentation and assessment of the investigation which they have conducted for their own site. Items in the investigation for which the PRPs are not responsible are the evaluation of the hydraulic connection between the intermediate and lowest aquifers and evaluation of potential impacts on the municipal well field.

The report will utilize graphical representations of data insofar as possible. All data and information, such as boring logs, field notes and chemical analyses, pertinent to the investigation will be included. These may be incorporated as appendices if more appropriate to the reading of the report. Participating PRPs will be provided with a format for the report which will be modelled after the Northernaire RI report. A draft report will be submitted for review by MDNR/Jordan. A final report will be due three weeks after the PRP receives comments from MDNR on the draft report. MDNR/Jordan will be responsible for combining all individual reports into a final draft RI.

This task completes the requirements for PRP participation in the study. PRPs may make recommendations for alternatives for remedial action. These will be taken under consideration by MDNR in the selection of alternatives for evaluation in the feasibility study.

Task 14: Development of Alternatives

Based on the understanding of the site obtained during Task 2, a reasonable number of applicable remedial alternatives for the Cadillac Area study will be developed. These alternatives will range from the no remedial action alternative to total aquifer restoration (i.e. purge and treatment of groundwater, removal of contaminated soils), and the associated impacts. The screening of the alternatives involves several steps which are described below.

a. Establish Remedial Response Objectives. The first step in developing the remedial alternatives for an uncontrolled hazardous waste site is the establishment of the response objectives or cleanup goals. Although this is not a CERCLA site, Jordan proposes to establish response objectives as the unifying theme to guide the logic of the remedial investigation field program and the development and evaluation of cleanup alternatives during the feasibility study. The response objectives of the proposed remedial action will be established in accordance with the guidance provided in the National Oil and Hazardous Substances Contingency Plan (NCP) (40 CFR Part 300.68), MDNR, the EPA interim guidance, and the requirements of any other applicable federal, state or local regulation, guidance or policy.

The remedial response objectives will provide a benchmark by which to evaluate the effectiveness of remedial alternatives. The recommendation of a cost-effective remedial alternative will be based on extent to which the alternative meets the response objectives along with consideration of cost, technology, and reliability. The response objectives will be based upon adequate protection of public health and the environment.

Proposed response objectives will be established based on data reviewed in Task 2. These objectives will be reviewed during the RI as new site data are generated. If it appears that the objectives need to be modified, recommended adjustments will be submitted to MDNR for review. The proposed response objectives will guide the identification, development and screening of remedial alternatives. Recommendations for response objectives for the remedial action at the Northernaire site will be developed in close consultation with MDNR.

b. <u>Identification of Remedial Technologies</u>. The objective of the FS is to evaluate proposed remedies. Remedial alternatives are developed from specific technologies which are applicable to the conditions at the site. Therefore, an early task in the FS is to identify remedial technologies applicable to the site. Potential remedial technologies are normally divided into two broad categories:

- <u>Management of Migration</u> - Actions that are taken to minimize and mitigate the migration of hazardous substances or pollutants or contaminants and the effects of such migration. Management of migration actions may be appropriate where the hazardous substances or pollutants or contaminants are no longer at or near the area where they were originally located or situations where a source cannot be adequately identified or characterized. Measures may include but are not limited to provision of alternative water supplies, management of a plume of contamination or treatment of drinking water aquifer.

- o Source Control Remedial Action - Measures that are intended to contain the hazardous substances or pollutants or contaminants where they are located or eliminate potential contamination by transporting the hazardous substances or pollutants or contaminants to a new location. Source control remedial actions may be appropriate if a substantial concentration or amount of hazardous substances or pollutants or contaminants remain at or near the area where they are originally located and inadequate barriers exist to retard migration of hazardous substances or pollutants or contaminants into the environment. Source control remedial actions may not be appropriate if most hazardous substances or pollutants or contaminants have migrated from the area where originally located or if the lead agency determines that the hazardous substances or pollutants or contaminants are adequately contained.

Based on our review of the conditions at the site, it appears that a combination of remedial technologies may be applicable to the site which are described below. Upon completion of the RI, the list of technologies will be refined to identify those most applicable to the site conditions and response objectives.

1. Source Control remedial actions that are considered applicable may include:

- o excavation of contaminated soils and waste deposits,

- covering of deposits of waste and areas of contaminated soil, and

- a clay cap or some other capping mechanism as a part of the final site closure.

2. Management of Migration remedial actions that may be used to control the wastes and chemical substances at identified sources include, for example:

- extraction via pumping, of on-site or off-site contaminated groundwater.

c. Identification of Remedial Alternatives. From the remedial technologies identified above, Jordan will develop site-specific remedial alternatives for the site. In addition to source control and management of migration alternatives, the remedial alternatives identified will include "no-action".

The development of appropriate remedial alternatives for an uncontrolled hazardous waste site requires a comprehensive knowledge and understanding of the physical setting and contaminant sources at the site. The NCP (300.68 - Remedial Action) identifies procedures to be followed.

Criteria for identifying appropriate, site-specific remedial alternatives will be established based on the NCP and applicable State of Michigan policies and statutes. These criteria include: acceptable engineering practice, effectiveness and cost.

In developing the remedial alternatives, Jordan will maintain close coordination with MDNR. The identified remedial alternatives will be reviewed with MDNR prior to proceeding with the screening phase. Once MDNR has concurred with the potential remedial alternatives identified for the site, Jordan will proceed with the initial screening task.

Task 15: Initial Screening of Alternatives

This task involves the initial screening of remedial alternatives identified in Task 14. The objective of the initial screening is to screen acceptable engineering practice, effectiveness, and cost.

- acceptable engineering practice is defined as the technical feasibility of implementing a given remedial action
- effectiveness is defined as the relative contribution of our alternative toward the protection of public health, welfare, and the environment
- cost is estimated within an accuracy of +100% to -50%

Upon completion of the initial screening, a limited number of remedial alternatives appropriate for the site will have been identified.

Task 16: Laboratory Studies (Optional)

In developing and evaluating remedial alternatives for the site, there may be insufficient data available to enable definitive conclusions to be reached concerning the applicability of certain technologies. The objective of laboratory and bench-scale treatability studies is to develop, if necessary, additional data for use in the detailed engineering evaluation of remedial alternatives.

To determine the need for laboratory studies, Jordan will review the adequacy of available site data and literature regarding the treatability of the waste substances found at the site by those technologies identified as being appropriate for use at this site. If the available information is found to be inadequate to allow evaluation of one or more specific technologies the need for and scope of bench-scale laboratory or pilot-scale field studies will be presented to and reviewed with MDNR and for approval. This work plan does not include any budget for laboratory studies.

Task 17: Detailed Analysis of Alternatives

The development and initial screening of alternatives will result in a limited number of alternatives which will be evaluated in sufficient detail according to the NCP. These evaluations will include the following:

A. Refinement and specification of alternatives in detail, with emphasis on use of established technology;

B. Evaluation in terms of engineering implementation, reliability, and constructability; and

C. Detailed cost estimation, including operation and maintenance costs and distribution of costs over time.

Additional consideration of each alternative will include:

1. An assessment of the extent to which the alternative is expected to effectively prevent, mitigate, or minimize threats to, and provide adequate protection of, public health, welfare, and the environment;

2. An analysis of whether recycle/reuse, waste minimization or destruction or other advanced, innovative or alternative technologies is appropriate to reliably minimize present or future threats to public health, welfare or the environment;

3. An analysis of any adverse environmental impacts, methods for mitigating these impacts, and costs of mitigation.

A working definition for each of the seven categorical considerations required by the NCP is presented below.

Engineering Constructability

An assessment based on actual case histories, as to whether or not the technology under consideration has actually been built, implemented or constructed under similar site conditions and by available contractors.

Engineering Reliability

An assessment based on actual case histories, as to how well the technology under consideration performed once constructed, generally evaluated against the occurrence and extent of repairs and maintenance required under similar site conditions.

Engineering Implementation

An assessment based on actual case histories as to the degree of difficulty involved in constructing a technology and the time required to complete construction and the time (design life) to achieve the response objectives (levels of cleanup). This parameter also includes institutional barriers to implementation such as zoning restrictions and permitting.

Cost

Cost estimates after the detailed screening will be refined to within an accuracy of +50 to -30%.

Level of Protection

An Assessment based on volume and concentration calculations associated with each alternative as to the cleanup level which will be obtained from implementing the technology. The levels are then compared to known standards and guidelines.

Volume Reduction

Based on current literature*, recycling/re-use, waste minimization or destruction or advanced, innovative or alternative technologies are researched and considered with respect to their direct application to the site. For the technologies screen during initial screening, volume reduction will be considered as a component of the detailed screening.

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* Reference: U.S. EPA Remedial Action Costing Procedures Manual, March 9, 1984.

Adverse Environmental Impacts

For each alternative undergoing detailed screening, adverse environmental impacts are evaluated with respect to the actual field implementation of each alternative. An example of this type of evaluation is: in constructing a temporary storage area for contaminated soils, could drainage be controlled sufficiently to prevent contaminated runoff from reaching a nearby wetland or stream?

Task 18: Draft RI/FS

A draft RI/FS will be prepared summarizing Tasks 8 through 17. The report will present remedial alternatives which meet the selected remedial response objectives with supporting information developed in Task 17 to allow MDNR to select the appropriate remedial response(s) based on public health and environmental protection concerns, economic and technical feasibility, and statutory requirements.

2.3 FINAL RI/FS REPORT (PHASE III)

Task 19: Conceptual Design

Following evaluation of the draft RI/FS report and the selection of the appropriate remedial action(s) by MDNR, Jordan will perform a conceptual design of the selected alternative. The conceptual design will include traditional engineering elements associated with conceptual designs, as well as other data necessary to prepare a budget level cost estimate and to provide adequate information to enable subsequent activities to be initiated. The conceptual design will include the following:

- A conceptual plan view drawing of the overall site, showing general locations for project actions and facilities;

- Conceptual layouts (plan and cross-sectional views where required) for the individual facilities, other items to be installed, or actions to be implemented;

- Conceptual design criteria and rationale;

- Description of types of equipment required, including approximate capacity, size and materials of construction;

- Process flow sheets, including chemical consumption estimates and a description of the process, if applicable;

- Operational description of process units or other facilities;

- Approximate piping sizes, capacities, and rationale for their selection;

- Estimate of quantities of material or equipment required and rationale;

- Description of construction techniques and operation and maintenance requirements;
- Construction material requirements and rationale;
- Utility requirements and rationale;
- Institutional requirements including environmental permits;
- Identification and evaluation of potential construction problems, associated risks, and the proposed solutions;
- Outline of technical specifications and protocol;
- Outline of operation and maintenance manual;
- Outline of safety plans including cost impact on implementation;
- Right-of-way requirements;
- Description of technical requirements or environmental mitigation measures;
- Additional engineering data required to proceed with design;
- Construction permit requirements;

- Temporary hazardous material storage and disposal requirements and rationale;

- Off-site disposal procedures, including transportation and vehicle constraints, and final disposal and treatment facility options;

- Closure and long-term monitoring requirements and rationale;

- Performance standards to define what levels of cleanup will be required to complete the remedial action;

- Criteria for subcontractor selection;

- Budget level cost estimate (including capital and operation and maintenance costs);

- Implementation schedule; and

- Additional narrative information required as the basis for the completion of the final remedial design.

Task 20: Final RI/FS Report

A final RI/FS report will be prepared at the completion of the conceptual design. Draft review comments provided by MDNR will be incorporated into the final RI/FS report. Maximum use will be made of graphic and tabular formats to simplify the presentation of project data. The report will contain an executive summary and pictorial graphics suitable for public presentation and/or distribution.

3.0 MANAGEMENT PLAN

This section of the work plan outlines the management plan which will be used to complete the Cadillac Area Groundwater Contamination RI/FS. The following information describes the project organization, project staffing plan and the project management procedures which will be followed in the conduct of this RI/FS.

3.1 PROJECT ORGANIZATION

Figure 3-1 outlines the project organization and project staffing plan for the Cadillac Area RI/FS.

As prime contractor, Jordan will have responsibility for subcontracted activities. Keck Consulting Services, Inc. (Keck) will be utilized as a drilling subcontractor during the conduct of this RI/FS. Keck will provide drilling equipment, material and drillers as well as conduct the soil gas testing of Task 8. Jordan will direct the driller's activity and supply drilling monitors to log, coordinate and document drilling operations. In addition, Jordan will provide drilling specifications, locate drilling sites and interpret geologic conditions. Keck will provide mapping services for generating base maps and also perform the soil gas test procedures.

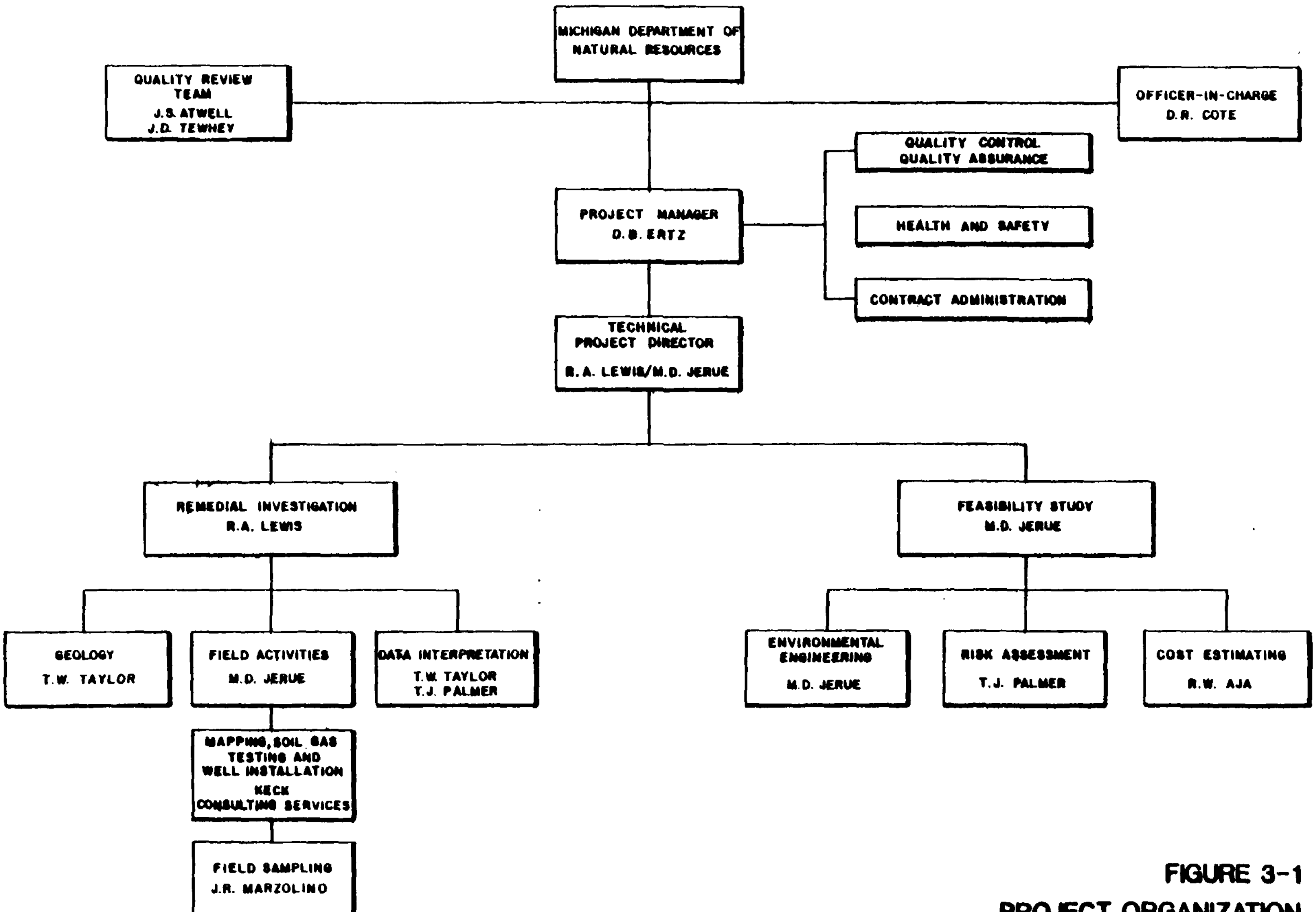

FIGURE 3-1
PROJECT ORGANIZATION
CADILLAC AREA PROJECT
MICHIGAN DEPARTMENT OF NATURAL RESOURCES

E.C. JORDAN CO.

4656-51

The project organization for the Cadillac Area site will ensure close interaction between MDNR and Jordan. Jordan, as prime contractor, will provide the project management and technical supervision on the project. David B. Ertz, P.E. will be the Project Manager and will be the primary contact between MDNR and the project team on issues relating to the contract, scope, schedule, budget and subcontractor relationships. Administratively, his responsibilities are to ensure that the project is proceeding on schedule and that the budget for the various phases is maintained.

The Technical Project Director assignment will be shared between Matthew D. Jerue and Ronald A. Lewis.

o Ronand A. Lewis has primary technical responsibility for the Northernaire project and is familiar with the geologic and environmental conditions surrounding the Cadillac Area. Mr. Lewis will have overall technical responsibility for the interpretation of project data, the description of site conditions, and the development of remedial alternatives for the project area.

o Matthew D. Jerue will have responsibility for the accomplishment of field activities for the Cadillac Area project. Because of the magnitude of the field program, we are assigning the responsibility of coordinating and conducting field tasks to Mr. Jerue in the Southfield, MI office. His responsibilities will include:

- coordination/scheduling of drilling activities;
- coordination/scheduling of field staff including geologists and sampling crews; and
- direction and management of field activities.

Resumes of the key personnel scheduled to participate in this RI/FS have been provided as Appendix F.

3.2 PROJECT MANAGEMENT

To assist in the overall management of the project and to track project and work assignment schedules, budgets and manpower requirements, including those of the subcontractors, Jordan will utilize computerized management information systems. Through the use of these systems, Weekly Project Status Reports will be produced for submission to MDNR.

To monitor manpower utilization and costs, separate account numbers will be assigned for each phase of the project. Labor expenditures will be allocated to the appropriate account on a weekly basis. This information will be compiled weekly in the Weekly Project Status Report. This report will include the following information for each phase:

o name of each employee and number of hours charged to the work phase that week;

- total hours charged to the work phase by technical discipline for that week;

- total hours charged to date to the work phase by technical discipline;

- total hours charged to the work phase that week;

- dollar value of labor expenditures for that week;

- cumulative labor cost for the work phase;

- cumulative labor rate per hour versus budgeted rate; and

- other direct costs for the month and on a cumulative basis.

These weekly data will be compared to project budgets to determine project status. These administrative tools will be used by Jordan in the scheduling of work assignments and allocating staff resources.

This report will be used to: 1) determine if sufficient resources are being committed to each assignment; and 2) identify staff or budget variances as they develop. The reports will also be valuable in determining the status of the project on a weekly basis for preparation of weekly progress reports.

The budget for each phase will provide the basis for tracking project expenditures. The weekly computerized project status reports will be submitted to the

MDNR Project Administrator. These data will be supplemented by a brief written summary of work accomplished during that week, problems that developed, and steps taken to resolve any problems.

Project invoices will be submitted to MDNR every four weeks. At a minimum, Jordan's Project Manager/Technical Director and MDNR's Project Administrator will meet bi-weekly, if necessary, to review project status.

3.3 PROJECT PERSONNEL AND STATUS:

Table 3.1, Labor Cost Summary for Jordan, has been prepared to identify project personnel, project hours and hourly rates. The table has been sealed in an enclosed envelope at the back of the Work Plan. The current status of all proposed field personnel in the Jordan health monitoring program will be listed in the HASP developed in Task 3. If it becomes necessary to replace or to add to the list of personnel, the MDNR will be notified of the changes or additions and supplied with equivalent information.

4.0 SCHEDULE AND COST

4.1 PROJECT SCHEDULE

The schedule for the Cadillac Area Groundwater Contamination RI/FS is shown in Figure 4-1. The schedule indicates that approximately 15 months are required to complete the RI/FS following approval of the Work Plan. The project schedule shows the development and initial screening of remedial alternatives occurring during the RI. This will enable the detailed evaluation to be initiated during the data interpretation phase when site-specific data are available.

Completion of the RI/FS on schedule is contingent upon 30-day turnaround of analytical results. Also, MDNR review must be completed in a timely manner to allow for completion of the RI/FS within the designated time period.

Deliverables (reports) will be submitted to MDNR at the conclusion of the following Tasks:

Task 1 - Work Plan and Quality Assurance Project Plan
Task 2 - Description of Current Situation
Task 3 - Health and Safety Plan
Task 4 - Preinvestigative Evaluation
Task 13 - Data Interpretation/Contamination Assessment
- o Response Objectives
- o Remedial Alternatives

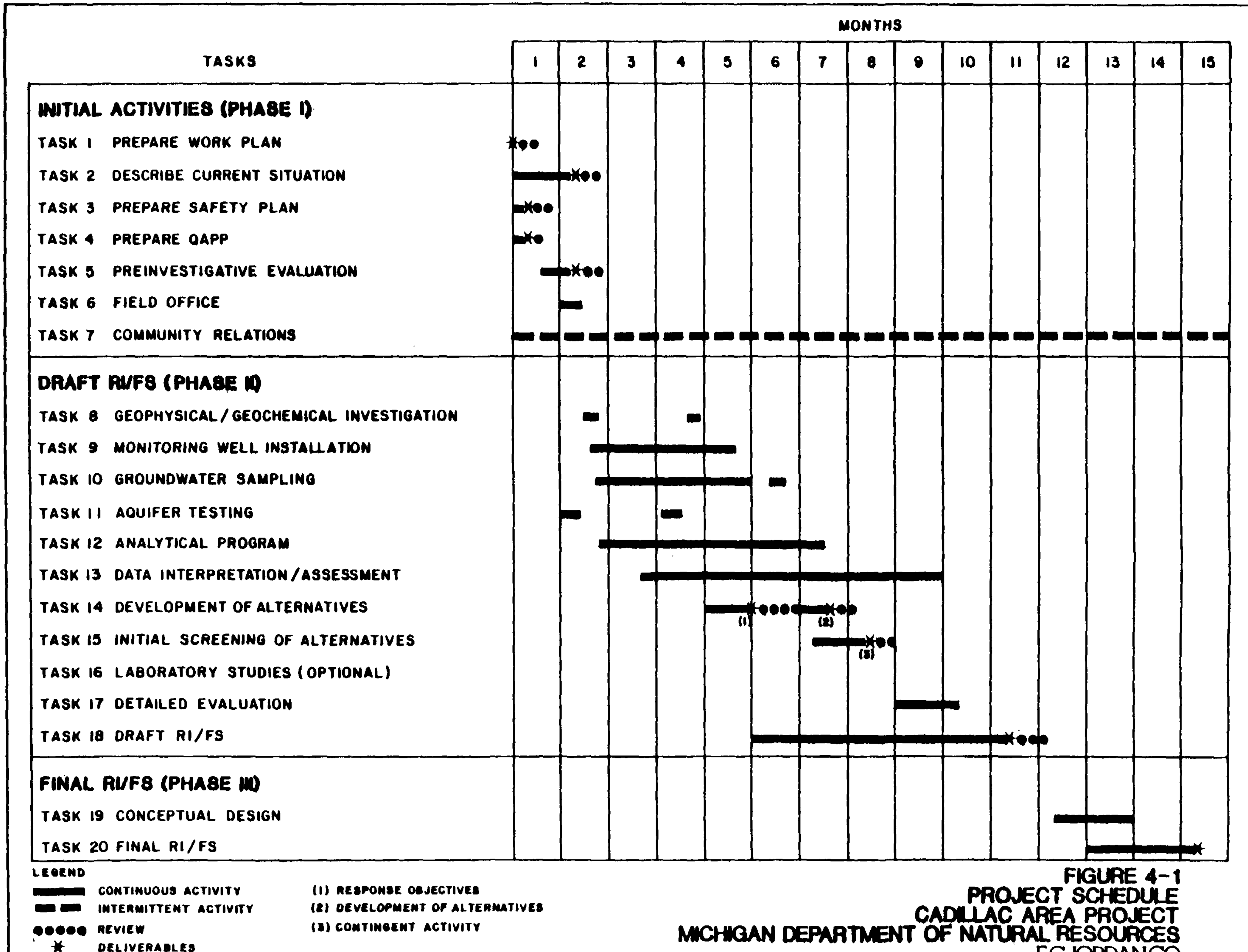

FIGURE 4-1
PROJECT SCHEDULE
CADILLAC AREA PROJECT
MICHIGAN DEPARTMENT OF NATURAL RESOURCES
E.C.JORDAN CO.

Task 15 - Initial Screening

Task 18 - Draft RI/FS Report

Task 20 - Final RI/FS Report

4.2 COST AND BUDGET

The total estimated cost of the site Remedial Investigation and Feasibility Study, including initial activities is $1,304,997.00.

Total man-hours required for the RI/FS have been estimated at 11,132. Manpower requirements by task are presented in Tables 4.1 to 4.3. Project cost summaries are presented in Table 4.4. Table 4.5 presents a summary of other direct project costs.

TABLE 4-1

CADILLAC AREA
REMEDIAL INVESTIGATION/FEASIBILITY STUDY
E.C. JORDAN CO. HOURS BY LEVEL
INITIAL ACTIVITIES (PHASE I)

TASK	PROFESSIONAL 4	PROFESSIONAL 3	PROFESSIONAL 2	PROFESSIONAL 1	TECHNICAL 2	TECHNICAL 1	TASK TOTAL
1. Prepare Work Plan	26	232	84	8	24	16	390
2. Describe Current Situation	24	86	40	--	12	12	174
3. Prepare Safety Plan	2	24	20	--	--	--	48
4. Prepare Quality Assurance Project Plan	6	26	20	--	4	--	56
5. Preinvestigative Evaluation	4	36	--	--	--	--	40
6. Site Office	4	8	--	16	--	8	36
7. Community Relations	24	120	30	--	16	10	200
Hours by Level	90	532	196	24	56	46	944
Secretarial/Clerical							100
Subtotal Hours							1,044

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TABLE 4-2

CADILLAC AREA
REMEDIAL INVESTIGATION/FEASIBILITY STUDY
E.C. JORDAN CO. HOURS BY LEVEL
DRAFT RI/FS (PHASE II)

TASK	PROFESSIONAL 4	PROFESSIONAL 3	PROFESSIONAL 2	PROFESSIONAL 1	TECHNICAL 2	TECHNICAL 1	TASK TOTAL
8. Geophysical and Geochemical Investigation	4	12	90	--	--	20	126
9. Contamination Investigation	120	240	1200	1200	1100	--	3,860
10. Groundwater Sampling	20	40	--	840	500	--	1,400
11. Aquifer Testing	10	56	80	200	200	--	546
12. Analytical Program	270	24	--	800	0	--	1,094
13. Data Interpretation/Contamination Assessment	60	260	440	--	--	40	800
14. Development of Alternatives	10	116	--	--	--	--	126
15. Initial Screening of Alternatives	20	146	--	40	--	--	206
16. Laboratory Studies (Optional)	--	--	--	--	--	--	0
17. Detailed Evaluation	20	168	--	120	--	--	308
18. Draft RI/FS	60	302	--	120	80	--	562
Hours by Level	594	1364	1810	3320	1880	60	9,028
Secretarial/Clerical							260
Subtotal Hours							9,288

TABLE 4-3

CADILLAC AREA
REMEDIAL INVESTIGATION/FEASIBILITY STUDY
E.C. JORDAN CO. HOURS BY LEVEL
FINAL RI/FS (PHASE III)

TASK	PROFESSIONAL				TECHNICAL		TASK
	4	3	2	1	2	1	TOTAL
19. Conceptual Design	20	270	--	180	20	--	490
20. Final RI/FS Report	36	144	--	60	20	--	260
Hours by Level	56	414	--	240	40	--	750
Secretarial/Clerical							50
Subtotal Hours							800

TABLE 4-4

CADILLAC AREA
REMEDIAL INVESTIGATION AND FEASIBILITY STUDY
BUDGET SUMMARY

		COSTS($)				
PHASE	HOURS[1]	LABOR[2]	OTHER DIRECT	SUB-CONTRACTS	FEE	TOTAL
Initial Activities	1,044	$ 39,426	$ 10,880--		$ 6,458	$ 56,764
Draft RI/FS - Phase II	9,288	269,954	122,405	$702,298	116,843	1,211,500
Final RI/FS - Phase III	800	29,248	2,950	--	4,535	36,733
TOTAL	11,132	$338,628	$136,235	$702,298	$127,836	$1,304,997

[1]Prime contractor hours.
[2]Direct labor and overhead.
[3]Subcontractor Costs Summary:

Keck Consulting Services	$694,298
Survey (Granger Engineering)	8,000
	$702,298

TABLE 4-5

CADILLAC AREA
REMEDIAL INVESTIGATION/FEASIBILITY STUDY
OTHER DIRECT COST SUMMARY

	Est. Cost
Transportation	
Airfare: 28 RT @ $350 per RT	$ 9,800
Vehicle Charges: 269 days @ $50/day	13,450
Subsistence	
487 days @ $65 per day	31,655
Other Direct Costs	
Word Processing: 198 hours @ $30 per hour	5,940
Computer Services:	7,605
Printing: 1,000 copies @ $0.10 per copy	100
40,857 copies @ $0.07 per copy	2,860
Telephone:	2,500
Expendables (Bottles, Coolers, Ice, Sampling Supplies):	1,500
Safety:	14,900
Equipment Charges:	
PI Meter	13,250
Sampling	3,200
Water Meter	1,000
Shipping:	5,570
Health Monitoring Program	4,540
Decontamination Trailer:	2,900
Performance Bond:	15,465
	$136,235

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CONTRACT PRICING PROPOSAL
(RESEARCH AND DEVELOPMENT)

Office of Management and Budget Approval No. 29-RO184

This form is for use when (i) submission of cost or pricing data (see FPR 1-3.807-3) is required and (ii) substitution for the Optional Form 59 is authorized by the contracting officer.

PAGE NO. 1 NO. OF PAGES 3

NAME OF OFFEROR: E. C. Jordan Co.

SUPPLIES AND/OR SERVICES TO BE FURNISHED: Michigan Department of Natural Resources Cadillac Area RI/FS Work Plan

HOME OFFICE ADDRESS: 562 Congress Street/PO Box 7050 Portland, ME 04112

DIVISION(S) AND LOCATION(S) WHERE WORK IS TO BE PERFORMED: Above and Southfield, MI

TOTAL AMOUNT OF PROPOSAL: $1,304,997

GOV'T SOLICITATION NO.: 1525

DETAIL DESCRIPTION OF COST ELEMENTS

				EST COST ($)	TOTAL EST COST	REFERENCE
1. DIRECT MATERIAL (Itemize on Exhibit A)						
a. PURCHASED PARTS						
b. SUBCONTRACTED ITEMS						
c. OTHER—(1) RAW MATERIAL						
(2) YOUR STANDARD COMMERCIAL ITEMS						
(3) INTERDIVISIONAL TRANSFERS (At other than cost)						
TOTAL DIRECT MATERIAL						
2. MATERIAL OVERHEAD (Rate %X$ base =)						
3. DIRECT LABOR (Specify)	ESTIMATED HOURS	RATE/HOUR		EST COST ($)		
See Labor Cost Summary	11,132	Various		$144,097		
TOTAL DIRECT LABOR					$ 144,097	
4. LABOR OVERHEAD (Specify Department or Cost Center)	O.H. RATE	X BASE =		EST COST ($)		
	135%	$144,097		$194,531		
TOTAL LABOR OVERHEAD					$ 194,531	
5. SPECIAL TESTING (Including field work at Government installations)				EST COST ($)		
TOTAL SPECIAL TESTING						
6. SPECIAL EQUIPMENT (If direct charge) (Itemize on Exhibit A) See Table 4-5					$ 17,450	
7. TRAVEL (If direct charge) (Give details on attached Schedule)				EST COST ($)		
a. TRANSPORTATION See Table 4-5				$23,250		
b. PER DIEM OR SUBSISTENCE See Table 4-5				$31,655		
TOTAL TRAVEL					$ 54,905	
8. CONSULTANTS (Identify—purpose—rate)				EST COST ($)		
Keck Consulting Services, Inc. (attached)				$694,298		
Granger Engineering (Survey) (estimate)				$ 8,000		
TOTAL CONSULTANTS					$ 702,298	
9. OTHER DIRECT COSTS (Itemize on Exhibit A) See Table 4-5					$ 63,880	
10. TOTAL DIRECT COST AND OVERHEAD					$1,177,161	
11. GENERAL AND ADMINISTRATIVE EXPENSE (Rate % of cost element No.)						
12. ROYALTIES						
13. TOTAL ESTIMATED COST					$1,177,161	
14. FEE OR PROFIT					$ 127,836	
15. TOTAL ESTIMATED COST AND FEE OR PROFIT					$1,304,997	

OPTIONAL FORM 60
October 1971
General Services Administration
FPMR 101-45.1
5060-101

This proposal is submitted for use in connection with and in response to (Describe RFP, etc.)

MDNR - Cadillac Area RI/FS Work Plan

and reflects our best estimates as of this date, in accordance with the Instructions to Offerors and the Footnotes which follow

TYPED NAME AND TITLE	SIGNATURE
Donald R. Cote, P.E. Senior Vice President	STANLEY E. WALKER, V.P. FOR D.R. COTE

NAME OF FIRM	DATE OF SUBMISSION
E.C. Jordan Co.	May 7, 1986

EXHIBIT A—SUPPORTING SCHEDULE (Specify. If more space is needed, use reverse)

COST EL NO.	ITEM DESCRIPTION (See footnote 5)	EST COST ($)
3	See Labor Cost Summary	$144,097

I. HAS ANY EXECUTIVE AGENCY OF THE UNITED STATES GOVERNMENT PERFORMED ANY REVIEW OF YOUR ACCOUNTS OR RECORDS IN CONNECTION WITH ANY OTHER GOVERNMENT PRIME CONTRACT OR SUBCONTRACT WITHIN THE PAST TWELVE MONTHS?

X YES ☐ NO (If yes, identify below.) Mr. Milton Rapaport, Supervisory Auditor

NAME AND ADDRESS OF REVIEWING OFFICE AND INDIVIDUAL	TELEPHONE NUMBER/EXTENSION
Defense Contract Audit Agency Northeast Branch Office 12 New England Executive Park Burlington, MA 01803	617-229-6450

II. WILL YOU REQUIRE THE USE OF ANY GOVERNMENT PROPERTY IN THE PERFORMANCE OF THIS PROPOSED CONTRACT?

☐ YES X NO (If yes, identify on reverse or separate page)

III. DO YOU REQUIRE GOVERNMENT CONTRACT FINANCING TO PERFORM THIS PROPOSED CONTRACT?

X YES ☐ NO (If yes, identify.) ☐ ADVANCE PAYMENTS ☐ PROGRESS PAYMENTS OR ☐ GUARANTEED LOANS

IV. DO YOU NOW HOLD ANY CONTRACT (Or, do you have any independently financed (IR&D) projects) FOR THE SAME OR SIMILAR WORK CALLED FOR BY THIS PROPOSED CONTRACT?

☐ YES X NO (If yes, identify.)

V. DOES THIS COST SUMMARY CONFORM WITH THE COST PRINCIPLES SET FORTH IN AGENCY REGULATIONS?

X YES ☐ NO (If no, explain on reverse or separate page)

INSTRUCTIONS TO OFFERORS

1. The purpose of this form is to provide a standard format by which the offeror submits to the Government a summary of incurred and estimated costs (*and attached supporting information*) suitable for detailed review and analysis. Prior to the award of a contract resulting from this proposal the offeror shall, under the conditions stated in FPR 1-3.807-3 be required to submit a Certificate of Current Cost or Pricing Data. See FPR 1-3.807-3(h) and 1-3.807-4).

2. In addition to the specific information required by this form, the offeror is expected, in good faith, to incorporate in and submit with this form any additional data, supporting schedules, or substantiation which are reasonably required for the conduct of an appropriate review and analysis in the light of the specific facts of this procurement. For effective negotiations, it is essential that there be a clear understanding of:

a. The existing, verifiable data.
b. The judgmental factors applied in projecting from known data to the estimate, and
c. The contingencies used by the offeror in his proposed price.

In short, the offeror's estimating process itself needs to be disclosed.

3. When attachment of supporting cost or pricing data to this form is impracticable, the data will be described (*with schedules as appropriate*) and made available to the contracting officer or his representative upon request.

4. The formats for the "Cost Elements" and the "Proposed Contract Estimate" are not intended as rigid requirements. These may be presented in different format with the prior approval of the Contracting Officer if required for more effective and efficient presentation. In all other respects this form will be completed and submitted without change.

5. By submission of this proposal the offeror grants to the Contracting Officer, or his authorized representative, the right to examine, for the purpose of verifying the cost or pricing data submitted, those books, records, documents and other supporting data which will permit adequate evaluation of such cost or pricing data, along with the computations and projections used therein. This right may be exercised in connection with any negotiations prior to contract award.

FOOTNOTES

1 *Enter in this column those necessary and reasonable costs which in the judgment of the offeror will properly be incurred in the efficient performance of the contract. When any of the costs in this column have already been incurred (e.g., on a letter contract or change order), describe them on an attached supporting schedule. Identify all sales and transfers between your plants, divisions, or organizations under a common control, which are included at other than the lower of cost to the original transferor or current market price.*

2 *When space in addition to that available in Exhibit A is required, attach separate pages as necessary and identify in this "Reference" column the attachment in which the information supporting the specific cost element may be found. No standard format is prescribed; however, the cost or pricing data must be accurate, complete and current, and the judgment factors used in projecting from the data to the estimates must be stated in sufficient detail to enable the Contracting Officer to evaluate the proposal. For example, provide the basis used for pricing materials such as by vendor quotations, shop estimates, or invoice prices; the reason for use of overhead rates which depart significantly from experienced rates (reduced volume, a planned major re-arrangement, etc.); or justification for an increase in labor rates (anticipated wage and salary increases, etc.). Identify and explain any contingencies which are included in the proposed price, such as anticipated costs of rejects and defective work, or anticipated technical difficulties.*

3 *Indicate the rates used and provide an appropriate explanation. Where agreement has been reached with Government representatives on the use of forward pricing rates, describe the nature of the agreement. Provide the method of computation and application of your overhead expense, including cost breakdown and showing trends and budgetary data as necessary to provide a basis for evaluation of the reasonableness of proposed rates.*

4 *If the total cost entered here is in excess of \$250, provide on a separate page the following information on each separate item of royalty or license fee: name and address of licensor; date of license agreement; patent numbers, patent application serial numbers, or other basis on which the royalty is payable; brief description, including any part or model numbers of each contract item or component on which the royalty is payable; percentage or dollar rate of royalty per unit; unit price of contract item; number of units; and total dollar amount of royalties. In addition, if specifically requested by the contracting officer, a copy of the current license agreement and identification of applicable claims of specific patents shall be provided.*

5 *Provide a list of principal items within each category indicating known or anticipated source, quantity, unit price, competition obtained, and basis of establishing source and reasonableness of cost.*

CONTINUATION OF EXHIBIT A—SUPPORTING SCHEDULE AND REPLIES TO QUESTIONS 1 AND 4

CONTRACT PRICING PROPOSAL
(RESEARCH AND DEVELOPMENT)

Office of Management and Budget
Approval No. 29-RO184

This form is for use when (i) submission of cost or pricing data (see FPR 1-3.807-3) is required and (ii) substitution for the Optional Form 59 is authorized by the contracting officer.

PAGE NO | NO. OF PAGES

NAME OF OFFEROR: Keck Consulting Services, Inc.

SUPPLIES AND/OR SERVICES TO BE FURNISHED: Cadillac Area Groundwater Study

HOME OFFICE ADDRESS: 1099 West Grand River Avenue
Williamston, Michigan 48895

DIVISION(S) AND LOCATION(S) WHERE WORK IS TO BE PERFORMED:

TOTAL AMOUNT OF PROPOSAL: $ 694,298.00

GOV'T SOLICITATION NO.:

DETAIL DESCRIPTION OF COST ELEMENTS

Item	Estimated Hours / O.H. Rate	Rate/Hour / X Base =	EST COST ($)	TOTAL EST COST	REFERENCE
1. DIRECT MATERIAL (Itemize on Exhibit A)					
a. PURCHASED PARTS					
b. SUBCONTRACTED ITEMS					
c. OTHER—(1) RAW MATERIAL					
(2) YOUR STANDARD COMMERCIAL ITEMS					
(3) INTERDIVISIONAL TRANSFERS (At other than cost)					
TOTAL DIRECT MATERIAL					
2. MATERIAL OVERHEAD (Rate %X $ base =)					
3. DIRECT LABOR (Specify)	ESTIMATED HOURS	RATE/HOUR	EST COST ($)		
Various (refer to Attachment A)	Various	Various	79,752.		
TOTAL DIRECT LABOR				79,752.	
4. LABOR OVERHEAD (Specify Department or Cost Center)	O.H. RATE	X BASE =	EST COST ($)		
Payroll	57.68	79,752.	46,001.		
TOTAL LABOR OVERHEAD				46,001.	
5. SPECIAL TESTING (Including field work at Government installations)			EST COST ($)		
Drilling (refer to Attachment A)			167,210.		
TOTAL SPECIAL TESTING				167,210.	
6. SPECIAL EQUIPMENT (If direct charge) (Itemize on Exhibit A) (refer to Attachment A)				186,935.	
7. TRAVEL (If direct charge) (Give details on attached Schedule)			EST COST ($)		
a. TRANSPORTATION			22,000.		
b. PER DIEM OR SUBSISTENCE			59,345.		
TOTAL TRAVEL				81,345.	
8. CONSULTANTS (Identify—purpose—rate)			EST COST ($)		
TOTAL CONSULTANTS					
9. OTHER DIRECT COSTS (Itemize on Exhibit A)					
10. TOTAL DIRECT COST AND OVERHEAD					
11. GENERAL AND ADMINISTRATIVE EXPENSE (Rate 100.7% of cost element Nos.)				80,366.	
12. ROYALTIES					
13. TOTAL ESTIMATED COST					
14. FEE OR PROFIT 15% of line 3, 4, & 11; 5% of line 5, 6, 7				52,693.	
15. TOTAL ESTIMATED COST AND FEE OR PROFIT				$694,298.	

1

OPTIONAL FORM 60
October 1971
General Services Administration
FPR 1-16.806
1060-101

This proposal is submitted for use in connection with and in response to (Describe RFP, etc.)

and reflects our best estimates as of this date, in accordance with the Instructions to Offerors and the Footnotes which follow.

TYPED NAME AND TITLE	SIGNATURE
Robert D. Hilty, Vice President	Robert D. Hilty

NAME OF FIRM	DATE OF SUBMISSION
Keck Consulting Services, Inc.	April 30, 1986

EXHIBIT A—SUPPORTING SCHEDULE *(Specify. If more space is needed, use reverse)*

COST EL NO.	ITEM DESCRIPTION *(See footnote 5)*	EST COST ($)

I. HAS ANY EXECUTIVE AGENCY OF THE UNITED STATES GOVERNMENT PERFORMED ANY REVIEW OF YOUR ACCOUNTS OR RECORDS IN CONNECTION WITH ANY OTHER GOVERNMENT PRIME CONTRACT OR SUBCONTRACT WITHIN THE PAST TWELVE MONTHS?

☐ YES ☐ NO *(If yes, identify below.)*

NAME AND ADDRESS OF REVIEWING OFFICE AND INDIVIDUAL	TELEPHONE NUMBER/EXTENSION

II. WILL YOU REQUIRE THE USE OF ANY GOVERNMENT PROPERTY IN THE PERFORMANCE OF THIS PROPOSED CONTRACT?

☐ YES ☐ NO *(If yes, identify on reverse or separate page)*

III. DO YOU REQUIRE GOVERNMENT CONTRACT FINANCING TO PERFORM THIS PROPOSED CONTRACT?

☐ YES ☐ NO *(If yes, identify.):* ☐ ADVANCE PAYMENTS ☐ PROGRESS PAYMENTS OR ☐ GUARANTEED LOANS

IV. DO YOU NOW HOLD ANY CONTRACT *(Or, do you have any independently financed (IR&D) projects)* FOR THE SAME OR SIMILAR WORK CALLED FOR BY THIS PROPOSED CONTRACT?

☐ YES ☐ NO *(If yes, identify.):*

V. DOES THIS COST SUMMARY CONFORM WITH THE COST PRINCIPLES SET FORTH IN AGENCY REGULATIONS?

☐ YES ☐ NO *(If no, explain on reverse or separate page)*

APPENDIX A

NORTHERNAIRE HEALTH AND SAFETY PLAN

HEALTH AND SAFETY PLAN

NORTHERNAIRE PLATING COMPANY
CADILLAC, MICHIGAN

PREPARED BY

E.C. JORDAN CO.
PORTLAND, MAINE

JULY 1984

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1.0 BACKGROUND

SITE: Northernaire Plating Company	PROJECT NO.: 4465-32
Address: Cadillac, Michigan	EPA Region: V
EPA Contact: Mary Elaine Gustafson	Phone No.: 312-886-6144
State Contact: Gary Simons	Phone No.: 517-373-2811 or 800-292-4706
Plan Prepared by: D. Shah/B. Steeves	Date: July 1984

Objective: To conduct a Remedial Investigation of the Northernaire Site to determine the nature and extent of contamination.

Proposed Date (s) of Work: July/August 1984

Overall Hazard is:				
	High:	____	Moderate:	X
	Low:	X	Unknown:	____

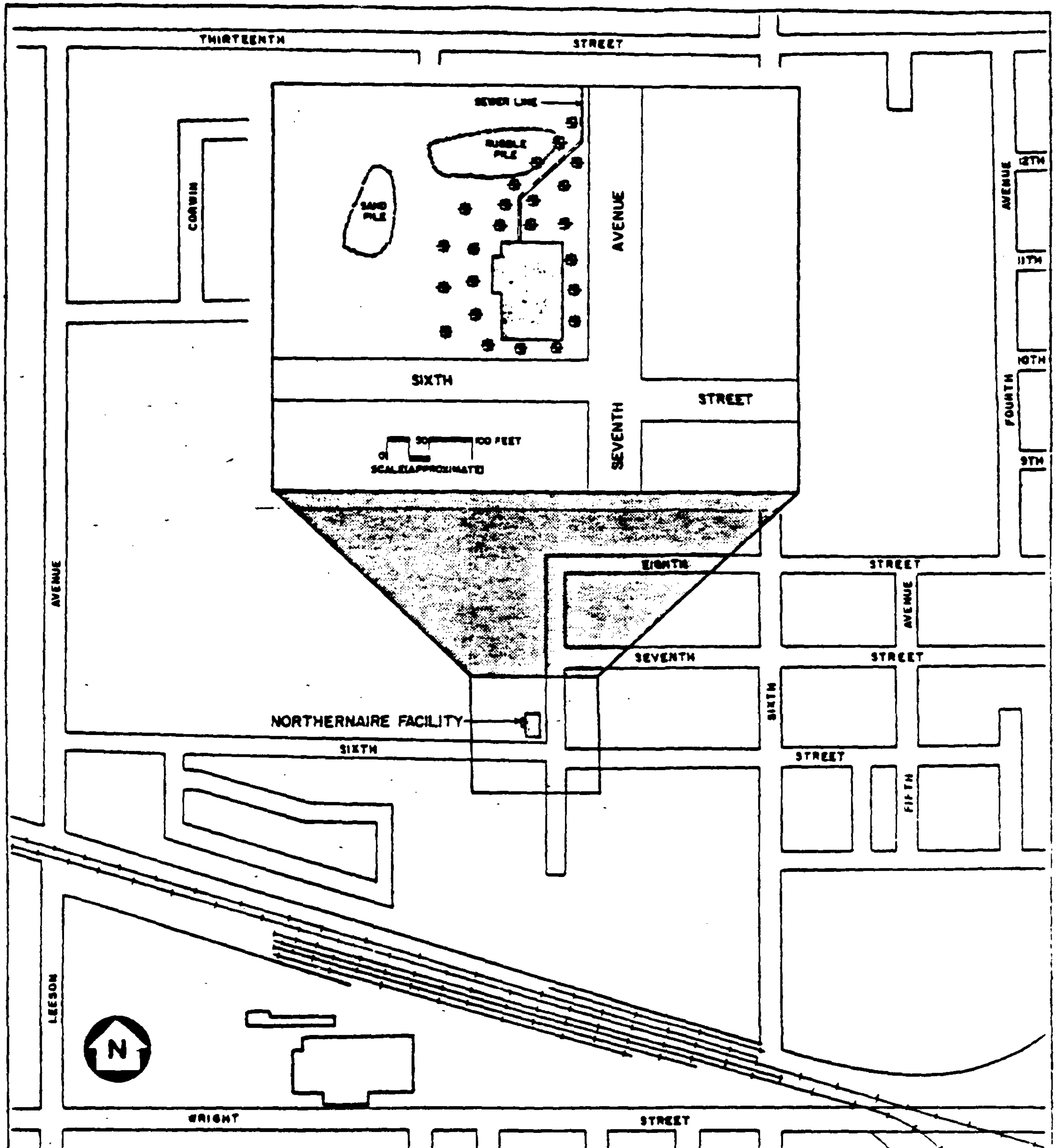
FACILITY DESCRIPTION: Northernaire Plating Company was a cadmium, chromium and zinc plating plant that began operations in 1971 and remained in business until the spring of 1981. The plating process wastewater was discharged from their facilities to a private sewer line which contained an infiltration pit. The private sewer was connected to the city sewer system downstream of this infiltration pit.

The company stored its chemicals in drums; plating operations were done in vats. The chemicals used were acids, cyanide and various salts of metals. A site investigation by the MDNR identified the following conditions:

- o The sewer manholes and pipe were not sealed, allowing wastes to enter the ground;
- o Improper storage of waste and visible signs of chemical spillage on the ground around the building; and
- o Condensation of plating vapors on exhaust ducts.

Figure 1-1 presents the Northernaire site plan.

PRINCIPAL DISPOSAL METHOD (type and location): Plating wastes were discharged through a private sewer line to the city sewer system. Since the private sewer line contained an infiltration pit, they were indirectly discharging some of the waste into the ground. It has been indicated that plating tank vapors were ventilated to the atmosphere through a scrubber vent. When the sewer connection was cut off by the city, all waste was stored in drums for off-site disposal.



EXPLANATION

PROPOSED TEST PIT

FIGURE 1-1

SITE PLAN

0 200 400
SCALE IN FEET

NORTHERNAIRE SITE

Michigan Department of Natural Resources

ECJORDANCO

UNUSUAL FEATURES (containers, buildings, dikes, power lines, terrain, etc.): This manufacturing facility is located in a mixed area of residential, commercial, and industrial facilities. The building was used for waste storage in the latter stage of facility operation, however, containerized wastes have been removed.

STATUS: inactive, abandoned site.

HISTORY: Northernaire Plating Company was a cadmium, chromium and zinc plating plant that began operation in 1971. Because of citizen complaints of well water quality in October of 1978, several private wells within the city of Cadillac were sampled and analyzed. Two of the private wells were found to contain high levels of chromium. The groundwater contamination problem was investigated by the Michigan Department of Natural Resources (MDNR).

In October 1978, the MDNR notified Mr. Willard Garwood, president of Northernaire, that Northernaire was suspected as the source of chromium contamination in the two private wells. Mr. Garwood was instructed to:

1. Inspect sewer lines for leaks;
2. Store all wastes in a contained area;
3. Remove for treatment and dispose of all waste in an approved manner;
4. Submit a pollution incident prevention plan (PIPP); and
5. Provide a plan for long-term control of waste material.

Also in October 1978 the city of Cadillac revoked Northernaire's sewer discharge permit and cut off their sewer connection. After this, Northernaire stored their wastes on-site and removed them by truck. In December 1978 the city of Cadillac installed a city water line to the two residences with contaminated wells. A hydrogeological study was requested but in July 1979 Mr. Garwood informed MDNR that Northernaire did not have the funds to proceed with the study. MDNR attempted to inspect the site in May 1981, but found the building locked and deserted. Many barrels of plating waste were left outside the building. In June 1981 the assets of Northernaire were sold to Top Locker Enterprises of Florida. This company has also ceased operations. In 1982, two children opened a drum and suffered chemical burns. The owner of the site, R.W. Meyer, Inc., subsequently moved all containers inside the building.

In October 1982 the USEPA notified all parties of their intent to begin activities under Superfund because of the potential for contamination of the municipal supply wells.

All of the hazardous waste from inside the building has been removed, as well as some of the contaminated soil and the sewer outside the building.

PROTECTION USED FOR PREVIOUS SITE WORK: __ A __ B __ C __ D _X_ Unknown

PREVIOUS SITE MONITORING: The vertical extent of contamination has not been determined; most of the soil samples were taken from the surface or near the surface. In October 1978 several private wells within the city of Cadillac were sampled. Two of the private wells were found to contain high levels of chromium. A number of soil and water samples were taken around Northernaire and the Cadillac city well field in April 1978 and October 1978 and sometime at the end of 1978. Sludge and standing water above the sludge in open containers were sampled in October 1978 and July 1982. Analysis of these samples verified that the plating wastes contain very high levels of the same heavy metals that were measured in the soil and in the well water.

In addition, soil samples were taken around the Northernaire plant in July 1982. These samples also indicated high levels of cyanide, chromium, cadmium and zinc.

2.0 SCOPE OF WORK

The scope is described in detail in the work plan for Remedial Investigation/ Feasibility Study for the Northernaire Plating Co. site. General site activities will include:

- surface and deep soil sampling and analysis;
- geophysical investigation;
- monitoring well installation;
- groundwater (monitoring wells) sampling and analysis; and
- aquifer testing.

3.0 PROJECT ORGANIZATION

Remedial investigation of the Northernaire Superfund Site is a state-lead (Michigan) project. Overall project control originates with the Michigan Department of Natural Resources (MDNR). E.C. Jordan Co., (ECJ) will perform the work utilizing the services of Walter Meinert Associates (hydrogeology and geophysics) and Keck Consulting Services (monitoring well installation). Organizational responsibilities are delineated below.

MDNR Contact:	G. Simons
Project Manager:	J.S. Atwell
Assistant Project Manager:	J. Dragun
Technical Director:	R.C. Minning
Remedial Investigation Task Leader	E.E. Everett
Geophysics:	G. Heney
Hydrogeology:	E.E. Everett
Analytical Services:	U.S. EPA Contract Laboratory Program
Site Safety Officer:	E.E. Everett
MDNR Site Manager:	M. Vanderlaan

4.0 HEALTH MONITORING AND SAFETY PROGRAM

To assure the health and safety of employees assigned to work at hazardous waste sites, the E.C. Jordan Co. has developed and implemented a Health Monitoring and Safety Program. This program is administered by a committee consisting of representatives of E.C. Jordan Co. technical department staffs with support from medical advisors. No personnel will be allowed on-site until enrolled in the Health Monitoring Program and trained as appropriate for their assigned function.

Jordan employees, subcontractors and consultants working on hazardous waste sites are to receive health and safety indoctrination prior to commencing work on the site and will be provided with periodic followup indoctrination and training as appropriate. Indoctrination and training will include:

- site history;
- inventory of chemicals known (will be updated and reviewed at each stage of the program);
- project organization;
- work plan review;
- project documentation;
- review of site safety rules (site safety rules will be updated as new information is available or after an accident or implementation of contingency plan;
- review of decontamination procedures;
- proper use and care of personal protective equipment;
- proper calibration and use of monitoring equipment; and
- emergency response procedures.

The site-specific information required to address the areas noted above is presented in this health and safety plan. The plan is intended to provide a framework within which information may be updated and ongoing decisions made regarding actual health and safety concerns at the site.

5.0 SITE SAFETY EQUIPMENT

In addition to the personnel protective gear designated for the assigned hazard level, various monitoring and safety equipment will be maintained on-site. Minimum on-site equipment will include:

- o HNU photoionization meter or Photovac Organic Vapor Analyzer;
- o Combustible gas indicator (explosimeter);
- o Oxygen meter or oxygen deficiency alarm;
- o Hydrogen cyanide gas monitor with alarm;
- o Fire extinguishers;
- o First Aid kits;
- o Radio Communications (if activities will occur outside line-of-sight); and
- o Radiation survey meter or radiation alert.

Additional equipment may be specified and obtained as field conditions dictate. An equipment list and field safety gear requirements are specified in the site safety summary (Appendix A).

6.0 PERSONAL PROTECTION LEVEL DETERMINATION

The level of personnel protective equipment required shall be determined by the type and levels of waste or spill material present at the site where project personnel may be exposed. In situations where the types of waste or spill material on-site are unknown or the hazards are not clearly established or the situation changes during onsite activities, the Site Safety Officer must make a reasonable determination of the level of protection that will assure the safety of investigators and response personnel until the potential hazards have been determined through monitoring, sampling, informational assessment, or other reliable methods. Once the hazards have been determined, protective levels commensurate with the hazards will be used. Protection levels will be evaluated on a continuous basis to reflect any new information acquired.

This plan has been prepared based on the site-specific information made available through RAMP and FIT reports, as well as other sources.

The levels of protection utilized by E.C. Jordan Co. are presented below:

Level A. Level A protection must be selected when the Site Safety Officer makes a reasonable determination that the highest available level of both respiratory and skin and eye contact protection is needed. It should be noted that while Level A provides maximum available protection, it does not protect against all possible hazards. Consideration of the heat stress that can arise from wearing Level A protection should also enter into the subtask leaders decision. (Comfort is not a decision factor, but heat stress will influence work rate, scheduling, and other work practices.)

Level B. The Site Safety Officer must select Level B protection when the highest level of respiratory protection is needed, but hazardous material exposure to the few unprotected areas of the body (i.e., the back of the neck) is unlikely.

Level C. The Site Safety Officer may select Level C when the required level of respiratory protection is known, or reasonably assumed to be, not greater than the level of protection afforded by full face air purifying respirators; and hazardous materials exposure to the few unprotected areas of the body (i.e., the back of the neck) is unlikely. Level C requires carrying an emergency escape respirator.

Level D. Level D is the basic work uniform. Investigators and response personnel must not be permitted to work in civilian clothes.

Respiratory protection criteria and suitable protection gear are summarized in Table 6-1. Fit testing of safety equipment will be an important part of establishing adequate respiratory and dermal protection. Fit testing will be accomplished prior to site explorations and each individual will be assigned a fitted respirator for the duration of the project. These will be tagged for identification.

TABLE 6-1
Protective Gear
(Air Quality Levels in ppm)

	Level D	Level C	Level B	Level A
Air Quality (Above Background)	0	0 to 5	5-500	500-1000
Respirator Type*	Escape	Full Face + Escape	SCBA	SCBA
Clothing				
o Boots	X	X	X	X
o Safety glasses or equivalent	X	X	X	
o Hard hat	X	X	X	
o Gloves, inner and outer	X	X	X	X
o Booties		X	X	X
o Coveralls	X	X	X	
o Chemical protective coveralls		X	X	
o Totally encapsulated suit				X

* Use of a respirator for organic vapors is allowed only where identification of organic vapor constituents has occurred and appropriate respirator cartridges have been obtained.

4.84.76
0014.0.0

It should be recognized that situations exist where different combinations of respiratory and dermal protective gear are appropriate, e.g., where splash protection is required but no respiratory hazard exists. The Site Safety Officer may elect a modification of the above specified combinations.

6.1 Potential Hazards On-Site

6.1.1 Hazardous/Toxic Material

The following categories of waste materials have been identified or are potentially present at this site. Amounts are unknown at this time.

SLUDGE	OIL	SOLVENTS	CHEMICALS	SOLIDS	OTHERS
Amount ______	Amount ______	Amount ______	Amount ______	Amount ______	Amount ______
	Oily Wastes	Halogenated Solvents	Inorganic Acids	Unknown	Unknown
Metal Sludges		Non-Halogenated Solvents	Pickling Liquors		
			Caustics		
			Cyanide		
			Metals		

6.1.2 Characteristics of Waste:

Corrosive: X Flammable: _____ Radioactive: ____
Toxic: X Volatile: _____ Reactive: X Inert: _____
Other: _____

6.1.3 Physical Hazards of Site

No hazards known at this time.

6.1.4 Specific Chemical Data (if available)

The site was a plating company. Primary pollutants are heavy metals, acids and cyanide.

Inorganic salts pose a threat to the environment through groundwater contamination and air pollution (in the form of dust). Non-volatile cyanide salts appear to be relatively non-toxic, so long as they are not ingested and care is taken to prevent the formation of hydrocyanic acid. Many cyanides evolve hydrocyanic acid rather easily. Hydrocyanic acid is a gas and is highly toxic. Carbon dioxide in the air is sufficiently acidic to liberate HCN from cyanide solutions. At Northernaire, most chemicals are mixed with soil and are not in liquid form. No volatile organic compounds have been identified at levels of concern, but organic solvents are typically used in this industry.

Table 6-2 includes known data concerning the toxicity and chemical properties of some of the identified constituents.

6.2 Personal Protection Requirements

Levels of Protection Required: A ___ B _X_ C _X_ D _X_

Modifications:

Level C respiratory protection is recommended during dusty conditions because of the presence of heavy metals. If hydrogen cyanide gas (HCN) generation is detected, Level A will be required because hydrogen cyanide can penetrate through skin. At this site, there is only a remote possibility that HCN gas will be generated. Level B may be required for entry to confined spaces.

TABLE 6-2
CHEMICAL TOXICITY AND OTHER INFORMATION*

Chemical	TLV mg/m³	ACC or STEL mg/m³	Physical State	Remarks
Cadmium** Salts dust and fumes	.05	0.2 0.1	Silver/white/blue tinged metal compounds have different appearances.	Continuous exposure to cadmium may cause irreversible lung injury, abnormal lung function and kidney disease. Increased incidence of prostatic cancer, kidney and and respiratory cancer in cadmium workers has been observed. Dust or powder is flammable, toxic gases may be released in a fire. Symptoms: Inhalation: irritation of nose and throat, 0.5 to 2.5 mg/m³ exposure can cause non-fatal lung inflamation. 4 to 10 hours exposure - severe chest pain, persistent cough, and difficulty in breathing. Eye: Irritation Ingestion: A dose of 15-30 mg of metal or soluble salt may cause increased salivation, chocking, vomiting, abdominal pain, etc. First Aid: Ingestion: Conscious person - give large amounts of water immediately and seek medical advice. Incompatabilities: Strong oxidizers, elemental sulfur, selenium, zinc, hydrobenzoic acid, ammonium nitrate.
Chromium (II) and (III) Hexavalent (VI)** oxide	0.5 0.05		Steel gray metal or silver metal powder.	The toxicity of chromium varies with different chromium compounds. Chromic acids and chromates appear to be more toxic than chromium metal dust, insoluble chromium salts, and soluble chromic and chromous salts. Exposure to certain hexavalent chromium compounds is associated with an increased lung cancer incidence in humans. Symptoms: Inhalation: Dust may cause irritation of nose, throat, respiratory passages, and lungs. Repeated or prolonged exposure to chromic acid or dust may cause ulceration and perforation of the nasal septum. Skin: Dermatitis, repeated exposure may cause an allergic skin rash. Incompatabilities: Alkalies, dil H_2SO4 & HCl.
Copper Fume Dust & mist as copper	 0.2 1.0	 2.0	Reddish Lustrous metal No odor	Symptoms: Inhalation: Copper and Copper oxide fumes may cause metal fume fever - chills, fever, aching muscles, dry mouth and throat, headache, nausea, vomiting, diarrhea and stomach pains. Skin: May cause irritation - metal solutions can cause swelling and itching. Ingestion: May cause stomach pain, nausea, vomiting and diarrhea from ingestion of 10 mg of copper by an adult and 8.5 mg by a child. Long Term: No long term effects from inhalation or ingestion reported. Copper fragment in cornea may cause cataracts. First Aid: Ingestion: Seek medical attention. (Pennillamine or triethylenetetramine dihydrochloride may be beneficial in reducing body burden.) Incompatabilities: Acetylene gas, magnesium metal, oxidizing agent.

TABLE 6-2 (cont.)
CHEMICAL TOXICITY AND OTHER INFORMATION

Chemical	TLV mg/m³	ACC or STEL mg/m³	Physical State	Remarks
Cyanide Compounds KCN NaCN	5 (Skin)		White solids with faint almond odor	Cyanide compounds of sodium and potassium can affect the body if inhaled or in contact with eyes and skin or ingested. Sufficient cyanide may be absorbed through the skin to cause fatal poisioning. Symptoms: Low level of exposure causes weakness, headache, confusion, may cause nausea and vomiting. Irritation to nose and skin. High exposure causes rapid loss of consciousness, stop breathing and death. First Aid: Obtain medical advice immediately and follow instruction in first aid kit. A first aid kit should contain minimum of 48 ampules each of 0.3 ml amyl nitrate and complete instructions for use. Trained medical personnel should have physician's kit which includes an addition to amyl nitrate, sterile sodium nitrite solution 3% and sterile sodium thiosulfate solution (25%). Incompatibilities: In closed containers it may form toxic concentration of HCN gas. Strong oxidizers such as chlorates, nitrates, acid or acid salts Cyanide salt may react with CO_2 in air to form HCN (toxic gas).
Hydrogen Cyanide (gas or liquid)			Colorless or pale blue liquid or gas, bitter almond odor.	Symptoms and First Aid are similar to cyanide. However, liquid or gas is very toxic and additional precautions are required. (HCN odor should be treated as poor warning. Although odor threshold of 0.1 ppm is below the permissible exposure limit the sense of smell is easily fatigued and wide individual variation in the minimum odor threshold is known.) First Aid: Same as cyanide. Incompatibilities: Bases - caustic and ammonia may cause violent polymerization and explosion. Liquid HCN will attack some forms of plastics, rubber and coatings.
Lead Lead Chloride** Lead Nitrate**	0.15	0.45	Bluish white or silvery gray solid.	Lead is a cumulative poison. Increasing amount builds up in the body and eventually a point is reached where symptoms and disability may occur. Lead dust carried home may cause symptoms in other family members. Symptoms: Long term exposure: decreased physical fitness, fatigue, sleep distrubances headache, aching bones, constipation, decreased appetite, and abdominal pain. Inhalation of large amounts of lead may lead to seizures, coma and death. First Aid: Get medical attention. Ingestion: If victim is conscious, give water. Incompatibilities: Reacts violently with potassium.

TABLE 6-2 (cont.)
CHEMICAL TOXICITY AND OTHER INFORMATION*

Chemical	TLV mg/m^3	STEL mg/m^3	Physical State	Remarks
Nickel and	1		Silvery White workable metal	Nickel is an insoluble metal, but most common salts are soluble.
Soluble Nickel Compounds	0.1	0.3	White or colored crystal or powder	Symptoms: (From nickel dust and salts.) Inhalation: Dust and mists can cause lung irritation, shortness of breath, coughing and wheezing. Skin: Itching, burning and sores referred to as "nickel itch". Eyes: Irritation and damage to cornea. Ingestion: Giddiness and nausea Long term exposure, in addition to symptoms listed above, impairment of sense of smell, chest pain, destruction of nasal tissues and asthmatic lung disease. Dust inhalation has been associated with an increased risk of lung and nasal cancer. First Aid: Ingestion: large amounts of water. Seek medical attention. Incompatibilities: Nickel dust is flammable. Reacts violently with fluorine, strong mineral acids; liberates H_2, ammonium nitrate, etc.
Zinc Zinc Chloride**			Blue powder	Zinc is considered an essential trace element, necessary for normal growth and development. Most zinc compounds have a relatively low order of toxicity however, occupational exposure to zinc chloride and zinc oxide has been associated with adverse health effects. Spontaneous combustion may occur if zinc dust is stored in a damp place. Zinc dust forms an explosive mixture with air. Symptoms: Inhalation: Inhalation of mists of fumes may cause respiratory or gastrointestional irritation, shortness of breath, a feeling of constriction in the chest and coughing with phlegm and bloody sputum. It may also produce a cyanosis, resulting in a blue color of the skin and lip. Exposure to freshly formed zinc oxide fumes can cause a flu-like illness called metal fume fever, with symptoms similar to those encountered with viral influenza. Skin: Skin contact with zinc chloride may produce dermatitis. Ingestion: 12 grams of zinc metal over two days has caused sluggishness, light headedness; a staggering gait and difficulty in writing. Incompatibilities: Acids, strong alkalies, amines, chlorides, chlorates, nitrates, oxides, fluroine, S, CB_2.

NOTES

* Toxic chemicals may be present in form of metal compounds. However, they are presented here as metals. Toxicity of particular compound may be different then indicated here. If the compound is known then individual toxicity should be referred to in the available literature. Use this table as a general guide.

** Suspected carcinogens, teratogens and mutagens.

7.0 WORKER SAFETY PROCEDURES

Workers will be expected to adhere to the established safety practices for their respective specialties (i.e., drilling, laboratory analysis, construction, etc.). The need to exercise caution in the performance of specific work tasks is made more acute due to weather conditions, restricted mobility and visibility caused by the protective gear itself, the need to maintain the integrity of the protective gear and the increased difficulty in communicating caused by respirators. Work at the site will be conducted according to established protocol and guidelines for the safety and health of all involved. Among the most important of these principals for working at a hazardous waste site are the following.

1. In any unknown situation, always assume the worst conditions and plan responses accordingly.

2. Employ the buddy system. Establish and maintain communication. Develop a set of hand signals as conditions may greatly impair verbal communications.

3. Minimize contact with excavated or contaminated materials. Plan work areas and procedures to accomplish this. Do not place equipment on drums or on the ground. Do not sit on drums or other materials.

4. Employ disposable items where possible to minimize risks in decontamination and cross-contamination. This will require a common sense approach to potential risks and costs.

5. Smoking, eating, or drinking after entering the work zone and before decontamination will not be allowed. Oral ingestion of contaminants is probably the second most likely means of introduction of the toxic substances into the body (inhalation being first).

6. Avoid heat and other work stresses related to wearing the protective gear. Work breaks should be planned to prevent stress related accidents or fatigue. Appendix D provides a summary heat stress casualty prevention plan.

7. Maintain monitoring systems. Conditions can change quickly if subsurface areas of contamination are penetrated.

8. Conflicting situations which may arise concerning safety requirements and working conditions must be addressed and resolved rapidly by the Site Safety Officer to relieve any motivations or pressures to circumvent established safety policy.

9. Unauthorized breaches of specified safety protocol will not be allowed. Personnel unwilling or unable to comply with the established procedures will be replaced. Any changes in established procedure should be documented on the form provided. The change should have very specific, valid reasoning and must be approved by the Site Safety Officer.

10. Be observant of not only one's own immediate surroundings but also that of others. Everyone will be working under constraints to awareness and it is a team effort to notice and warn of impending dangerous situations.

11. Use of contact lenses will not be allowed on-site. These prevent proper flushing should corrosive or lachrymous substances enter the eyes.

12. Sites potentially requiring Level C, B or A protection will require the removal of facial hair (except moustaches) to allow a proper facepiece fit.

13. The site leader, the Site Safety Officer and sampling personnel shall maintain records in a bound notebook recording daily activities, meetings, facts, incidents, data, etc., relating to the project. These record books will remain on the site during the full duration of the project so that replacement personnel may add information in the same record book, maintaining continuity. The notebooks and daily records will be available at the completion of the project. Examples of forms, records and logs to be used at each site are given in Appendix B and C.

8.0 MEDICAL SURVEILLANCE PROCEDURES

8.1 Health Monitoring Program

All on-site personnel and laboratory staff must be enrolled in an equivalent of the E.C. Jordan Co. Health Monitoring Program, developed in conjunction with the Maine Poison Control Center, Maine Medical Center in Portland, Maine. This program consists of an initial medical examination to establish the employee's general health profile and provides important baseline laboratory data for comparative study. The scope of the initial comprehensive physical examination and laboratory testing routine is detailed in Table 8-1. Follow-up examinations are completed for all personnel enrolled in the health monitoring progam on an annual basis, or more frequently if project assignments warrant testing following specific field activities. The level of potential exposure that Jordan personnel are subjected to in carrying out hazardous waste work assignments is recorded by the individual and reviewed by the site supervisor on a daily basis. A copy of the Personal Hazardous Waste Exposure Record is included in Appendix B.

8.2 Review of Exposure Symptoms

Symptoms of exposure to hazardous materials will be reviewed for each site in order to indicate to personnel the recognized signs of possible exposure to those materials. This information will be supplemented with a discussion of the need for objectivity in the personal health assessment to account for normal reaction to stressful situations. The Site Safety Officer will be watchful for outward evidences of changes in worker health. These outward symptoms may include skin irritations, skin discoloration, eye irritation, muscular soreness, fatique, nervousness or irritability, reduced libido, intolerance to heat or cold or loss of appetite. Employees will routinely be asked to assess their general state of health during the project.

To date, no special medical monitoring has been identified for this site.

TABLE 8-1

HEALTH MONITORING PROGRAM
INITIAL EXAMINATION

Physical Examination

- medical history survey
- medical examination
- vision: near and distance vision
  color vision
- hearing: audiometry
- radiologic: PA:LAT
- electrocardiogram: 12 lead
- spirometry

Lab Studies

- hematology
    - red blood count
    - white blood count
    - hemoglobin
    - hematocrit
    - platelet
    - indices
    - sedimentation rate
- blood chemistry
    - SMA 17
    - electrolytes
    - creatinine
    - SGPT
    - carbon dioxide
    - cholesterol
    - serum iron
    - urinanalysis
    - Papanicolaou (PAP) test (females)
    - cholinesterase level
    - thyroid function test T3/T4

9.0 EMERGENCY PLANNING

9.1 Emergency Medical Services

Dr. Frank Lawrence, M.D. and Bruce Campbell, R. Ph. will act as medical liasons between project staff and hospitals. Prior to site investigation or activity on hazardous sites, nearby health facilities will be evaluated to determine their capabilities in relation to the needs of on-site project staff. Criteria such as emergency department physician coverage, decontamination capabilities and available medical specialists will be evaluated.

- On-site First Aid

  - An industrial first-aid kit will be provided at the work site. The contents of the kit will be checked weekly and restocked as necessary. Contents will also include: oxygen, backboard and straps, splints, and a cervical collar.

  - At least one person qualified to perform first aid will be present onsite at all times during work activity. This person will have earned a certificate in first-aid from the American Red Cross or will have received equivalent training. Designated first aides will receive regular review training from the American Red Cross or an equivalent session.

  - An emergency eye-wash/shower station will be provided at the work site, as well as flushing water for decontamination of boots, gloves, clothing, tools, etc.

- Transportation to Emergency Treatment:

  - A vehicle will be available at all times for use in transporting personnel to the hospital. The vehicle will contain decontamination fluid and/or a containment suit for the patient.

  - Personnel stretchers will be located at the work site for use in transporting personnel to the vehicle. Alternate transportation routes to area hospitals will be established prior to on-site activity.

9.1.1 Emergency Routes. Mercy Hospital on the Corner of Hobart Street and Oak Avenue is the closest medical facility. It can be reached by taking Sixth Street east to Eighth Avenue, north on Eighth Avenue to Seventh Street, east on Seventh Street to N. Mitchell or U.S. 131, then south on U.S. 131 to Cobbs St., east on Cobbs Street and follow large blue "H" signs to emergency entrance. The site manager and other response personnel will also drive the route to the hospital before site activities commence.

9.2 Contingency Planning

Prior to commencement of on-site activities, field personnel will review safety considerations with the Site Safety Officer. The Site Safety Officer is responsible for adherence to the designated safety precautions and assumes the role of Jordan on-site coordinator with Mary Vanderlaan, MDNR Site Manager, in an emergency response situation.

The Site Safety Officer for the Northernaire Site is Ed Everett.

All personnel will be familiarized with the route to the nearest hospital (Figure 9-1) and telephone or radio communication device location and operation, and will receive a list of emergency phone numbers (Table 9-1).

The local hospital and emergency response team will be advised of the work to be performed. The hospital will also be briefed on the availability of personnel health data and technical support through the Maine Medical Center's Poison Control Center (MMC-PCC).

Emergency communication will be required to ensure positive pre-planned notification of emergency authorities in the event of episodes requiring initiation of contingency plans.

- The communication will be coordinated with local agencies, fire department, police, ambulance and hospital emergency room.

- Two-way radio communication may need to be established in the field. A site alarm capable of warning site personnel and summoning assistance will also be maintained (e.g. air horns).

- Considering the nature of contaminants, little chance exists for emergency evacuation for residents of nearby homes, but a person will be designated on-site to be responsible for such an event. This person will be made aware of the total number of households within a radius of 2,000 feet. Table 9-1 provides a listing of emergency contacts that will be required.

- Prior to any activity, personnel should investigate possible routes of evacuation.

A copy of an accident report form is provided with this plan. It should be filled out by the Site Safety Officer and filed with the project records if an accident occurs.

9.3 Potential Hazards

The potential hazards associated with hazardous waste site investigation include 1) accidents; 2) contact, inhalation, or ingestion of hazardous materials; 3) explosion; and 4) fire.

9.3.1 Accidents. Accidents must be handled on a case by case basis. Minor cuts, bruises, muscle pulls, etc., will still allow the injured person to undergo reasonably normal decontamination procedures prior to receiving direct first aid. More serious injuries may not permit complete decontamination procedures to be undertaken, particularly if the nature of the injury is such that the victim should not be moved. The nature and degree of surface contamination at a site is generally low enough that emergency vehicles could reach the victim on-site without undue hazard. However, in the event that access on-site is limited, accident victims may be transported by Jordan personnel trained for this response.

9.3.2 Contact and/or Ingestion of Hazardous Materials. Properly prescribed and maintained protective clothing and adherence to established safety procedures are designed to minimize these hazards. However, it is still a possibility that contact or ingestion of materials may occur. One possibility for contamination is the puncture of a buried drum of liquid during drilling operations which might cause the random distribution of the drum contents. Standard first aid procedures should be followed. The drilling rig will have a tank of water which may be useful in some circumstances, particularly to flush off any exposed skin areas. Eye wash bottles will also be maintained at the site in case of emergencies. In cases of ingestion or other than minor contact with known substances, the Poison Control Center and local hospital should be contacted and the victim brought there immediately for further treatment and observation.

9.3.3 Explosion. The drilling crew should be keenly aware of combustible gas meter readings and withdraw at an indication of imminently hazardous conditions. The detection of such conditions shall be reported to local agencies for potential execution of the evacuation plan should the situation be assessed as warranting such response.

9.3.4 Fire. The combustible gas meter will also warn of imminent fire hazards at borings. The greatest fire hazard at the site should be recognized as handling the methanol used for decontamination. No smoking or open flames are allowed in this area. Carbon dioxide fire extinguishers will be kept at the drilling rig, and the decontamination area/field office. The Fire Department, previously informed of site activities, will be called as needed.

9.4 Evacuation Response Levels

Evacuation responses will occur at three levels: (1) withdraw from immediate work area (100+ feet upwind); (2) site evacuation; (3) evacuation of surrounding area. Anticipated conditions which might require these responses are described below:

Withdrawal Up-Wind (100 or more feet).

- Sensing ambient air conditions as containing greater contaminant concentrations than guidelines allow for the type of respiratory protection being worn. The work party may return upon donning greater respiratory protection and/or assessing the situation as transient or past.

- Breach in protective clothing or minor accident. The party may return when tear or other malfunction is repaired and first aid or decontamination has been administered.

- Respirator malfunctions and must be replaced.

Site Evacuation.

- o Sensing ambient air conditions as containing explosive and persistent levels of combustible gas or excessive levels of toxic gases, i.e., in excess of 100 ppm.
- o Fire or major accident
- o Imminent explosion or explosion

Surrounding Area Evacuation.

- o Persistent, unsuppressable release of toxic or explosive vapors (e.g. possible pressure release of punctured drums). Air quality should be monitored at several distances downwind to assess danger to surrounding area before initiating this response.
- o Fire
- o Explosion

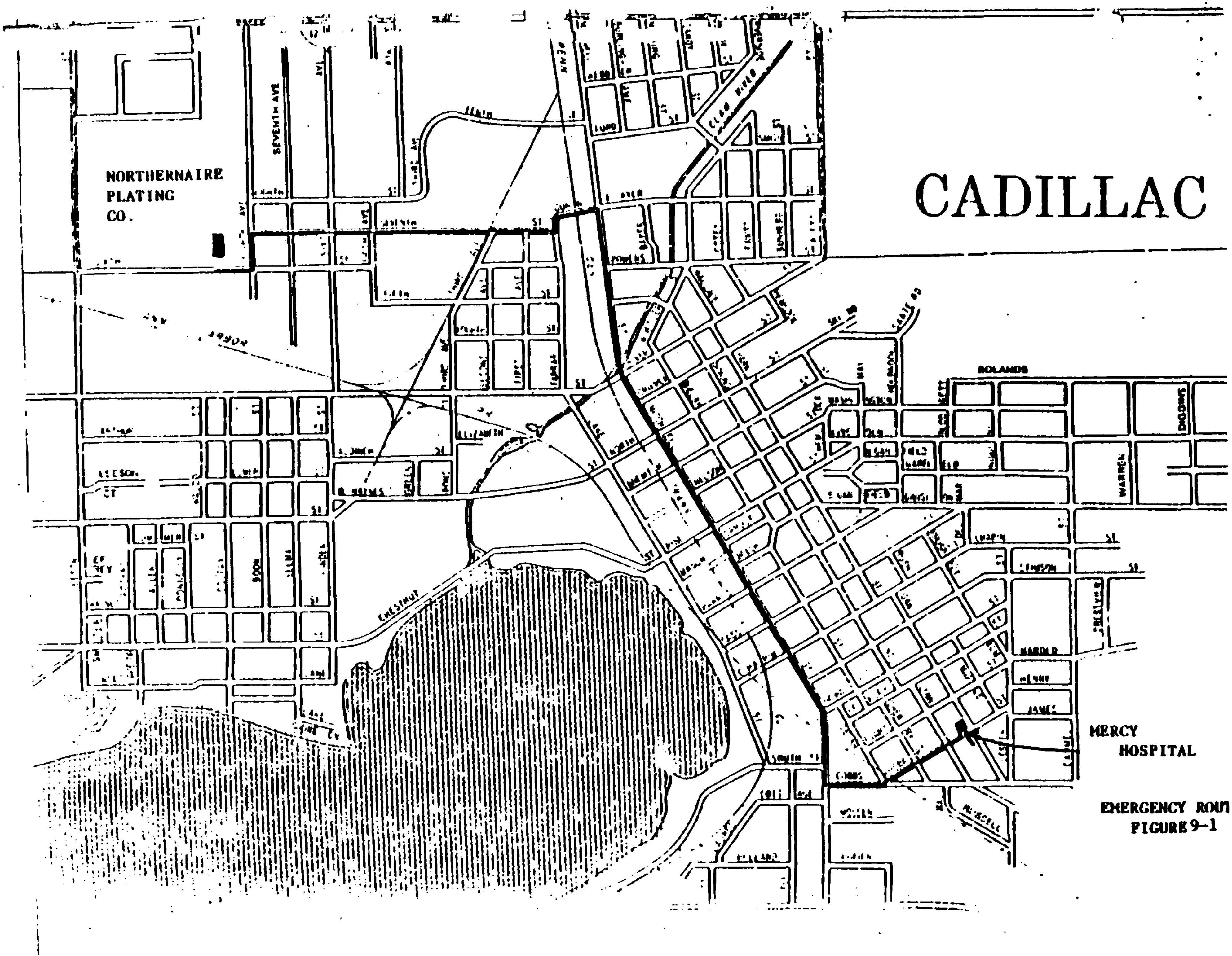
9.5 Evacuation Procedures

9.5.1 Withdrawal Upwind. The work party will note general wind directions while on-site. (A simple wind sock may be set up for visual determinations.) Upon noting the conditions warranting movement away from the borehole, the crew will move upwind a distance of approximately 100 feet or further as indicated by the PI meter. Donning SCBA and a safety harness and line, the Site Safety Officer or a member of the crew may return to the borehole to determine if the condition noted was transient or persistent. If persistent, then an alarm should be raised to notify on-site personnel of the situation and the need to leave the site or don SCBA. An attempt should be made to decrease emissions only if greater respiratory protection is donned. The Site Safety Officer and client will be notified of conditions. Considering access to the site, if this situation occurs the crew may be instructed to evacuate the site rather than move upwind.

9.5.2 Site Evacuation. Upon determination of conditions warranting site evacuation, the work party will proceed upwind and notify the security force, Site Safety Officer and the field office of site conditions. If the decontamination area is upwind and greater than 300 feet from the source, the crew will pass quickly through decontamination to remove contaminated outer suits. If the hazard is toxic gas, respirators will be retained. The crew will proceed to the field office to assess the situation. There the respirators may be removed (if monitoring indicates an acceptable condition). As more facts are determined from the field crew, these will be relayed to the appropriate agencies. The advisability and type of further response action will be coordinated and carried out by the Site Safety Officer and Site Manager.

TABLE 9-1
EMERGENCY CONTACT RESOURCES
NORTHERNAIRE SITE

AGENCY/ORGANIZATION		Telephone Number and Address
Ambulance (Waxford County Sheriff		779-9211
Hospital Emergency Room (Mercy Hospital)		775-1331
Poison Control Center (Children's Hospital Detroit)		(313) 494-5711
Police (local, Lyle Reddy, Chief)		(313) 774-3491
Fire Department (local, Harold Bassett, Chief)		(313) 775-2411
Waxford County Emergency Preparedness, Merritt Chandler		775-7602
EPA Contact (regional)		
Michigan Department of Natural Resources		800-292-4706 or 517-373-7660
Site Water Supply		
Site Telephone		
Site Radio		
E.C. Jordan Co.		207-775-5401
Maine Medical Center	Dr. Frank Lawrence/	207-871-2617
	Bruce Campbell	207-871-2950
National Response Center		800-482-8802
Chemtrec		800-424-9300



EMERGENCY ROUT
FIGURE 9-1

9.5.3 Evacuation of Surrounding Area. When the Site Safety Officer and Site Manager determine that conditions warrant evacuation of downwind residences and commercial operations, the local agencies will be notified and assistance requested. Designated onsite personnel will initiate evacuation of the immediate off-site area without delay.

9.6 Training

The following matrix (Figure 9-2) indicates training received by on-site personnel. All personnel should become familiar with this matrix to minimize response times.

FIGURE 9-2
ON-SITE PERSONNEL TRAINING

TOPIC	HRS														
INTRODUCTION															
FIRST AID															
CPR															
DECONTAMINATION															
OVA															
HNU															
SCBA REVIEW															
SAMPLING															
HEALTH MONITORING															
NUS COURSE															
NWWA COURSE															

○ INDICATES REQUIRED TOPIC. ✓ INDICATES REQUIRED TRAINING COMPLETED

CERTIFIED BY ______________________________

10.0 DECONTAMINATION

10.1 Personnel Decontamination Procedure

A decontamination procedure will be carried out by all personnel leaving hazardous waste sites. Under no circumstances (except emergency evacuation) will personnel be allowed to leave the site prior to decontamination. Procedures for removal of protective clothing are as follows:

- o Drop tools, monitors, samples and trash at designated drop stations. These will be plastic containers or drop sheets.
- o Step into designated shuffle pit area and scuff feet to remove gross amounts of dirt from outer boots. If necessary, wash boots down with clear water in designated wash pit area.
- o Remove tape from boots and remove boots. Discard in drum container.
- o Remove outer gloves and place in container.
- o Remove hard hat and respirator and hang in the designated area.
- o Remove coveralls and discard in container.
- o Remove inner gloves and discard in container.
- o If the site required utilization of a decontamination trailer, all personnel would also shower before leaving the site at the end of the work day.

Note: Disposable items (coveralls, inner gloves, and latex overboots) will be changed on a daily basis unless there is reason for changing sooner. Dual respirator canisters will be changed twice weekly unless more frequent changes are deemed appropriate by site surveillance data or personnel assessment.

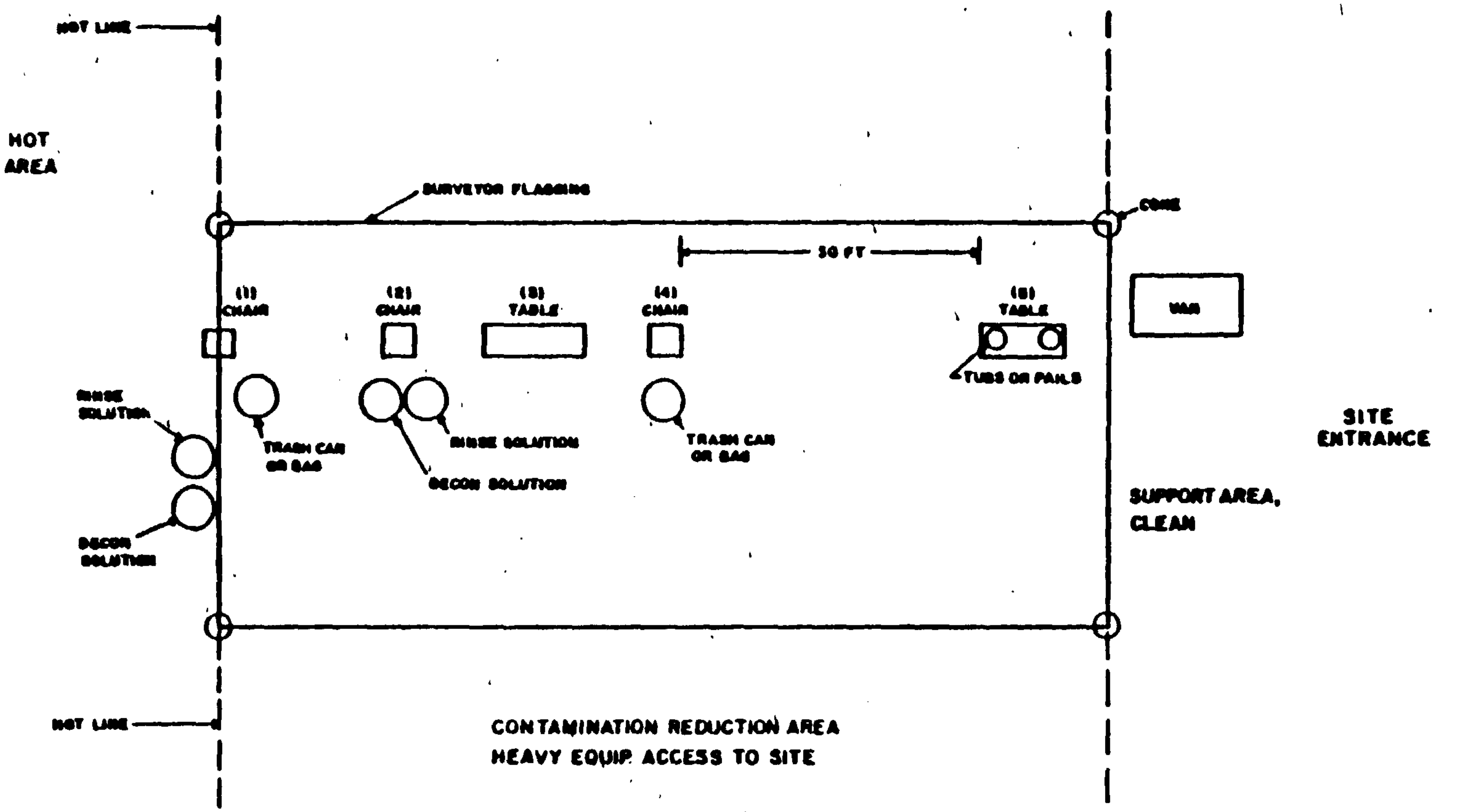
A water hose and/or designated wash area will be available for wash down and cleaning purposes.

A schematic of a typical decontamination area is shown in Figure 10-1.

10.2 Equipment Decontamination

Equipment to be decontaminated during the project may include: (1) drilling rig and tools; (2) sample containers; (3) monitoring equipment; (4) respirators; (5) decontamination trailer and (6) laboratory equipment.

All decontamination will be done by personnel in protective gear appropriate for the level of decontamination, determined by the Site Safety Officer. The decontamination work tasks will be split or rotated among support and work crews. Decontamination procedures within the trailer (if used) should take



NOT TO SCALE

TASK

(1) Wash boots
(2) Wash gloves—Rinse gloves
(3) SCBA Tank change over table w/spare tanks
(4) Remove boots, gloves
(5) Remove SCBA, wash mask in pails or tubs

FIGURE 10-1
TYPICAL PERSONNEL DECONTAMINATION STATION

E.C.JORDAN CO.

place only after other personnel have cleared the "hot area", moved to the clean area and the door between the two areas closed.

Miscellaneous tools and samplers will be dropped into a plastic pail, tub or other container. They will be brushed off and rinsed (outside, if possible) and transferred into a second pail to be carried to further decontamination stations. They will be washed with a trisodium phosphate or detergent solution, rinsed with acetone or methanol, rinsed with a trisodium phosphate or detergent solution and finally rinsed with clean water.

10.2.1 Drilling Rig and Tools. It is anticipated that both backhoe and drill rigs will be contaminated during test pit/borehole activities. They will be cleaned with high pressure water or portable high pressure steam followed by soap and water wash and rinse. Loose material will be removed by brush. The person performing this activity will usually be at Level C protection.

10.2.2 Sample Containers. Exterior surfaces of sample bottles will be decontaminated prior to packing for transportation to the analytical laboratory. Sample containers will be wiped clean and placed in individual Zip-Loc bags at the sample site. It will be difficult to keep the sample containers completely clean. The samples will be taken to the decontamination area. Here they will be further cleaned if necessary and transferred to a clean carrier and the sample identities noted and checked off against the chain-of-custody record. The samples, now in a clean carrier, will be stored in a secure area prior to shipment.

10.2.3 Monitoring Equipment. Monitoring equipment will be protected as much as possible from contamination by draping, masking or otherwise covering as much of the instruments as possible with plastic without hindering the operation of the unit. The HNU meter, for example, can be placed in a clear plastic bag which allows reading of the scale and operation of the knobs. The HNU sensor can be partially wrapped, keeping the sensor tip and discharge port clear.

The contaminated equipment will be taken from the drop area and the protective coverings removed and disposed of in the appropriate containers. Any dirt or obvious contamination will be brushed or wiped with a disposal paper wipe and the used wipers disposed. The units will then be taken inside in a clean plastic tub, wiped off with damp disposable wipes and dried. The units will be checked, standardized and recharged as necessary for the next day's operation. They will then be covered with new protective coverings.

10.2.4 Respirators. Respirators will be decontaminated daily. Taken from the drop area, the masks will be disassembled, the cartridges set aside and the rest placed in a cleansing solution. (Parts will be precoded, e.g., #1 on all parts of mask #1). After an appropriate time within the solution, the parts will be removed and rinsed off with tap water. The old cartridges will be marked to indicate length of usage and will be discarded into the contaminated trash container for disposal when considered spent. In the morning the masks will be re-assembled and new cartridges installed if appropriate. Personnel will inspect their own masks to be sure of proper readjustment of straps for proper fit.

10.2.5 Decontamination Trailer or Truck and Staging Area. The decontamination trailer or truck, if used, will be cleaned daily. This will include vacuuming with a vacuum having a water filter to capture dust particles. The area will be wet mopped with cleanser and again with clean water. Work bench areas will be wiped down. Wash buckets and the cleaning area will be decontaminated and made ready for the next day's use.

10.2.6 Laboratory Equipment. Sample handling areas and equipment will be cleaned/wiped down daily. Disposable wipes will be used and discarded into a plastic bag. These will subsequently be taken to and placed in the disposal drum for final disposition. For final cleanup, all equipment will be disassembled and decontaminated. Any equipment which cannot be satisfactorily decontaminated will be disposed of (e.g., glassware, covers for surfaces) as previously indicated.

All decontamination will be done by personnel in protective gear appropriate for the level of decontamination, determined by the Site Safety Officer. The decontamination work tasks will be split or rotated among support and work crews. Decontamination procedures within the trailer (if used) should take place only after other personnel have cleared the "hot area", moved to the clean area and the door between the two areas closed.

11.0 DOCUMENTATION AND RECORDKEEPING

Samples of field activity documentation forms are attached (see Appendix B). Minimum documentation consists of:

- o daily field record kept by site technical leader or designee;
- o site surveillance record kept by the Site Safety Officer;
- o sampling - related records kept by sample collection team;
- o chain-of-custody records for each sample collected; and
- o personal hazardous material exposure record.

The Site Safety Officer is also responsible for immediate notification of Jordan's Health and Safety Coordinator in the event of personnel injury.

12.0 UPDATING OF HEALTH AND SAFETY PLAN

The Site Safety Officer is responsible for maintaining proper documentation regarding daily safety log sheets. If any deficiency is encountered in the health and safety plan, a report will be prepared and forwarded to the health and safety coordinator at Jordan and copies sent to the project manager and technical director. The Site Safety Officer will immediately initiate necessary changes to improve protection of field staff.

During the remedial investigation process or after initial field investigation, any new chemical encountered will be evaluted and safety plans modified to reflect the effect of that chemical. Similarly, any physical hazards that are discovered will be addressed by the Site Safety Officer and reported.

13.0 REFERENCE GUIDES FOR HAZARDOUS MATERIALS

Reference guides for material classification determinations are:

1) The Merck Index, 9th Edition, Merck, Sharp & Dohme Ltd., 1980.

2) Handbook of Chemistry & Physics, 56th Edition, CRC Press, 1979.

3) Pocket Guide to Chemical Hazards, 1980 Edition, NIOSH/OSHA, DHEW (NIOSH) Publication No. 78-120.

4) Registry of Toxic Effects of Chemical Substances, 8th edition NIOSH, 1978.

5) Dangerous Properties of Industrial Materials, Sax, N.I., Fifth edition, Van Nostrand Reinhold Co., 1979.

6) Threshold Limit Values for Chemical Substances and Physical Agents in the Work Environment, 1983-84. Adopted by ACGIH.

APPENDIX A

SITE SAFETY SUMMARY

SITE SAFETY SUMMARY

DATES PLAN IN USE: July/August 1984 DATE PREPARED: July 1984

PREPARED BY: D. Shah/J. McMullen SITE/AREA NAME: Northernaire Plating Co.

LOCATION: Cadillac, Michigan

EXISTING INFORMATION FOR SITE: Detailed___ Preliminary___ Sketchy X None___

HAZARDOUS MATERIAL FORM: Gas___ Liquid___ Sludge X Solid X

CONTAINMENT: Drum___ Pit___ Pond___ Lagoon___ Tank___ Soils X Debris X

OTHER: CONDITIONS: Contaminated soils around facility and infiltration manholes.

CHARACTERISTICS: Corrosive X Ignitable___ Radioactive___ Volatile___ Toxic X Reactive X Other_______________ Unknown___

SITE/AREA SPECIFICS: High Hazard Material

Compound	Anticipated Concentration In Surface Soils (ppm)	Warning Properties
Organics*		
Trichloroethylene	--	Sweet odor
Inorganics		
Cadmium	2,110	----
Chromium	51,000	----
Hexavalent Chromium	--	----
Copper	160	----
Iron	5,900	----
Nickel	12	----
Lead	23,000	----
Zinc	5,200	----
Cyanide (Total)	67	----
Hydrogen Cyanide (gas)		Bitter Almond+ 1 ppm

* Analysis for organic constituents was not available. The most likely compound that could be present is indicated.

\+ Hydrogen cyanide gas, if generated or present, has poor warning properties. The odor threshold is below permissible exposure limit, but this is a poor guide because of its highly toxic property and wide individual variation for odor threshold.

SITE HISTORY

STATUS: Open___ Closed X Limited Access___ Unknown___

HISTORY: Northernaire Plating Co. began operation in 1971 and remained in business until the spring of 1981 when they abandoned the property. The company left behind barrels of plating waste, sludges and unused chemicals in a 50' x 100' building. In July and August of 1983, wastes were removed and the building cleaned. The sewerline and manhole at the rear of the building were removed.

UNUSUAL FEATURES: The City of Cadillac municipal drinking water well fields lie one-half mile northwest of the site.

HAZARD ASSESSMENT

EVALUATION OF EXPECTED HAZARD (work assignments, operational considerations, routes of exposure, health effects, material stabilty): Study area activities include surveying, soil sampling and drilling of on-site and off-site wells. The areas will range from high to low contamination. The Northernaire property and groundwater are potentially contaminated with acid or caustic plating wastes containing lead, nickel, zinc, cadmium, chromium, and cyanide. There may also be solvents present, although none have been identified to date. These wastes present a possible hazard of exposure by inhalation, skin contact and ingestion. Dusts which may be generated by drilling and other study area activities may be contaminated with heavy metals which pose a potential hazard by inhalation. Direct dermal contact poses a hazard for skin absorption of heavy metals or burns from caustic or acid contaminated soils. Associated hazards for the known contaminants are in Table 6-2.

PERIMETER CONTROL: Will be designated as work progresses, with an established boundary in temporary work area.

STAGING AREA: Access Road

PERSONNEL PROTECTION
Levels of Protection Required: A___ B X C X D X

MODIFICATION OR SPECIALIZED EQUIPMENT: The level of protection will be D with a provision for both C and B. A reserve air pack will be supplied to each individual in case emergency escape is required due to cyanide gas. Level C may be specified under dusty or high volatile organics conditions. Level B may be required for confined space entry.

DETECTION EQUIPMENT (survey, meters, dosimeters): HNU Photoionization meter, O_2 and CH_4 meter, hydrogen cyanide gas monitor and method of wind direction detection (e.g., wind socks).

COMMUNICATIONS (type, range, frequencies, alternates, hand signals): Telephone communication will be established on-site. Air horn and hand signals will be selected and coordinated on-site.

AUTHORIZED TEAM PERSONNEL

Name	Position
J.S. Atwell	Project Manager
R.C. Mining	Technical Director
G. Heney	Geophysics
E.E. Everett	Hydrogeology
E.E. Everett	Site Safety Officer
B. Fontes	Environmental Technician
M. Friedericks	Environmental Technician

OTHER PERSONNEL (prearranged visitors, support personnel):

Name	Agency/Company	Restrictions
M. Vanderlaan	MDNR	None
R. Steeves	ECJ	None

MONITORING PROCEDURES: (use and employment of fixed, portable, real-time, continuous and/or periodic monitoring devices): Monitoring of cyanide gas and volatile organics will be continuous for on-site earth moving activities. After initial survey, the frequency of monitoring (exclusive of that described above) will be determined by initial results and activity.

PERSONNEL DECONTAMINATION PROCEDURES: Gross contamination brush off, soap and water rinse, disposal of outer suits.

SAMPLING EQUIPMENT DECONTAMINATION: High pressure water or stream, acetone/methanol rinse, trisodium phosphate/soap and water rinse.

SUPPORT EQUIPMENT DECONTAMINATION: High pressure water or steam, soap and water rinse.

DECON MATERIALS REQUIRED: See Table A-2.

HIGH HAZARD MATERIALS (known or anticipated):

Name	Acute Exposure Symptoms
Hydrogen Cyanide (gas)	weakness, headache, confusion and occasionally nausea or vomiting.

LOCATION OF NEAREST WORKING PHONE: On-site at staging/support trailer.

OTHER EMERGENCY COMMUNCIATIONS: Telephone communications/coordination with local sheriff's office.

EMERGENCY PHONE NUMBERS:

	Name/Location	Phone #	Standby
Ambulance	Waxford County Sheriff	779-9211	
Fire	Harold Bassett, Chief	775-2411	
Police	Lyle Reddy, Chief	775-3491	
Hospital	Mercy Hospital	775-1331	
EOD			
State Agency (DNR)		1-800-292-4706	
Poison Control Center	Children's Hospital Detroit	(313)494-5711	

ADDITIONAL RESOURCES:

Name	Agency/Company	Phone #
Meritt Chandler	Waxford County Emergency Preparedness	775-7601

ROUTE TO HOSPITAL: Mercy Hospital on the Corner of Hobart Street and Oak Avenue is the closest medical facility. It can be reached by taking Sixth Street east to Eighth Avenue, north on Eighth Avenue to Seventh Street, east on Seventh Street to N. Mitchell or U.S. 131, then south on U.S. 131 to Cobbs St., east on Cobbs Street and follow large blue "H" signs to emergency entrance.

After Action Report to: Robert A. Steeves Date:
Post Site Medicals: Only for exposure incidents
Plan Approved by: Date:

FIRST AID FOR OVERT PERSONNEL EXPOSURE

SKIN CONTACT: Dust off immediately and flush the contaminated skin with water. Remove clothing and flush skin with water. Get medical attention immediately.

INHALATION: Move person to fresh air at once. If breathing has stopped, perform artificial respiration and get medical attention as soon as possible.

INGESTION: Get medical attention.

EYE CONTACT: Eye should be irrigated with large amount of water, occasionally lifting lower and upper lids. Get medical attention immediately. No contact lenses will be worn when working in field.

IONIZING RADIATION

Normal background .01 to .02 mR/hr
If less than 2 mR/hr, continue investigation with caution
If 2 to 10 mR/hr, map 2 mR/hr contour
If greater than 10 mR/hr, evacuate site

TABLE A-1

PERSONNEL SAFETY EQUIPMENT CHECK LIST

Quantity Required	Protective and Safety Equipment	Model and Material
2	SCBA	MSA 401
2	Spare Cylinders	
5	Escape Mask	ELSA
5	Full Face	MSA
10 pr	Cartridge	MSA GMC-H
5	Hardhat w/ Face Shield	
5	Safety Glasses	
10 pr	Ear Protection	
3 boxes	Gloves - Surgical	
5	Gloves, outer:	
5	Chem Resist Coveralls	
10	Disposable Coveralls	Coated Tyvek
5	Splash Aprons	Vinyl
5	Boots: Safety Boots	
0	Fully Encapsulated Suits	
5	Dosimeters	TLD
	First Aid Equipment	
0	Utility first aid kit	
1	Industrial first aid kit	
2	Fire Extinguisher	CO_2
1	Safety Harness	
2	Eye Wash Station	Portable
1	Safety Shower	

TABLE A-2

DECONTAMINATION EQUIPMENT/MATERIALS

Quantity	Type	Remarks
4	wash tubs	
1	low pressure sprayer	
	high pressure water sprayer	
--	cold	
1	hot	
--	steam sprayer	
4	scrub brushes	
2	sponges	
4	containers to hold contaminated liquid	if required
20 gal/day	water	provided on site
1 box	trisodium phosphate	
--	methanol	
5 gal	acetone	
1 roll	plastic sheets	

TABLE A-3

ASSIGNED LEVELS OF PROTECTION

Activity	Level of Protection	Modification
Reconnaissance	D	C
Drilling	C	D or B
Soil sampling	D	C
Test Pit sampling	C	D or B
Water sampling		
Ground water	D	C

APPENDIX B

PERSONAL FORMS AND LOGS

PERSONAL HAZARDOUS WASTE EXPOSURE RECORD

NAME: ______________________________

SITE: ______________________________ DATE: ____________________

NUMBER OF HOURS ON SITE: ____________________

CONDITION OF SITE: ______________________________

AMBIENT AIR/SOIL/WATER INDICATORS:

LABORATORY: G.C. ____________________ ON-SITE: G.C. ____________________

P.I. METER ____________________ P.I. METER ____________________

OTHER ____________________ OTHER ____________________

TYPE OF EXPOSURE (i.e. inhalation, soil/water contact): ______________________________

____

OPERATION PERFORMED (i.e. test pit inspection, sampling, drilling): ______________________________

____

CHEMICALS BURIED OR KNOWN PRESENT: ______________________________

____

PROTECTIVE EQUIPMENT WORN:

- ☐ SAFETY SHOES (STEEL TOE & SHANK)
- ☐ CHEMICAL RESISTANT BOOTS, TYPE: ____________________
- ☐ HALF-FACE RESPIRATOR
  - TYPE OF CARTRIDGE: ____________________
- ☐ FULL-FACE RESPIRATOR
  - TYPE OF CARTRIDGE: ____________________
- ☐ INNER LAB GLOVES
- ☐ OUTER CHEMICAL RESISTANT GLOVES
  - TYPE: ____________________
- ☐ COVERALLS
  - TYPE: ____________________
- ☐ ____________________

DECONTAMINATION MEASURES TAKEN:

- ☐ CHANGE OF CLOTHES
- ☐ SHOWER
- ☐ CHANGE OF PROTECTIVE EQUIPMENT
- ☐ ____________________
- ☐ ____________________
- ☐ ____________________

PERSONAL PROTECTIVE DECONTAMINATION PROCEDURES: ______________________________

____

____

UNUSUAL SITE CONDITIONS/OCCURANCES: ______________________________

____

OBSERVED HEALTH EFFECTS: ______________________________

____

NOTES: ______________________________

____

____

PERSONAL SAFETY LOG

NAME: ______________________ AFFILIATION: ______________________

PROJECT: ______________________ PROJECT #: ______________________

DATES OF THIS SHEET: ______________________ PAGE ____ OF ______

	DAY									
	1	2	3	4	5	6	7	8	9	0
WORK AREA										
HOURS ON SITE										
PROTECTIVE GEAR										
COVERALLS										
TYVEK										
GLOVES, IN										
GLOVES, OUT										
BOOTIES										
HARD HAT										
RESP, DUST										
RESP, HALF										
RESP, FULL										
SCBA										
RESP, ESC										
DOSIMETER										
AIR MONITOR										
DECONTAMINATION										
COMPLETE										
INCOMPLETE										
PERSONAL ASSESS.										
OK										
COMPLAINT										

NOTES: ____

____

BY: ______________________

DAILY SAFETY LOG

PROJECT NAME: ______________________________ DATE: ______________

PROJECT NUMBER: ______________________________ DAY NO: ______________

E.C. JORDAN WORK PARTY: ______________________________

____

____

SUBCONTRACTOR ( ): ______________________________

____

____

VISITORS: ______________________________

____

WORK SITE LOCATION: ______________________________

____

____

SUMMARY OF CONDITIONS ENCOUNTERED: ______________________________

____

____

____

____

FIRST AID ADMINISTERED? ______________________________

____

____

INFRACTIONS OBSERVED: ______________________________

____

BY: ______________________________

PAGE ___ OF ___

CHAIN OF CUSTODY RECORD

PROJECT NO.	PROJECT NAME
SAMPLERS (SIGNATURE)	

STA. NO.	DATE	TIME	COMP.	GRAB	STATION LOCATION	NO. OF CON-TAINERS	SAMPLE TYPE												REMARKS INDICATE SOIL/WATER/AIR

RELINQUISHED BY: (SIGNATURE)	DATE / TIME	RECEIVED BY: (SIGNATURE)	RELINQUISED BY: (SIGNATURE)	DATE / TIME	RECEIVED BY: (SIGNATURE)
RELINQUISHED BY: (SIGNATURE)	DATE / TIME	RECEIVED BY: (SIGNATURE)	RELINQUISHED BY: (SIGNATURE)	DATE / TIME	RECEIVED BY: (SIGNATURE)
RELINQUISHED BY: (SIGNATURE)	DATE / TIME	RECEIVED FOR DISPOSAL BY: (SIGNATURE)	DATE / TIME	REMARKS	

APPENDIX C

MISCELLANEOUS REPORTS

SITE SAFETY FOLLOW UP REPORT

(To be completed for each field change in plan.)

Was the Safety Plan followed as presented? ___yes ___no

DESCRIBE IN DETAIL ANY CHANGES TO THE SAFETY PLAN:

REASON FOR CHANGES:

APPROVED BY SITE MANAGER: DATE:
SITE SAFETY OFFICER: DATE:

EVALUATION OF SITE SAFETY PLAN

Was the Safety Plan adequate? ___yes ___no

WHAT CHANGES WOULD YOU RECOMMEND?

ACCIDENT REPORT

Report No.:

SITE:
LOCATION:

PROJECT NO.:

DATE OF REPORT:
NAME AND ADDRESS OF INJURED:

PREPARERS NAME:
SSN:
AGE:
SEX:

YEARS OF SERVICES: TIME ON PRESENT JOB: TITLE/CLASSIFICATION:
DIVISION/DEPARTMENT: DATE OF ACCIDENT: TIME:

ACCIDENT CATEGORY:
__ Motor Vehicle
__ Property Damage
__ Fire
__ Chemical Exposure
__ Near Miss
__ Other

SEVERITY OF INJURY OR ILLNESS:
__ Non-disabling
__ Disabling
__ Medical Treatment
__ Fatality

AMOUNT OF DAMAGE: $

PROPERTY DAMAGED:

ESTIMATED NUMBER OF DAYS AWAY FROM JOB:
NATURE OF INJURY OR ILLNESS:

CLASSIFICATION OF INJURY

__ Fractures	__ Heat Burns	__ Cold Exposure
__ Dislocations	__ Chemical Burns	__ Frostbite
__ Sprains	__ Radiation Burns	__ Heat Stroke
__ Abrasions	__ Bruises	__ Heat Exhaustion
__ Lacerations	__ Blisters	__ Concussion
__ Punctures	__ Toxic Respiratory Exposure	__ Faint/Dizziness
__ Bites		__ Toxic Respiratory
__ Respiratory Allergy	__ Toxic Ingestion	__ Dermal Allergy

PART OF BODY AFFECTED:
DEGREE OF DISABILITY:
DATE MEDICAL CARE WAS RECEIVED:
WHERE MEDICAL CARE WAS RECEIVED:
ADDRESS (if off-site):

ACCIDENT LOCATION

Causative agent most directly related to accident (object, substance, material, machinery, equipment, conditions):

WAS WEATHER A FACTOR?

UNSAFE MECHANICAL/PHYSICAL/ENVIRONMENTAL CONDITION AT TIME OF ACCIDENT (be specific):

UNSAFE ACT BY INJURED AND/OR OTHERS CONTRIBUTING TO THE ACCIDENT (be specific, must be answered):

PERSONAL FACTORS (improper attitude, lack of knowledge of skill, slow reaction, fatigue):

LEVEL OF PERSONAL PROTECTION EQUIPMENT REQUIRED IN SITE SAFETY PLAN:

MODIFICATIONS:

WAS INJURED USING REQUIRED EQUIPMENT?

IF NOT, HOW DID ACTUAL EQUIPMENT USE DIFFER FROM PLAN?

WHAT CAN BE DONE TO PREVENT A RECURRENCE OF THIS TYPE OF ACCIDENT (modification of machine; mechanical guards; correct environment; training):

DETAILED NARRATIVE DESCRIPTION (how did accident occur, why; objects, equipment, tools used, circumstance assigned duties. Be specific.):

(Use back of sheet as required)

WITNESSES TO ACCIDENT:

Signature of Preparer:
Signature of Site Manager:

APPENDIX D

HEAT STRESS CASUALTY PREVENTION PLAN

HEAT STRESS CASUALTY PREVENTION PLAN

Due to the increase in ambient air temperatures and the effects of protective outer wear decreasing body ventilation, there exists an increase in the potential for injury, specifically, heat casualties. Site personnel will be instructed in the identification of a heat stress victim, the first-aid treatment procedures for the victim and the prevention of heat stress casualties.

A. IDENTIFICATION AND TREATMENT

1. Heat Exhaustion

a) Symptons: Usually begins with muscular weakness, dizziness, nausea, and a staggering gait. Vomiting is frequent. The bowels may move involuntarily. The victim is very pale, his skin is clammy, and he may perspire profusely. The pulse is weak and fast, his breathing is shallow. He may faint unless he lies down. This may pass, but sometimes it remains and death could occur.

b) First Aid: Immediately remove the victim to the Decomtamination Reduction Zone in a shady or cool area with good air circulation. Remove all protective outer wear. Call a physician. Treat the victim for shock. (Make him lie down, raise his feet 6-12 inches, and keep him warm but loosen all clothing.) If the victim is conscious, it may be helpful to give him sips of a salt water solution (1 teaspoon of salt to 1 glass of water). Transport vicitm to a medical facility.

2. Heat Stroke

a) Symptons: This is the most serious of heat casualties due to the fact that the body excessively overheats. Body temperatures often are between 107°- 110°F. First ther is often pain in the head, dizziness, nausea, oppression, and a dryness of the skin and mouth. Unconsciousness follows quickly and death is imminent if exposure continues. The attack will usually occur suddenly.

b) First Aid: Immediately evacuate the victim to a cool and shady area in the Decontamination Reduction Zone. Remove all protective outer wear and all personal clothing. Lay him on his back with the head and shoulders slightly elevated. It is imperative that the body temperature be lower immediately. This can be accomplished by applying cold wet towels, ice bags, etc., to the head. Sponge off the bare skin with cool water or rubbing alcohol, if available, or even place him in a tub of cool water. The main objective is to cool him without chilling him. Give no stimulants. Transport the victim to a medical facility as soon as possible.

B. PREVENTION OF HEAT STRESS

1) One of the major causes of heat casualties is the depletion of body

fluids. On the site there will be plenty of fluids available. Personnel should replace water and salts loss from sweating. Salts can be replaced by either a 0.1% salt solution, more heavily salted foods, or commercial mixes such as Gatorade. The commercial mixes are advised for personnel on low sodium diets.

2) A work schedule will be established so that the majority of the work day will be during the morning hours of the day before ambient air temperature levels reach their highs.

3) A work/rest schedule will be implemented for personnel required to wear Level B protection. A sufficient period will be allowed for personnel to "cool down". This may require shifts of workers during operations in addition to the breaks provided by required air tank changes (at least every 25 minutes). Maximum daily time at Level B shall be:

Ambient Temperatures	Maximum Number of Air Tanks per Day
Above 90°F	3
80°-90°F	4
60°-80°F	6
40°-60°F	7
30°-40°F	8

Notes:

1. This schedule does not apply to Level A work.
2. This schedule must be modified for strenuous activities.

APPENDIX B

SOIL GAS TESTING

Soil Gas Sampling Method

The following outlines the rationale, theory and field procedures for the soil gas sampling method. The method is described in detail in a paper presented at a 1983 National Water Well Association conference ("Detection of Groundwater Contamination by Shallow Soil Gas Sampling in the Vadose Zone," Eric G. Lappala and Glen M. Thompson).

Groundwater occurs as four zones within soils. The deepest zone occurs beneath the water table and is called the phreatic zone. The phreatic zone is saturated with water at a pressure greater than atmospheric pressure. Above the phreatric zone is the capillary zone where water is drawn into the soil voids by surface tension. Soil within the capillary zone ranges from being saturated with water at a pressure less than atmospheric pressure to having a water content nearly equal to the specific moisture capacity of the soil. The tension saturated portion of the capillary zone is called the capillary fringe. The portion of the capillary zone above the capillary fringe where water content grades from 100 percent to residual water is called the funicular zone. Above the capillary zone lies the unsaturated vadose zone with residual moisture equal to the specific moisture capacity of the soil.

When a product loss occurs in granular soils above an unconfined aquifer, vertical migration via gravity through the vadose zone develops. A lighter-than-water liquid contaminant (gasoline) will migrate through the unsaturated soils until it reaches the capillary zone where it will begin to flow outward. Lateral flow occurs through the unoccupied voids in the funicular zone. A lens of nearly pure product develops with a primary direction and rate of movement governed by the hydraulic gradient, soil permeability and rate of product loss. A gasoline product plume will move slightly faster than one-half the velocity of the groundwater ("Practical Geohydrological Aspects of Groundwater Contamination," William D. Shepard). Some of the product components will be solubilized into the groundwater and will travel at a greater velocity than the free product plume. Therefore, given a sufficient product spill, a plume of product will develop within the capillary zone with a halo of soluble components preceding it all flowing in the general direction of groundwater flow.

Water table fluctuations along with wicking actions of other transport mechanisms (hydrodynamic dispersion, molecular diffusion) cause contaminants to be introduced into the unsaturated portions of the capillary zone. Once volatile contaminants (gasoline) are present in and above the capillary fringe, they will volatilize (separate into liquid and gas phases). The more volatile the compound, the more tendency there is for it to go into the gaseous phase. Subsequent upward movement of the gas through the air filled voids of the vadose zone occurs through gaseous diffusion. Gas concentrations within the vadose zone are related to the concentration in the capillary zone and the distance from the source. Soil gas concentrations over a free product plume will be higher than those over a soluble product plume and concentrations two feet above a plume will be higher than concentrations 50 feet above a plume. A small spill on the surface may produce high soil gas concentrations near the surface which decrease with depth. This decrease results from the product being tied up the soil's specific moisture retention capacity. The result is less and less product with depth that is available to partition into the gaseous phase. Conversely, if the soil gas is being introduced into the vadose zone from the groundwater, soil gas concentrations will increase with depth until the groundwater source is encountered.

Soil gas sampling field procedures start by establishing a survey grid that extends upgradient of suspected sources and off to the sides of any suspicious plume. This grid will usually be extended to include any known contaminated groundwater points. Soil gas samples are obtained at each grid point by driving a hollow metal probe to a desired reference elevation into the soil. Work by Eric Lappala et al has shown that a depth of approximately 10 percent of the vertical distance to the water table is sufficient. A gas sample is obtained by attaching Tygon tubing to the probe and applying a vacuum to the system. A sample is drawn directly from the tubing by a needle syringe which is then injected into a gas chromatograph for analysis and quantification.

Soil gas concentrations are entered on a grid map of the area at each respective sample point and subsequently contoured. The contour map will outline the contamination plume by depicting the areas of highest concentrations. In this manner, plume contours can be used to define the contamination origin, lateral extent and direction of movement.

Based on the soil gas sample contour map, monitor wells are installed upstream of the pinpointed source, on either side of the outlined plume, and within the plume. During the drilling process to install the monitor wells, vertical soil gas samples are obtained to confirm that the mapped soil gas concentrations are the result of a groundwater contamination plume and not from a minor localized surface spill. Vertical sampling is conducted by driving the sampling probe ahead of the hollow stem lead auger which has been advanced to just above the sampling point. Samples are taken at regular intervals from the surface down to the water table.

Protocol for Soil Gas Testing

The analysis of soil gas for organic constituents requires the collection of a representative sample followed by accurate, precise detection. Because many variables are involved in the process a precise analytical protocol must be developed to minimize the influence of these variables.

The detection component of the technique is accomplished using either a Photovac 10A10 or a AID 210 portable gas chromatograph (GC). Each instrument is designed to provide detection based on the operation of chromatographic separation techniques and the differential photoionization potential of organic compounds. The sample is injected into a nitrogen gas stream which carries the sample through a column packed with a porous solid coated with a stationary liquid phase. The volume provides a surface which attracts and absorbs molecules. The extent of attraction and retention is determined by the size and polarity of the molecules, and the liquid solvent. As a result of these differential attractions and retentions each organic compound completes passage through the column at a different reproducible rate. Thus, this process functions to effectively separate each compound within the gas stream.

TABLE 1

IONIZATOIN POTENTIALS FOR SELECTED MOLECULES

Compound	I.P. (eV)
Diethyle sulfide	8.43
m-Xylene	8.56
Dimethyl sulfide	8.68
Toluene	8.82
Cyclohexene	8.94
1,3-Butadiene	9.07
Benzene	9.25
Pyridine	9.32
Tetrachloroethylene	9.32
Trichloroethylene	9.45
Allyl Alcohol	9.67
Acetone	9.69
Methyl ethyl ketone	9.53
Tetrahydrofuran	9.54
Cyclohexene	9.88
Vinyl Chloride	9.95
Carbon disulfide	10.08
Acrolein	10.10
Hexane	10.17
Ethyl alcohol	10.48
Ethylene	10.51
Choroethane	10.98
1,1-Dichloroethane	11.06
T-1,2-Dichloroethene	11.12
Methylene chloride	11.28

As the sample stream is eluded from the column it enters a detector chamber. An adjacent ultraviolet lamp emits photons at an energy of 10.0 or 10.6 electron volts (interchangeable lamps are available) which are directed into the gas stream. A percentage of the organic molecules in the gas stream having sufficient low ionization potentials will be ionized by this energy source. The ions produced are collected, forming a small current which is amplified and eventually measured by a potentiometric recorder. Table 1 contains a list of selected organic compounds having a photoionization potential less than 12eV.

Through this combination of separation and ionization it is possible to detect the presence of numerous organic compounds. When organic compounds of known concentrations are similarly analyzed it becomes possible to quantify the presence of organic compounds in unknown samples through comparison of their relative potentiometric responses.

The operation of the GC is largely independent of sampling techniques. To insure optimal operation the method of injection, injection volume, flow rate and temperature are identical for samples and standards. Fresh standards are prepared daily for meter calibration using headspace vapor techniques.

Glass Teflon syringes are used for all standards and samples. Quality control is assured by the injection of blank samples with syringes prior to their use in sampling or with standards.

The collection of samples requires the ability to withdraw representative soil gas from a known depth in the soil column and maintain the integrity of the sample during transport to the GC. This process is accomplished with a soil gas probe in conjunction with a vacuum pump. This probe is constructed of hollow 7/8-inch O.D. stainless steel with a solid, screw-on tip. The lower one-foot section contains 16 1/6-inch holes which permit entry to the soil gas. The upper section has a 1/16-inch hose barb to which 1/16-inch I.D. urethane tubing is attached.

The probe is driven to the appropriate depth and the tubing attached to a battery operated vacuum pump. The pump design limits air contact to the urethane tubing and the stainless steel probe. The vacuum pump is operated for 30 seconds to purge the probe casing and insure a representative sample. the sample is collected by inserting a Pressure-Lock syringe into the air vacuum tubing and withdrawing a one ml sample. The sample is secured in the syringe with the locking mechanism and transported to the GC for injection. The section of tubing perforated by the syringe is removed and the line reconnected to the probe for the next sample.

<u>APPENDIX C</u>

CHAIN OF CUSTODY FORM

PAGE ___ OF ___

CHAIN OF CUSTODY RECORD

PROJECT NO	PROJECT NAME					NO OF CONTAINERS	SAMPLE TYPE												REMARKS INDICATE SOIL/WATER/AIR SEDIMENT/SLUDGE
SAMPLERS (SIGNATURE)																			
STA NO	DATE	TIME	COMP.	GRAB	STATION LOCATION														

RELINQUISHED BY: (SIGNATURE)	DATE / TIME	RECEIVED BY: (SIGNATURE)	RELINQUISHED BY: (SIGNATURE)	DATE / TIME	RECEIVED BY: (SIGNATURE)
RELINQUISHED BY: (SIGNATURE)	DATE / TIME	RECEIVED BY: (SIGNATURE)	RELINQUISHED BY: (SIGNATURE)	DATE / TIME	RECEIVED BY: (SIGNATURE)
RELINQUISHED BY: (SIGNATURE)	DATE / TIME	RECEIVED FOR DISPOSAL BY: (SIGNATURE)	DATE / TIME	REMARKS	

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APPENDIX D

WELL INSTALLATION GUIDELINES

AND SPECIFICATIONS

Monitoring Well Installation

Should a PRP choose to participate in the hydrogeologic study of the Cadillac area, there will be a need for a high degree of standardization of all phases of field activities to ensure the consistency and validity of acquired information. The following section describes the procedures and design specifications to be followed for the installation of monitoring wells. Inasmuch as these monitoring wells will provide the primary source of geologic and hydrologic information, it is essential that the specified protocols and guidelines be followed.

The first consideration is that a registered well driller, familiar with small-diameter, monitoring well installations be retained. The required drilling method will be to use hollow stem augers (HSA) that are approximately 6 inches outside diameter (O.D.) and 4 inches inside diameter (I.D.). The driller should be able to attain a drilling depth of at least 180 feet below ground surface although most wells will be shallower. (Note: A depth of 193 feet was achieved using HSA at the Northernaire site.)

The capability to collect groundwater samples at 10 foot intervals during drilling using screened augers or other suitable methods is necessary so that the vertical distribution of contaminants and the final placement of the well screen can be determined in the field. This method of groundwater sampling with depth will be used inasmuch as mulitiple-screen, single well installations will not be allowed.

During advancement of the borehole, the drill cuttings should be logged and attention paid to the depth at which strata changes occur. A geologic log noting all observations during drilling should be maintained in the field by an experienced geologist for future reference. The Unified Soil Classification System shall be used for all sediment identification. Attached is the standard log form to be completed by the geologist for each well installation. An additional well construction form will also be completed by the geologist showing the final specifications and vertical position of the monitoring well.

Once the borehole has been advanced to the desired depth and the position of the screen determined, the monitoring well can be constructed through the hollow stem augers. The monitoring well will have a 2 inch I.D. galvanized steel riser pipe with a 5 foot length of stainless steel, No. 10 slot (.010 inch), screen. The well sections will be steam cleaned prior to well construction. The riser pipe should have an above-ground stick-up of approximately 2.0 to 2.5 feet after the well is completed. With the well set at the desired depth, a sand filter pack will be placed around the well screen as the augers are withdrawn. In most instances, a natural sand pack can be used by allowing the borehole to collapse around the well screen as the augers are withdrawn. The sand pack should continue above the top of the well screen for about 5 feet. In the unlikely event the the well screen is being set in a zone high in silt or clay, a clean, coarse silica sand will be poured between the well pipe and augers and allowed to settle slowly until it can be determined by a weighted tape measure that the entire well screen is surrounded by the sand pack. With the sand pack in place, a bentonite grout using a tremmie line will be pumped into the annulus to provide a seal to prevent any unnatural migration of contaminants through the disturbed area around the borehole. If the well screen is to be set within only a few feet of the water table, dry bentonite pallets will be used to ensure a tight seal prior to filling the remainder of the annulus with the grout.

Upon completion of grouting and extraction of all the auger flights, all down hole drilling equipment must be steam cleaned prior to use at another well site to prevent cross contamination.

Once installed, the new monitoring well will be developed by hydraulic surging using the air lift technique. The air supply should be equipped with an oil filter to prevent the introduction of hydrocarbons into the well and aquifer. The filter should be changed frequently and an indicating filter (one that changes color with the presence of hydrocarbons) is recommended. Well development can either be done after all of the wells have been installed or individually at the completion of each well. Development should continue until all fine sediments have been removed from the well.

Because the bentonite grout used to seal the borehole will settle considerably over a 48 hour period, re-grouting to the surface will be necessary. After regrouting, a protective steel casing with locking cap will be securely cemented into place. The protective casing should extend a minimum of 2 feet below ground surface. The distance between the top of the protective casing to the top of the well riser should not exceed 3 inches.

The elevation of the top of the rise pipe and the ground surface near the monitoring well must be surveyed by a licensed surveyor based on the National Geodetic Vertical (NGV) datum. The horizontal position of each well must also be determined for location on a area wide base map. A standard base map will be provided to participating PRPs by MDNR for use in planning, recording and report presentations. Each monitoring well will be permanently labeled and identified based on a standardized identification system which MDNR will relate to the PRP.

CADILLAC AREA GROUNDWATER STUDY	JOB NO.________________	BORING I.D.__________
	DRILL RIG/METHOD__________	LOCATION__________
FIELD BORING LOG	OPERATOR__________	DATE STARTED__________
COMPANY__________	INSPECTOR__________	DATE COMPLETED__________

SAMPLE DEPTH		TYPE OF SAMPLE	CLP/ECJ I.D.	STRATA CHANGE	UNIFIED SOIL CLASSIFICATION: COLOR: GRADATION(PLASTICITY):DENSITY(CONSISTENCY): MOISTURE:STRUCTURE:GEOLOGIC NAME	FIELD PARAMETERS					
FROM	TO					HEAD SPACE	HEX CR.	pH	SPEC. COND.	TEMP.	OTHER

GROUNDWATER ENCOUNTERED AT _______ FT.

UNITS

NOTES:

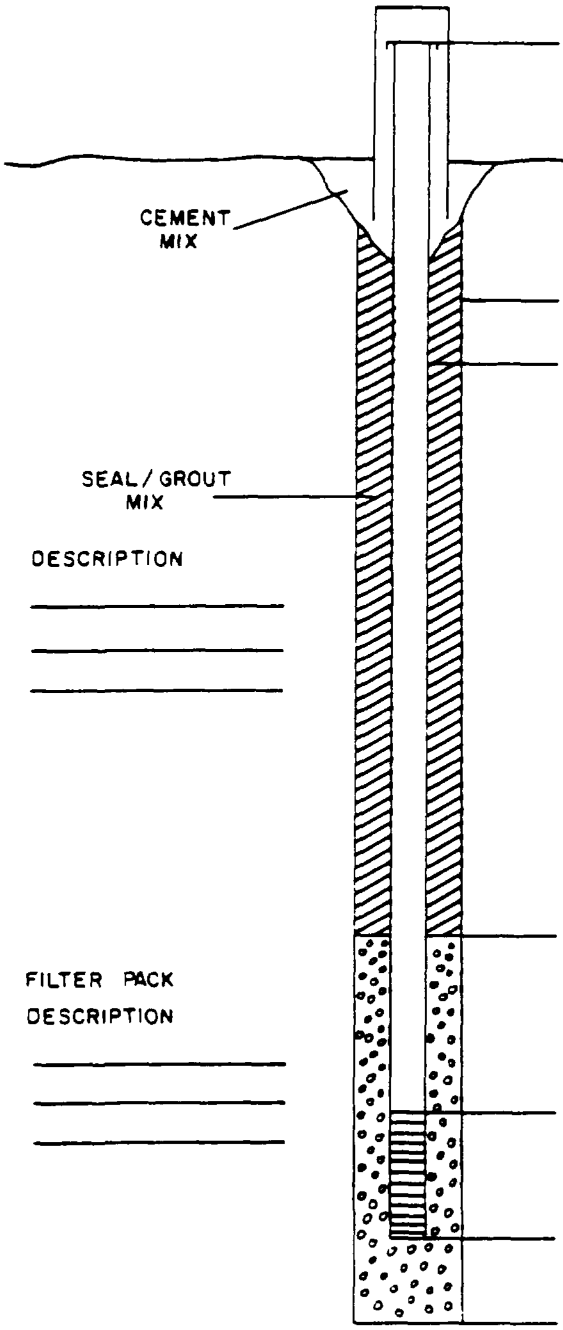
TOTAL DEPTH

PAGE OF

TRACE 0-10% LITTLE 10-20% SOME 20-35% AND 35-50%

MONITORING WELL INSTALLATION RECORD

MONITORING WELL ____________



TOP OF RISER PIPE
ELEVATION, (UNPLUGGED) ____________

GROUND SURFACE ELEVATION

CEMENT MIX

DIAMETER OF BOREHOLE ____________

I.D. OF RISER PIPE ________

MATERIAL ____________

SEAL / GROUT MIX

DESCRIPTION

____

____

____

DEPTH, TOP OF FILTER PACK ____________

FILTER PACK DESCRIPTION

____

____

____

DEPTH, TOP OF SCREEN ____________

MATERIAL ____________

SLOT SIZE ____________

DEPTH, BOTTOM OF SCREEN ____________

DEPTH, BOTTOM OF BORING ____________

AQUITARD INFORMATION

DEPTH TO UPPER ____________ THICKNESS ____________

DEPTH TO LOWER ____________

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4656-51

UNIFIED SOIL CLASSIFICATION SYSTEM

GRADATION AND PLASTICITY CHARACTERISTICS (Excluding particles larger than 3 in.)				GROUP SYMBOL	SOIL NAMES
PREDOMINANTLY COARSE-GRAINED More than 50% of soil retained by No. 200 Sieve	GRAVEL More than 50% of coarse-grained fraction retained by No. 4 sieve	CLEAN Less than 5% fines	$C_u = \frac{D_{60}}{D_{10}}$ greater than 4; $C_c = \frac{(D_{30})^2}{D_{10} \times D_{60}}$ between 1 and 3	GW	Well-graded GRAVEL or SANDY GRAVEL
			Not meeting all gradation requirements for GW	GP	Poorly graded, uniform, or gap-graded GRAVEL or SANDY GRAVEL
		WITH FINES More than 12% fines*	Atterberg limits below A-line or PI less than 4	GM	SILTY GRAVEL
			Atterberg limits above A-line with PI greater than 7†	GC	CLAYEY GRAVEL
	SAND More than 50% of coarse-grained fraction passes No. 4 sieve	CLEAN Less than 5% fines	$C_u = \frac{D_{60}}{D_{10}}$ greater than 6; $C_c = \frac{(D_{30})^2}{D_{10} \times D_{60}}$ between 1 and 3	SW	Well-graded SAND or GRAVELLY SAND
			Not meeting all gradations requirements for SW	SP	Poorly graded, uniform, or gap-graded SAND or GRAVELLY SAND
		WITH FINES More than 12% fines*	Atterberg limits below A-line or PI less than 4	SM	SILTY SAND
			Atterberg limits above A-line with PI greater than 7†	SC	CLAYEY SAND
PREDOMINANTLY FINE-GRAINED More than 50% of soil passes No. 200 sieve	Plasticity characteristics of fraction passing No. 40 sieve: SILT Atterberg limits below A-line or PI less than 4	INORGANIC	Liquid limit less than 50	ML	Nonplastic, slightly plastic, or moderately plastic SILT, CLAYEY SILT, SANDY SILT, or GRAVELLY SILT
			Liquid limit greater than 50	MH	Highly plastic or very highly plastic SILT, CLAYEY SILT, SANDY SILT, or GRAVELLY SILT
		ORGANIC	Liquid limit less than 50	OL	Slightly plastic or moderately plastic ORGANIC SILT (ORGANIC CLAY if Atterberg limits above A-line)
			Liquid limit greater than 50	OH	Highly plastic or very highly plastic ORGANIC SILT (ORGANIC CLAY if Atterberg limits above A-line)
	CLAY Atterberg limits above A-line with PI greater than 7†		Liquid limit less than 50	CL	Slightly plastic or moderately plastic CLAY, SILTY CLAY, SANDY CLAY, or GRAVELLY CLAY
			Liquid limit greater than 50	CH	Highly plastic or very high plastic CLAY, SILTY CLAY, SANDY CLAY, or GRAVELLY CLAY
HIGHLY ORGANIC (identified by spongy feel)				Pt	PEAT

*Predominantly coarse-grained soil with 5% to 12% fines is borderline case requiring use of dual symbol, such as GW-GC or SM-SP.

†Atterberg limits above A-line with PI between 4 and 7 is borderline case requiring use of dual symbol, GC-GM or SC-SM for predominantly coarse-grained soil or CL-ML for predominantly fine-grained soil.

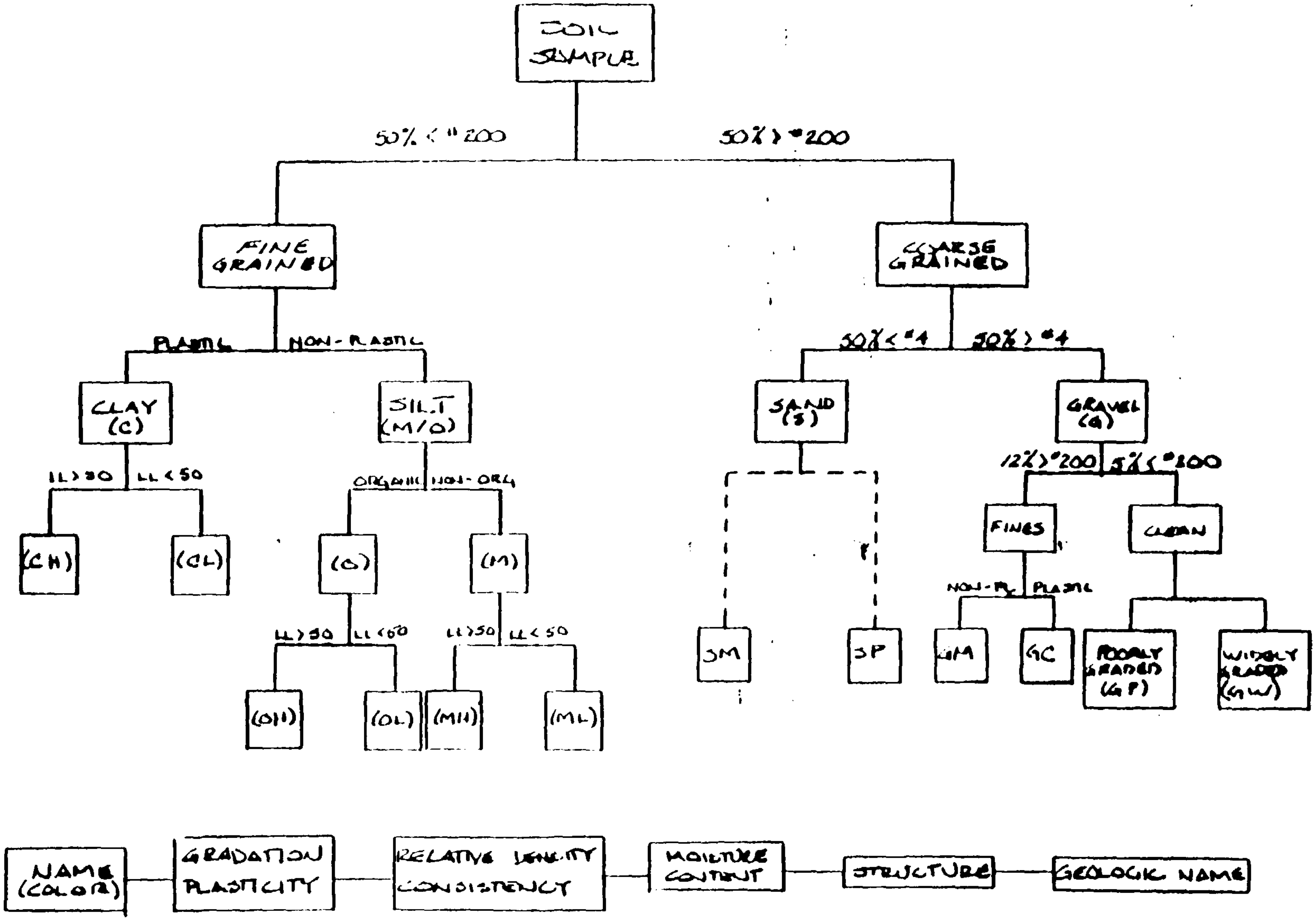
JOB NO.

DATE [redacted] 14, 1986

COMP. BY

CHK. BY

PROJECT



SOIL DESCRIPTION FLOW CHART

UNIFIED SOIL CLASSIFICATION SYSTEM

FORMAT FOR DESCRIPTION OF SOILS

1. NAME (*basic name* indicating predominant constituent and having a *modifier*, where required, indicating major subordinate constituent, both preceded, where required by *introductory name* for unusual or mixed soils)

   a. Introductory name: Fill, Topsoil, Varved Clay, Layered Sand and Silt, etc.
   b. Modifier: Gravelly, Sandy, Silty, Clayey, or Organic
   c. Name of predominant constituent: Gravel, Sand, Silt, Clay, or Peat

2. GRADATION AND/OR PLASTICITY

   For predominantly coarse-grained soil:

   a. Evaluation of *overall* gradation of soil (excluding fraction coarser than 3 in.): well-graded, poorly graded, or uniform for clean soil; widely graded, uniform, or omitted for soil with fines
   b. Characteristics of individual constituents, starting with coarsest fraction and proceeding to finest, omitting percent of predominant constituent; where required, indication of shape, angularity, and either mineralogical composition or soundness of individual particles:

      (1) Percent (by volume occupied before excavation) and maximum particle size of *boulders* (coarser than 12 in.) and *cobbles*
      (2) Percent (by weight of entire fraction finer than 3 in.) and either limitation in range of particle sizes (coarse, fine, or a combination of both) or maximum particle size of *gravel*; where poorly graded, indication of excessive size range
      (3) Percent and limitation in range of particle sizes (coarse, medium, fine, or a combination of two) of *sand*; where poorly graded, indication of excessive size range
      (4) Percent and overall plasticity of *fines*

   For predominantly fine-grained soil:

   a. Plasticity: nonplastic, slightly plastic, moderately plastic, highly plastic, very highly plastic, or a combination of two
   b. Percent and range of particle sizes of coarse-grained consituents

3. RELATIVE DENSITY OR CONSISTENCY (for undisturbed soil only)

   For noncohesive or low-cohesion soil:

   Relative density: very loose, loose, compact, dense, or very dense

   For cohesive soil:

   Consistency: very soft, soft, firm, stiff, very stiff, or hard

4. NATURAL MOISTURE CONDITION

   Dry, damp, moist, or saturated, where required

5. COLOR (noted at natural moisture condition)

6. STRUCTURE (including discontinuities and inclusions)

7. GEOLOGICAL ORIGIN AND/OR LOCAL NAME (where applicable)

8. UNIFIED SOIL CLASSIFICATION SYSTEM GROUP SYMBOL

Table 3.6 Soil Components and Fractions

Soil	Soil Component	Symbol	Grain Size Range and Description	Significant Properties
Coarse-grained components	Boulder	None	Rounded to angular, bulky, hard, rock particle, average diameter more than 12 in.	Boulders and cobbles are very stable components, used for fills, ballast, and to stabilize slopes (riprap). Because of size and weight, their occurrence in natural deposits tends to improve the stability of foundations. Angularity of particles increases stability.
	Cobble	None	Rounded to angular, bulky, hard, rock particle, average diameter smaller than 12 in. but larger than 6 in.	
	Gravel	G	Rounded to angular bulky, hard, rock particle, passing 3-in. sieve (76.2 mm) retained on No. 4 sieve (4.76 mm)	Gravel and sand have essentially same engineering properties differing mainly in degree. The No. 4 sieve is arbitrary division, and does not correspond to significant change in properties. They are easy to compact, little affected by moisture, not subject to frost action. Gravels are generally more perviously stable, resistant to erosion and piping than are sands. The well-graded sands and gravels are generally less pervious and more stable than those which are poorly graded (uniform gradation). Irregularity of particles increases the stability slightly. Finer, uniform sand approaches the characteristics of silt: i.e., decrease in permeability and reduction in stability with increase in moisture.
	Coarse		3- to ¾-in.	
	Fine		¾-in. to No. 4	
	Sand	S	Rounded to angular, bulky, hard, rock particle, passing No. 4 sieve (4.76 mm) retained on No. 200 sieve (0.74 mm)	
	Coarse		No. 4 to 10 sieves	
	Medium		No. 10 to 40 sieves	
	Fine		No. 40 to 200 sieves	
Fine-grained components	Silt	M	Particles smaller than No. 200 sieve (0.74 mm) identified by behavior: that is, slightly or non-plastic regardless of moisture and exhibits little or no strength when air dried	Silt is inherently unstable, particularly when moisture is increased, with a tendency to become quick when saturated. It is relatively impervious, difficult to compact, highly susceptible to frost heave, easily erodible and subject to piping and boiling. Bulky grains reduce compressibility; flaky grains, i.e., mica, diatoms, increase compressibility, produce an "elastic" silt.
	Clay	C	Particles smaller than No. 200 sieve (0.74 mm) identified by behavior; that is, it can be made to exhibit plastic properties within a certain range of moisture and exhibits considerable strength when air dried	The distinguishing characteristic of clay is cohesion or cohesive strength, which increases with decrease in moisture. The permeability of clay is very low, it is difficult to compact when wet and impossible to drain by ordinary means, when compacted is resistant to erosion and piping, is not susceptible to frost heave, is subject to expansion and shrinkage with changes in moisture. The properties are influenced not only by the size and shape (flat, plate-like particles) but also by their mineral composition; i.e., the type of clay-mineral, and chemical environment or base exchange capacity. In general, the montmorillonite clay mineral has greatest, illite and kaolinite the least, adverse effect on the properties.
	Organic matter	O	Organic matter in various sizes and stages of decomposition	Organic matter present even in moderate amounts increases the compressibility and reduces the stability of the fine-grained components. It may decay causing voids or by chemical alteration change the properties of a soil, hence organic soils are not desirable for engineering uses.

From Wagner, 1957.

Note. The symbols and fractions were developed for the Unified Classification System. For field identification, ¼ in. is assumed equivalent to the No. 4, and the No. 200 is defined as "about the smallest particle visible to the unaided eye." The sand fractions are not equal divisions on a logarithmic plot; the No. 10 was selected because of the significance attached to that size by some investigators. The No. 40 was chosen because the "Atterberg limits" tests are performed on the fraction of soil finer than the No. 40.

EXAMPLES OF SOIL DESCRIPTIONS

Sand, uniform, fine, well-rounded, 2-5% fines, loose, damp, light brown (SP)

Clay, highly plastic, stiff, dark gray (CH)

Silt, nonplastic, 5-10% very fine sand, very loose, saturated, light greenish gray, some mica, lacustrine (ML)

Gravelly sand, well-graded, 15-20% subangular gravel to 0.6 in. max, subangular sand, <5% fines, dense, moist, reddish brown, alluvial (SW)

Silty clay, moderately plastic, tough at plastic limit, 5-10% fine sand, hard undisturbed, becomes stiff when remolded, saturated, medium brown, calcareous throughout (CL)

Sand, uniform, medium to fine, mostly medium, 5-10% nonplastic fines, compact, white, noncalcareous (SP-SM)

Organic silt, highly plastic, weak at plastic limit, 3-8% coarse to fine sand, firm to stiff but spongy undisturbed, becomes soft and sticky when remolded, dull dark olive streaked with dark brown, many fine roots, trace mica (OH)

TILL: Silty gravel, widely graded to 1.2 in. max, subrounded, 30-40% mostly medium to fine sand, 15-20% slightly plastic fines, very dense, damp, brownish red (GM-SM)

LAYERED SILT AND CLAY: Silt, nonplastic, sudden reaction to shaking, 2-4% fine sand, medium greenish gray, layers mostly 1.5 to 8.3 in. thick, 85% of sample; clay, highly plastic, firm and brittle undisturbed, becomes very soft and sticky when remolded, dull dark gray, layers 0.2 to 1.2 in. thick (ML and CH)

VARVED CLAY: Silty clay, slightly to moderately plastic, firm, medium gray, grades downward within varve to sandy silt, nonplastic, 10-15% very fine sand, light gray; varves 0. 3 to 0.4 in. thick (CL to ML)

BOULDER TILL: Silty sand, widely graded, 10-15% rounded boulders and cobbles to 3 ft max, 20-25% subrounded gravel, 15-20% nonplastic fines, dense, saturated, olive gray (SM)

Silty sand, widely graded, 15-25% freshly broken fragments well-rounded flat particles weak sandstone to 1.0 in. max, mostly fine sand, 20-30% slightly plastic fines, compact, moist, light yellowish brown, very few small particles coal, fill (SM)

Sandy clay, moderately plastic, 15-20% medium to fine sand, hard undisturbed, crumbles when remolded, bright medium gray marbled with light yellowish brown, some 0.3 to 0.6 in. layers or lenses light brown fine silty sand (CL)

Clayey sand, coarse to fine, mostly medium to fine, 35-45% moderately plastic fines, very stiff, dark greenish gray, some mica, many shells (SC-CL)

Sandy silt, highly plastic, 25-30% fine sand, stiff, saturated, very light gray to white, weak calcareous cementation, marl (MH)

FIELD IDENTIFICATION PROCEDURES FOR FINE-GRAINED SOILS OR FRACTIONS

TYPE OF TEST		TOUGHNESS (Consistency near plastic limit)	DILATANCY (Reaction to shaking)	DRY STRENGTH (Crushing characteristics)
		Procedures performed on fraction passing No. 40 sieve; for field classification, simply remove by hand coarser particles that interfere.		
TEST PROCEDURE AND EVALUATION OF RESULTS		Mold a lump of soil about 1/2 in. in size to the consistency of putty. If too dry, add water. If too wet, spread out into a thin layer and allow it to lose some water by evaporation. Roll the lump by hand on a smooth surface or between the palms to a thread about 1/8 in. in diameter. Fold the thread and reroll repeatedly until the thread crumbles at a diameter of about 1/8 in. when the plastic limit is reached. After the thread crumbles, the pieces should be lumped together and kneaded until the lump crumbles. The tougher the thread near the plastic limit and the stiffer the lump when it finally crumbles, the more active are the clay minerals in the soil. Weakness of the thread at the plastic limit and quick loss of coherance of the lump below the plastic limit indicate plastic silts, kaolin-type clays, or organic soils which occur below the A-line on the plasticity chart.	Mold a lump of soil about 1/2 in. in size with enough water, if necessary, to make the soil soft but not sticky. Smooth the soil pat in the palm of one hand and shake horizontally, striking the back of the hand vigorously against the other hand several times. A positive reaction consists of the appearance of water on the surface of the pay which changes to a livery consistency and becomes glossy. When the pat is squeezed between the fingers, the water and gloss disappear, the pat stiffens, and finally it cracks or crumbles. The reaction is **sudden** if instantly produced by a single blow, **fast** if produced by less than 5 blows, **slow** if requiring more than 5 blows, and **none** if no change can be seen after many blows. Nonplastic silts give a sudden or fast reaction whereas clays which occur above the A-line show no reaction.	Mold a lump of soil about 1/2 in. in size to the consistency of putty, adding water if necessary. Allow the lump to dry completely in the sun, air, or oven, and test its strength by crushing between the fingers. The crushing strength is as follows: **none** or **very low** if the lump crumbles with the mere pressure of handling. **low** if the lump crumbles to powder with little finger pressure. **medium** if considerable finger pressure is required to powder the lump, but a smear of powder can be easily rubbed off a smooth surface of the lump. **high** if the lump cannot be crushed to powder by finger pressure, even though it may be broken, and it is not even possible to rub off a smear of powder from a smooth surface of the lump. **very high** if lump cannot be broken between thumb and a hard surface.
IDENTIFICATION OF GROUPS	ML	None to weak	Sudden to slow	None to low
	MH	Very weak to firm	Slow to none	Low to medium
	OL	Very weak to weak	Slow	Low to medium
	OH	Very weak to firm	Slow to none	Medium to high
	CL	firm to tough	Slow to none	Medium to high
	CH	Firm to very tough	None	High to very high

PLATE 4

Table 4.5 Unified Soil Classification

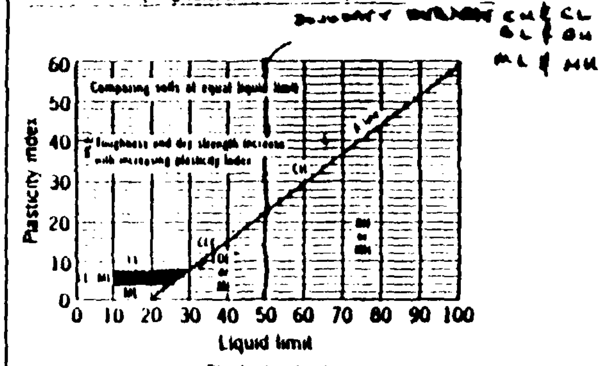
Major divisions			Field Identification Procedures (Excluding particles larger than 3 in. and basing fractions on estimated weights)	Group Symbols[a]	Typical Names	Information Required for Describing Soils	Laboratory Classification Criteria
Coarse-grained soils: More than half of material is larger than No. 200 sieve size[b] (The No. 200 sieve size is about the smallest particle visible to naked eye)	Gravels: More than half of coarse fraction is larger than No. 7 sieve size (For visual classification, the ¼ in. size may be used as equivalent to the No. 7 sieve size)	Clean gravels (little or no fines)	Wide range in grain size and substantial amounts of all intermediate particle sizes	GW	Well graded gravels, gravel-sand mixtures, little or no fines	Give typical name; indicate approximate percentages of sand and gravel, maximum size; angularity, surface condition, and hardness of the coarse grains; local or geologic name and other pertinent descriptive information; and symbols in parentheses	$C_u = \frac{D_{60}}{D_{10}}$ Greater than 4; $C_c = \frac{(D_{30})^2}{D_{10} \times D_{60}}$ Between 1 and 3
			Predominantly one size or a range of sizes with some intermediate sizes missing	GP	Poorly graded gravels, gravel-sand mixtures, little or no fines		Not meeting all gradation requirements for GW
		Gravels with fines (appreciable amount of fines)	Nonplastic fines (for identification procedures see ML below)	GM	Silty gravels, poorly graded gravel-sand-silt mixtures	For undisturbed soils add information on stratification, degree of compactness, cementation, moisture conditions and drainage characteristics	Atterberg limits below "A" line, or PI less than 4 — Above "A" line with PI between 4 and 7 are borderline cases requiring use of dual symbols
			Plastic fines (for identification procedures, see CL below)	GC	Clayey gravels, poorly graded gravel-sand-clay mixtures		Atterberg limits above "A" line, with PI greater than 7
	Sands: More than half of coarse fraction is smaller than No. 7 sieve size	Clean sands (little or no fines)	Wide range in grain sizes and substantial amounts of all intermediate particle sizes	SW	Well graded sands, gravelly sands, little or no fines	Example: *Silty sand, gravelly*; about 20% hard, angular gravel particles ½ in. maximum size; rounded and subangular sand grains coarse to fine; about 15% non-plastic fines with low dry strength; well compacted and moist in place; alluvial sand; (SM)	$C_u = \frac{D_{60}}{D_{10}}$ Greater than 6; $C_c = \frac{(D_{30})^2}{D_{10} \times D_{60}}$ Between 1 and 3
			Predominantly one size or a range of sizes with some intermediate sizes missing	SP	Poorly graded sands, gravelly sands, little or no fines		Not meeting all gradation requirements for SW
		Sands with fines (appreciable amount of fines)	Nonplastic fines (for identification procedures, see ML below)	SM	Silty sands, poorly graded sand-silt mixtures		Atterberg limits below "A" line or PI less than 5 — Above "A" line with PI between 4 and 7 are borderline cases requiring use of dual symbols
			Plastic fines (for identification procedures, see CL below)	SC	Clayey sands, poorly graded sand-clay mixtures		Atterberg limits below "A" line with PI greater than 7

Laboratory classification (coarse-grained soils): Determine percentages of gravel and sand from grain size curve. Depending on percentage of fines (fraction smaller than No. 200 sieve size) coarse grained soils are classified as follows: Less than 5% GW, GP, SW, SP; More than 12% GM, GC, SM, SC; 5% to 12% *Borderline* cases requiring use of dual symbols.

Use grain size curve in identifying the fractions as given under field identification.

Identification Procedures on Fraction Smaller than No. 40 Sieve Size

Major divisions		Dry Strength (crushing characteristics)	Dilatancy (reaction to shaking)	Toughness (consistency near plastic limit)	Group Symbols[a]	Typical Names	Information Required for Describing Soils
Fine-grained soils: More than half of material is smaller than No. 200 sieve size (The No. 200 sieve size is about the smallest particle visible to naked eye)	Silts and clays, liquid limit less than 50	None to slight	Quick to slow	None	ML	Inorganic silts and very fine sand, rock flour, silty or clayey fine sands with slight plasticity	Give typical name; indicate degree and character of plasticity, amount and maximum size of coarse grains; colour in wet condition, odour if any, local or geologic name and other pertinent descriptive information, and symbol in parentheses
		Medium to high	None to very slow	Medium	CL	Inorganic clays of low to medium plasticity, gravelly clays, sandy clays, silty clays, lean clays	
		Slight to medium	Slow	Slight	OL	Organic silts and organic silt-clays of low plasticity	For undisturbed soils add information on structure, stratification, consistency in undisturbed and remoulded states, moisture and drainage conditions
	Silts and clays, liquid limit greater than 50	Slight to medium	Slow to none	Slight to medium	MH	Inorganic silts, micaceous or diatomaceous fine sandy or silty soils, elastic silts	
		High to very high	None	High	CH	Inorganic clays of high plasticity, fat clays	Example: *Clayey silt*, brown; slightly plastic; small percentage of fine sand; numerous vertical root holes; firm and dry in place; loess; (ML)
		Medium to high	None to very slow	Slight to medium	OH	Organic clays of medium to high plasticity	
Highly Organic Soils		Readily identified by colour, odour, spongy feel and frequently by fibrous texture			Pt	Peat and other highly organic soils	



Plasticity chart
for laboratory classification of fine grained soils

From Wagner, 1957.

[a] *Boundary classifications*: Soils possessing characteristics of two groups are designated by combinations of group symbols. For example GW-GC, well graded gravel-sand mixture with clay binder.

[b] All sieve sizes on this chart are U.S. standard.

Field Identification Procedure for Fine Grained Soils or Fractions

These procedures are to be performed on the minus No. 40 sieve size particles, approximately 1/64 in. For field classification purposes, screening is not intended, simply remove by hand the coarse particles that interfere with the tests.

Dilatancy (Reaction to shaking)

After removing particles larger than No. 40 sieve size, prepare a pat of moist soil with a volume of about one half cubic inch. Add enough water if necessary to make the soil soft but not sticky.

Place the pat in the open palm of one hand and shake horizontally, striking vigorously against the other hand several times. A positive reaction consists of the appearance of water on the surface of the pat which changes to a livery consistency and becomes glossy. When the sample is squeezed between the fingers, the water and gloss disappear from the surface, the pat stiffens and finally it cracks or crumbles. The rapidity of appearance of water during shaking and of its disappearance during squeezing assist in identifying the character of the fines in a soil.

Very fine clean sands give the quickest and most distinct reaction whereas a plastic clay has no reaction. Inorganic silts, such as a typical rock flour, show a moderately quick reaction.

Dry Strength (Crushing characteristics)

After removing particles larger than No. 40 sieve size, mould a pat of soil to the consistency of putty, adding water if necessary. Allow the pat to dry completely by oven, sun or air drying, and then test its strength by breaking and crumbling between the fingers. This strength is a measure of the character and quantity of the colloidal fraction contained in the soil. The dry strength increases with increasing plasticity.

High dry strength is characteristic for clays of the CH group. A typical inorganic silt possesses only very slight dry strength. Silty fine sands and silts have about the same slight dry strength, but can be distinguished by the feel when powdering the dried specimen. Fine sand feels gritty whereas a typical silt has the smooth feel of flour.

Toughness (Consistency near plastic limit)

After removing particles larger than the No. 40 sieve size, a specimen of soil about one-half inch cube in size, is moulded to the consistency of putty. If too dry, water must be added and if sticky, the specimen should be spread out in a thin layer and allowed to lose some moisture by evaporation. Then the specimen is rolled out by hand on a smooth surface or between the palms into a thread about one-eight inch in diameter. The thread is then folded and re-rolled repeatedly. During this manipulation the moisture content is gradually reduced and the specimen stiffens, finally loses its plasticity, and crumbles when the plastic limit is reached.

After the thread crumbles, the pieces should be lumped together and a slight kneading action continued until the lump crumbles.

The tougher the thread near the plastic limit and the stiffer the lump when it finally crumbles, the more potent is the colloidal clay fraction in the soil. Weakness of the thread at the plastic limit and quick loss of coherence of the lump below the plastic limit indicate either inorganic clay of low plasticity, or materials such as kaolin-type clays and organic clays which occur below the A-line.

Highly organic clays have a very weak and spongy feel at the plastic limit.

APPENDIX E

LIST OF EPA CLP LABORATORIES

CURRENT CLP SMO
IFB ANALYSIS LABORATORIES, January 1986

INORGANICS

Accu-Labs Research (ACCU)
11485 West 48th Avenue
Wheat Ridge, CO 80033
303/423-2766

Anacon, Inc. (ANACON)
2950 S. Shaver, A-14
Pasadena, TX 77502
713/941-6703

California Analytical Laboratories, Inc. (CAL)
2544 Industrial Boulevard
West Sacramento, CA 95691
916/372-1393

Cambridge Analytical Associates (CAMBRG)
1106 Commonwealth Avenue
Boston, MA 02215
617/232-2207

Centec Analytical Services (CENTEC)
2160 Industrial Drive
Salem, VA 24153
703/387-3995

Chemtech Consulting Group, Ltd. (CHEM)
360 West 11th Street
New York, NY 10014
212/255-2100

CompuChem Laboratories (COMPU)
3308 Chapel Hill/Nelson Highway
Research Triangle Park, NC 27709
919/549-8263

Hittman-Ebasco Associates, Inc. (HITMAN)
9151 Ramsey Road
Columbia, MD 21045
301/730-8525

JTC Environmental Consultants, Inc. (JTC)
Four Research Place, Suite L-10
Rockville, MD 20850
301/921-9790

Mack Laboratories (MACK)
2199 Dartmore Avenue
Pittsburgh, PA 15210
412/885-2900

Radian Corporation (RADIAN)
8501 Mo-Pac Boulevard
Austin, Travis County, TX 78759
512/454-4797
(Mailing Address: P.O. Box 9948; Austin, TX 78766)

Rocky Mountain Analytical Lab, Inc. (RMAL)
5530 Marshall Street
Arvada, CO 80002
303/421-6611

Spectrix Corporation (SPECS)
3911 Fondren, Suite 100
Houston, TX 77063
713/266-6800

Toxicon Laboratories (TOXI)
3213 Monterrey Boulevard
Baton Route, LA 70814
504/925-5012

U.S. Testing Company, Inc. (USTEST)
1415 Park Avenue
Hoboken, NJ 07030
201/792-2400

Versar, Inc. (VERSAR)
6850 Versar Center
Springfield, VA 22151
703/750-3000

Weyerhaeuser Company (WEYER)
Weyerhaeuser Technology Center
2B-25
Tacoma, WA 98477
206/924-6373

INORGANICS (cont'd)

Wilson Laboratories (WILSON)
525 North Eighth Street
Salina, KS 67402
913/825-7186
(Mailing Address: P.O. Box 1884

ORGANICS

Acurex Corporation (ACUREX)
Energy & Environmental Division
405 Clyde Avenue Mountain View, CA 94042
415/964-3200

Analytical Resources, Inc. (ARI)
3008B Sixteenth Avenue West
Seattle, WA 98119
206/285-1577

Analytical Technologies, Inc. (ATI)
225 West 30th Street
National City, CA 92050
619/477-4173 or 4174

Aquatec, Inc. (AQUAI)
75 Green Mountain Drive South Burlington, VT 05401
802/658-1074

California Analytical Laboratories, Inc. (CAL)
2544 Industrial Boulevard
West Sacramento, CA 95691
916/372-1393

Cambridge Analytical Associates (CAMBRG)
1106 Commonwealth Avenue
Boston, MA 02215
617/232-2207

Century Refining Company (CENREF)
695 North Seventh St.
Brighton, CO 80601
303/659-0497
(Mailing Address: P.O. Box 68)

Clayton Environmental Consultants Inc. (CLAYTN)
25711 Southfield Road
Southfield, MI 48075
313/424-8860

CompuChem Laboratories (COMPU)
3308 Chapel Hill/Nelson Highway
Research Triangle Park, NC 27709
919/549-8263
(Mailing Address: P.O. Box 12652

EA Engineering Science & Technology Inc. (EAENG)
15 Loveton Circle
Sparks, MD 21152
301/771-4950

EAL Corporation (EAL)
2030 Wright Avenue
Richmond, CA 94804
415/235-2633

Ecology & Environment (E&E)
4285 Tennessee St.
Buffalo, NY 14225
716/631-0360

Enseco Corporation (ENSECO)
205 Alewife Brook Parkway
Cambridge, MA 02138-1101
617/661-3111

Envirodyne Engineers (ENV)
12161 Lackland Road
St. Louis, MO 63146
314/434-6960

Environmental Control Technology Corp. (ENCOT)
3985 Research Park Drive
Ann Arbor, MI 48104
313/761-1389

Environmental Monitoring & Services, Inc. (EMSI)
EESD - EMSC
2421 W. Hillcrest Drive, Newbury Park
Ventura County, CA 91320
805/498-6771

Environmental Research Group (ERG)
117 North First Street
Ann Arbor, MI 48104
313/662-3104

Environmental Testing & Certification Corp. (ETC)
284 Raritan Center Parkway
P.O. Box 7808
Edison, NJ 08818
201/225-5600

GCA Technology Division (GCA)
213 Burlington Road
Bedford, MA 01730
617/275-5444

Gulf South Research Institute (GULF)
5010 LeRoy Johnson Road
New Orleans, LA 70126
504/283-4223

Hazleton Laboratories (HAZLET)
3301 Kinsman Boulevard
Madison, WI 53704
608/241-4471
(Mailing Address: P.O. Box 7545; Madison, WI. 53707)

IT Analytical Services (ITSTU)
5815 Middlebrook Pike
Knoxville, TN 37921
615/588-6401

IT Corporation (IT)
17605 Fabrica Way, Suite D
Cerritos, CA 90701
714/523-9200
213/921-9831

Laucks Testing Laboratories, Inc. (LAUCKS)
940 South Harney Street
Seattle, WA 98108
206/767-5060

Nanco Laboratories, Inc. (NANCO)
Unity Street at Route 376
Hopewell Junction, NY 12533
914/221-2485
(Mailing Address: P.O. Box 10)

NUS Corporation (NUS)
Park West Two
Cliff Mine Road
Pittsburgh, PA 15275
412/788-1080

PEI Associates, Inc. (PEI)
11499 Chester Road
Cincinnati, OH 45246
513/782-4700

Radian Corporation (RADIAN)
10395 Old Placerville Road
Sacramento, CA 95827
916/362-5332

Rocky Mountain Analytical Lab, Inc. (RMAL)
5530 Marshall Street
Arvada, CO 80002
303/421-6611

Science Applications International Corp. (SAI)
Division of Applied Environmental Sciences
476 Prospect Street
LaJolla, CA 92038
619/456-7521

S-Cubed (S^3)
3398 Carmel Mountain Road
San Diego, CA 92121
619/453-0060

Southern Research Institute (SoRI)
2000 Ninth Avenue South Birmingham, AL 35255
205/323-6592
(Mailing Address: P.O. Box 55303)

Southwest Laboratory of OK Inc. (SWOK)
10926 E. 55th Place Tulsa, OK 74146
918/665-0680

Southwest Research Institute (SWRI)
6220 Culebra Rd.
San Antonio, TX 78238
512/684-5111

Spectrix Corporation (SPECS)
3911 Fondren, Suite 100
Houston, TX 77063-5821
713/266-6800

Toxicon, Inc. (TOXI)
3213 Monterrey Boulevard
Baton Rouge, LA 70814
504/925-5012

Utah Biomedical Test Laboratory, Inc. (UBTL)
520 Wakara Way
Salt Lake City, UT 84108
801/584-3232

University of Iowa (IOWA)
University Hygenic Laboratory
Iowa City, IA 52242
319/353-5990

U.S. Testing Company, Inc. (USTEST)
1415 Park Avenue
Hoboken, NJ 07030
201/792-2400

Versar, Inc. (VERSAR)
6850 Versar Center
Springfield, VA 22151
703/750-3000

Wadsworth Testing (WADSWO)
1600 Fourth Street, S.E.
Canton, Ohio 44701
216/454-5809

West Coast Analytical Service, Inc. (WCAS)
9840 Alburtis Avenue
Santa Fe Springs, CA 90670
213/948-2225

Western Research Institute (WRI)
P.O. Box 3395
Laramie, WY 82071
307/721-2217
(Shipping Address: Ninth & Louis Streets)

Weston Analytical Lab, Inc. (WESTON)
256 Welsh Pool Road
Lionville, PA 19353
215/524-0180

York Laboratories (*YORK*)
200 Monroe Turnpike
Monroe, Connecticut 06468
203/261-4458

APPENDIX F

RESUMES

R. WILLIAM AJA, CHIEF COST ESTIMATOR

Education

Champlain College - Associates Degree in Business Administration, 1966

Professional Licenses

Certified Professional Estimator - General Construction

Affiliations

American Society of Professional Estimators - President, Maine Chapter

Professional Experience

Mr. Aja has 17 years of experience in preparing preliminary and final capital cost estimates for environmental and industrial projects. Specific experience related to hazardous waste projects includes cost estimates at a Superfund site in Coventry, Rhode Island for the State of Rhode Island, and numerous sludge landfill projects for municipalities, utilities, and pulp and paper firms. He is currently working on a remedial capping system for a hazardous waste landfill for the Emhart Corporation. Mr. Aja was responsible for the complete estimating for Jordan's work at the Vermont Agency of Transportation Superfund site in Burlington, including earthwork, concrete, building structure, site work, equipment, piping, mechanical, electrical and instrumentation.

For the Coventry, Rhode Island project, which has been identified by the state as its top priority for cleanup, Mr. Aja estimated costs of removal alternatives for industrial waste solvents, liquids, and sludges. He also was responsible for cost estimates for an ash disposal facility for the Central Maine Power Company and coal ash disposal for Boston Edison.

Mr. Aja's experience also includes preparation of estimates for numerous waste treatment facilities and for the Jordan Company's technical assistance to the U.S. Environmental Protection Agency Effluent Guidelines Division. These responsibilities involved cost estimates for pulp, paper, and paperboard industry compliance with proposed and revised effluent limitations.

JAMES S. ATWELL, MANAGER OF ENVIRONMENTAL SERVICES

Education

University of Maine - B.S. in Civil Engineering, 1965
University of Maine - M.S. in Civil Engineering, 1966

Professional Licenses

Professional Engineer - Maine, Massachusetts, Michigan

Affiliations

American Society of Civil Engineers, Solid Waste Management Committee
National Solid Waste Management Association
Water Pollution Control Federation

Professional Experience

As Manager of the Environmental Services division of E.C. Jordan Co., Mr. Atwell oversees the firm's geotechnical, solid and hazardous waste, wastewater, water, environmental laboratory and planning departments.

Mr. Atwell is responsible for the development of hazardous and nonhazardous waste management programs for public and private sector clients. Management programs include: 1) evaluation and design of remedial action at Superfund and other uncontrolled hazardous waste sites; 2) planning and design of land disposal systems including secure landfills and land treatment systems; 3) permitting for the treatment, storage and disposal of hazardous wastes in accordance with 40 CFR 122 and 40 CFR 264; and 4) planning and design of resource recovery facilities. Recent projects have included remedial investigations/feasibility studies at Superfund and other hazardous waste sites in Michigan for the Michigan Department of Natural Resources; evaluation and design of remedial action at the Pine Street Canal Superfund site in Burlington, Vermont; evaluation and implementation of closure plans for several industrial hazardous waste facilities including design of capping systems and monitoring programs; and a geologic investigation and contamination assessment at a site formerly used for chemical storage and now under consideration for purchase by the Boston Edison Company.

He has served on Jordan's Quality Review Committee for such projects as a remedial action assessment and long-term environmental monitoring at Love Canal in Niagara Falls, New York; an assessment of the extent of solvent, resin and inorganic contamination in soils and surface and groundwaters at the Silresim Superfund site in Lowell, Massachusetts; data evaluation and remedial action assessment at the North Hollywood Dump Superfund site in Memphis, Tennessee; and the site evaluation, selection and design of three ash disposal systems for Central Maine Power Co. These landfills incorporated dual liner systems (flexible synthetic liner and compacted clay) and leachate collection. He also played a major role in Jordan's support to the U.S. EPA in review and recommendations for hazardous waste regulation guidance documents.

During the past five years, Mr. Atwell has been responsible for several design projects including secure landfills with low permeability liners and leachate collection, industrial waste lagoon closures, and remedial actions at closed hazardous and non-hazardous waste sites.

Mr. Atwell served as project manager for EPA Contract No. 68-01-5772 from February 1979 until April 1980. This contract involved several subcontracts and required the establishment of large-scale sampling programs for the collection and analysis of solid and liquid samples for priority pollutant analyses. The contract also required assessment of waste management technologies and the preparation of industry profiles, regulatory support packages, and technical guidance documents.

He was also responsible for solid waste management projects which emphasized the evaluation and development of resource or energy recovery alternatives including waste/resource recovery projects for the North Kennebec Regional Planning Commission; the municipalities of Biddeford, Saco, Old Orchard Beach, Orono, and Old Town; and the University of Maine at Orono.

Partial List of Publications and Presentations

"Boiler Ash Disposal." Presented at TAPPI Annual Meeting, Chicago, Illinois; March 3, 1981. To be published in TAPPI Magazine, July 1981.

"Practical Approaches to Coal Ash Disposal." Presented at annual meeting of the Associated Industries of Maine, Portland, Maine; October 1980.

"Design of Hazardous Waste Landfills." Presented at Fall Meeting of New England Water Pollution Control Association, Falmouth, Massachusetts; October 1980.

Atwell, James S. and Walker, Stanley E., "Site Selection and Design of Utility Oil Ash Landfill." Presented at Third Annual Madison Conference of Applied Research and Practice on Municipal and Industrial Waste; September 10-12, 1980.

Solid Waste Disposal for the Pulp and Paper Industry." Presented at the American Institute of Chemical Engineers Annual Meeting, Houston, Texas; April 1979.

"Wastewater Sludge Incineration." Published by and presented at the Spring TAPPI Meeting, New York; March 1979.

DONALD R. COTE, SENIOR VICE PRESIDENT

Education

Northeastern University - 1967, B.S. in Sanitary Engineering
Northeastern University - 1969, M.S. in Sanitary Engineering

Professional Licenses

Professional Engineer - Maine, New Hamsphire, Vermont, Massachusetts, Connecticut, Pennsylvania, New York, Delaware, Rhode Island, Illinois, Michigan, Maryland, and New Jersey

Professional Experience

Mr. Cote has overall technical and administrative responsibility for the firm's multidisciplinary consulting services in design engineering, earth and water resources, pulp and paper design, and project management. These responsibilities include the direction of Jordan projects in the solid and hazardous waste field, which incorporate services offered by the firm's environmental and geotechnical engineers, hydrogeologists, soil scientists, and chemists.

Recent solid/hazardous waste-related projects under his management include: a remedial action assessment and long-term environmental monitoring at Love Canal in Niagara Falls, New York; EPA Effluent Guideline Division's Best Available Technology Programs for food, and pulp, paper, and paperboard point source categories; toxic pollutant identification and treatment research and development for EPA's IERL; development of remedial action plans for three uncontrolled hazardous waste disposal sites; design of four secure waste disposal sites; development of a process water reclamation system for removal of nonconventional and toxic pollutants; and the planning and design of a 200-tpd incineration/energy recovery system.

DAVID B. ERTZ, REGIONAL MANAGER, CENTRAL UNITED STATES

Education

University of Vermont - B.S. in Civil Engineering, 1973
Cornell University - M.S. in Waste Management, 1975

Professional Licenses

Professional Engineer - Maine

Affiliations

Water Pollution Control Federation
New England Water Pollution Control Association

Professional Experience

Mr. Ertz is responsible for overseeing Jordan's projects in the Central United States. He brings to this position 12 years of experience in the management of major multidisciplinary projects assessing industrial generation and disposal of hazardous wastes. He has managed Superfund site remedial investigations/feasibility studies and remedial design, conducted under the auspices of the U.S. EPA (or its contractors) or state agencies under cooperative agreement. As a result, he has developed a working knowledge of environmental regulations, including those promulgated under CERCLA, RCRA, and TSCA.

Mr. Ertz served as project manager for a site contamination audit and heavy metal sludge landfill closure at two separate sites, as well as additional contamination assessments that were conducted for confidential clients. These projects emphasized hydrogeologic investigations and contamination assessments to define existing and potential environmental and health problems and provided a basis for selecting and implementing a remedial alternative. Detailed design of a remedial capping system was involved at two of these sites.

For private clients, he has managed site audits and contamination assessments, landfill and lagoon closures, and projects that developed remedial alternatives, including design of alternative water supply, sewer removal and disposal, and remedial capping systems.

Specific Superfund and other uncontrolled hazardous waste site project experience includes remedial investigations/feasibility studies at an abandoned hydrocarbon facility, the Cannons Engineering Corporation site in Bridgewater, Massachusetts, and the Acme Solvent site near Rockford, Illinois, where solvent recovery still-bottoms sludge were disposed. For the North Hollywood Dump site in Memphis, Tennessee, Mr. Ertz managed Jordan's evaluation of the extent and nature of environmental problems caused by the dump and development of remedial action alternatives to address environmental and health problems. In another project, design and construction of a permanent water supply was completed to serve residents threatened by contamination of domestic wells near the Winthrop Landfill in Winthrop, Maine.

Mr. Ertz served as project manager for a $4-million contract with the U.S. EPA's Effluent Guidelines Division that addressed generation of hazardous wastes and discharge of toxic pollutants by specific food industries. For this contract, Mr. Ertz coordinated and managed technical programs for multiple industries, many of which required major sampling events to detect and quantify toxic pollutants in liquid and solid media. A major portion of these sampling programs related to a study to determine the fate of toxic pollutants in publicly owned treatment works (POTW). Technology was assessed for the control and treatment of toxic and conventional pollutants, followed by the preparation of conceptual designs and attendant cost estimates. These Jordan activities involved several functions and up to 20 personnel from several disciplines.

While employed in the industrial sector, Mr. Ertz had the responsibility of monitoring and evaluating proposed and enacted regulations dealing with solid and hazardous wastes, air, potable water, and wastewater. He also managed a $100,000 remedial action program at a site where leaks from several PCB-filled transformers had taken place. Mr. Ertz organized and instituted a program to determine the extent of contamination and then initiated work to alleviate the problem. Remedial action involved individual contractors for transformer removal, PCB disposal and cleanup. A clean-up effort followed to remove contamination from the adjacent areas. In addition, Mr. Ertz developed a program for characterization and disposal of waste treatment residues at a textile facility. This work was conducted in cooperation with the responsible state agency to comply with hazardous waste regulations.

Publications and Presentations

"Dissolved Air Flotation Treatment of Seafood Processing Wastes -- An Assessment." Presented at the Eighth National Symposium on Food Processing Wastes, March 30 - April 1, 1977, Seattle, Washington.

"Caution - EPA Contractor at Work." Presented at the Conference on Seafood Waste Management in the 1980s, September 23-25, 1980, Orlando, Florida.

"Development of BCT Effluent Guidelines for the Food Processing Industry." Presented at the Eleventh Conference on Environmental and Energy Engineering in the Food Processing Industry, February 22-27, 1981.

MATTHEW D. JERUE, SENIOR CHEMICAL ENGINEER

Education

University of Michigan - B.S. in Chemical Engineering, 1976
University of Michigan - B.S. in Environmental Engineering, 1976

Professional Experience

Mr. Jerue is experienced in the conduct of remedial investigation/feasibility studies as well as other hazardous waste management studies (prospective and retrospective), including those of an industrywide nature.

He has performed hazardous waste management inspections and audits to assess potential liabilities, and made recommendations for waste management and/or remedial activities programs at a variety of industrial facilities and landfills. He has performed evaluations of unit operations to assess their potential to produce hazardous waste, as well as determinations of potential hazardous wastes generated during various demolition projects.

He has assisted in the management of Jordan's multi-site contract with the Michigan Department of Natural Resources (MDNR). For this assignment, Mr. Jerue has coordinated Jordan's activities with MDNR staff and has provided on-site safety coordination, monitoring and data interpretation and assessment. This coordination has involved the simultaneous coordination of field investigations at several sites.

Prior to joining Jordan, Mr. Jerue had project management responsibilities for multi-task hazardous waste site investigations and site audits performed for private and public clients. He has been involved in field activities including sample collection and field analysis and is experienced in the operation of field analytical equipment such as the Photovac 10A10 organic vapor analyzer.

Mr. Jerue has prepared health and safety procedure manuals for onsite inspections and drilling operations, and technical feasibility and economic impact data for proposed governmental regulations. He is thoroughly familiar with EPA and DOT hazardous materials regulations and has participated in hazardous management and DOT hazardous material (regulations) orientation and training seminars for both clients and parent company personnel.

RONALD A. LEWIS, SENIOR CHEMICAL ENGINEER

Education

University of Maine - B.S. in Chemical Engineering, 1964
University of Maine - M.S. in Chemical Engineering, 1966

Professional Experience

Mr. Lewis has almost 12 years' experience in waste management and hazardous waste site assessment. His broad range of experience includes acting as liaison between industry and state and Federal agencies on environmental matters, environmental data assessment, contaminant and groundwater modeling, and supervising a physical/chemical wastewater treatment plant.

Mr. Lewis has performed a variety of environmental project responsibilities including onsite safety coordination, monitoring and data interpretation of extensive soils, surface water and groundwater sampling conducted at Superfund sites in Massachusetts, New York, Rhode Island, and Tennessee. He provided onsite safety coordination, monitoring and data interpretation and assessment at the Silresim site in Lowell, Massachusetts. Monitoring of organics in the air, soil and water were conducted with portable sensors to protect the investigation team and to assess problems.

He has become increasingly involved with assessments of contaminant transport through groundwater movement by the use of numerical models. Applications have ranged from the use of analytical models at the Silresim site to finite element and finite difference models for groundwater flow for industrial and state clients. He is currently engaged in the application of both finite difference and finite element models to the hydrogeology of Love Canal in the design of a long-term monitoring program for that site. His responsibilities on the Love Canal project have also included sample collection, field assays, and sample preservation, storage and transport. In addition, he assisted in the contamination assessment by determining the present and potential impacts of contaminants migrating from the site.

He was responsible for environmental data compilation and review and served as part of Jordan's site remedial action assessment team for the North Hollywood Dump Superfund site in Memphis, Tennessee. He also participated in the preparation of a remedial action plan for a Superfund site in Coventry, Rhode Island. This project involved identifying and evaluating alternative remedial actions and recommending the most cost-effective alternative.

Mr. Lewis has also assisted several industries in determining their liability and procedures for complying with RCRA and CERCLA regulations.

JAMES R. MARZOLINO, JUNIOR ENGINEER

Education

B.S. in Environmental Science - Lake Superior State College, 1982
A.A. in Water Quality Technology - Lake Superior State College, 1979
A.A. in Natural Resources Technology - Lake Superior State College, 1979

Special Courses

Certificate - Seminar for Hazardous Material Handling

Professional Affiliations

Michigan Association Environmental Professionals
Air Pollution Control Association

Professional Experience

Mr. Marzolino serves as a member of the Jordan field investigation team which conducts sampling programs from groundwater, surface water, soil, sediment, and air. His current field responsibilities include participation in the sampling of water, soil and air at various industrial sites and uncontrolled hazardous waste sites, including Michigan as part of Jordan's statewide RI/FS contract with the Michigan department of Natural Resources. Mr. Marzolino has also been involved in field investigation activities for the following projects:

- Detroit Edison Company - Site Assessment Audit, Marysville, Michigan Groundwater and Soil Investigation.

- Illinois Environmental Protection Agency -
  Acme Solvents - Superfund Site
  Winnebago County, Illinois Sampling for a hydrogeologic Investigation of the Deep Groundwater regime.

- New York State Department of Environmental Conservation - Love Canal Perimeter Survey and Long Term Monitoring Program Implementation for Love Canal - Phase I, Niagra Falls, New York.

He has conducted testing, monitoring and analysis of air quality samples and has a thorough knowledge of U.S. EPA Regulations. In addition, he has managed field investigation groups and acted as group leader for air quality monitoring projects. Mr Marzolino has also performed water quality field sampling projects.

Before joining Jordan, Mr. Marzolino developed sampling and analytical methodology for source emission testing products and was responsible for training technicians in source sampling methodologies. He participated in program development and implementation of projects that required air quality monitoring.

TIMOTHY J. PALMER, SCIENTIST

Education

Michigan Technological University - B.S. in Chemistry, 1973

Affiliations

Michigan Association of Environmental Professionals, Charter Member
National Association of Environmental Professionals
Air Pollution Control Association

Professional Experience

Mr. Palmer has 12 years of multidisciplinary environmental consulting experience relating to pollution assessment, control, and monitoring. He has had project management responsibilities for environmental investigations developed to evaluate compliance with applicable air, water and waste regulations. He has also directed and conducted the testing and evaluation of many air and water pollution control devices. He is experienced as a Quality Assurance Coordinator for all calibration and field sampling programs.

Mr. Palmer is currently involved in site work at various RI/FS sites for the Michigan Department of Natural Resources through Jordan's multi-site RI/FS contract. Responsibilities during the site exploration portion of the remedial investigation include air, groundwater, and soil sampling, site safety, and monitoring well installations.

He has performed hazardous waste management inspections and audits to assess potential liabilities at industrial facilities and conducted field monitoring investigations to determine the presence of hazardous wastes generated during demolition projects and to implement proper disposal practices.

Air monitoring background includes measurement of stationary source emissions from boilers, coke ovens, incinerators, chemical industries, tar and solvent storage plants, and other industrial processes. He has been involved in research and development to establish emission standards for such processes as coke oven waste, heat exhausts, and ethylene oxide sterilization units.

Mr. Palmer has extensive experience in collection of industrial water and wastewater samples and flow monitoring of industrial wastewater bodies. Environmental investigations have included insitu measurements for pH, dissolved oxygen, temperature, salinity and chlorine.

Analytical laboratory experience includes grammetric, liquid and gas chromatography, and atomic absorption and sample analyses for particulate, sulfur, and nitrogen oxides, hydrocarbons, metals, and oil and greases.

THEODORE W. TAYLOR, GEOLOGIST

Education

Lehigh University - M.S. in Geology, 1983
Colby College - B.A. in Geology, 1981

Affiliations

American Geophysical Union
Mineralogical Society of America
Sigma Xi, The Scientific Research Society

Professional Experience

Mr. Taylor has been involved in numerous contaminated site assessment studies. He has been responsible for the data collection, analysis, and interpretation of hydrologic conditions in order to assess the boundaries and environmental impact of soil and rock contamination and the mode of contaminant transport. Mr. Taylor has conducted projects as part of plant site closure programs and has designed and implemented groundwater monitoring investigations. He has also been the site hydrogeologist for assessment studies at operating plant facilities for large industrial corporations. Mr. Taylor is experienced in monitoring well installation and development, permeability tests, pump tests, groundwater sampling, geophysical surveys, and in-field geologic investigations. He is proficient in the operation of computer automated x-ray fluorescent and diffraction equipment for quantitative and qualitative geochemical analyses.

With the Lawrence Livermore National Laboratory (U.S. Dept. of Energy), Livermore, California, Mr. Taylor was a field geologist investigating the mineralogical and structural characteristics of an igneous stock being used as a test site for the burial of nuclear waste.

With the AMAX Mount Tolman Project, Mr. Taylor was an assistant to the geologic staff that was exploring and evaluating the Mount Tolman porphyry molybdenum deposit near Grand Coulee, Washington. He participated in field investigations, map compilation, and core logging.

Publications

"Petrological and Geochemical Study of the O.K. Copper-Molybdenum Deposit, Beaver County, Utah," Utah Geological and Mineral Survey, Special Studies, No. 67 (1985).

"A Petrological and Geochemical Study of the O.K. Copper-Molybdenum Deposit, Beaver County, Utah," GSA Abstracts with Programs, Vol. 16, No. 4, p/ 257 (1984). (With C.B. Sclar.)

"An Application of Magnetic and Electromagnetic Surveys to the Interpretation of Bedrock Geology, Sidney, Maine, (abst.)," The Maine Geologist, Vol. 7, No. 3 (1980). (With T.D. Leary.)

JOHN D. TEWHEY, SENIOR SCIENTIST

Education

Colby College - B.A. in Geology and Chemistry, 1965
University of South Carolina - M.S. in Geology, 1968
Brown University, Ph.D. in Geology, 1975

Professional Licenses

Certified Geologist - Maine, California

Affiliations

Geological Society of America
American Geophysical Union
Society of Mining Engineers of AIME
Geological Society of Maine, President
Maine Mineral Resources Association, Secretary

Professional Experience

Dr. Tewhey served as project technical director for the hydrogeologic and soils investigation of the Silresim uncontrolled hazardous waste site in Lowell, Massachusetts. The project was completed on schedule and within the estimated budget. The project report was well received by the EPA (Region 1), the Commonwealth of Massachusetts and interested citizen groups. EPA has concurred that Jordan's remedial action recommendations be implemented at the site.

Dr. Tewhey was also the project technical director on a contract to evaluate technical data and recommend remedial alternatives for the North Hollywood Dump, a Superfund site in Memphis, Tennessee. He is currently serving as a member of a quality assurance team for Jordan's remedial action assessment and long-term environmental monitoring at Love Canal in Niagara Falls, New York.

Dr. Tewhey has had extensive project management experience at the Lawrence Livermore Laboratory of the University of California. His experience and responsibilities included development, characterization, stability assessment, and production technology of SYNROC, a ceramic waste form matrix for the incorporation of nuclear waste. The project was a $2-million per year development project for disposal of high-level radioactive wastes located at U.S. defense sites. Other project responsibilities included radioactive waste isolation program studies of glass leaching, rock sorption, field tracer studies in fractured geologic media, and the use of geochemical equilibrium codes to solve waste isolation problems. Dr. Tewhey led the engineering geology group and the field geology group at Lawrence Livermore Laboratory prior to becoming a project manager.

Dr. Tewhey is listed in the 13th edition of American Men and Women of Science and 1979 edition of Outstanding Young Men of America.

JOHN D. TEWHEY (continued)

Partial List of Publications and Presentations

"Site Conditions and Corrective Action at the North Hollywood Dump," presented at the 5th National Conference on Management of Uncontrolled Hazardous Waste Sites, November, Washington, D.C. (1984). (With A.F. McClure and T.K. Cothron.)

"Bridge Construction Procedures to Mitigate the Impact of Subsurface Hazardous Waste Deposits - A Case Study," presented at the 27th Annual Meeting of the Association of Engineering Geologists, October, Boston, Massachusetts (1984). (With A.K. Ahlers and E.G. Hill.)

"Silresim: A Hazardous Waste Case Study," presented at the National Conference on Management of Uncontrolled Hazardous Waste Sites, November, Washington, D.C. (1982). (With J.E. Sevee and R.L. Fortin.)

Associate Editor for "Scientific Basis for Nuclear Waste Management VI." Published proceedings of the Materials Research Society Symposia Volume 15, Boston, Massachusetts (1982).

Associate Editor for "Scientific Basis for Nuclear Waste Management." Published proceedings of the Materials Research Society Annual Meeting, Boston Massachusetts (1981).

"The Importance of Geosciences in Developing In-Situe Technologies for Energy Resource Recovery," American Association for the Advancement of Science, Abstracts for the 144th Annual Meeting, p. 1979 (1978). (With L.L. Schwartz, D.O. Emerson and W.J. Beiringer.)

"The Application of SYNROC to High-Level U.S. Defense Wastes." In press, Waste Management 1980 (1980). (With W.B. Durham, C.L. Hoenig, and F.J. Ryerson.)

"Development of Injection Criteria for Geothermal Resources," Geothermal Resources Council Transactions, 2, 649-652 (1978). (With M.A. Chan, P.W. Kasameyer, and L.B. Owen.)

"The Distribution of Uranium in Sediment Samples as Determined by Multi-Element Analysis," U.S. Geological Survey, Circulat 753, J.A. Campbell, Editor (1977).

"Geology of the Irmo Quadrangle, Richland and Lexington Counties, South Carolina," published as folio MS-22 with four maps by the South Carolina Geological Survey (1977).

"Geochemical Studies of Sorption and Transport of Radionuclides in Rock Media," University of California Report, UCRL 52929 (1980). (With D.G. Coles and H.C. Weed.)

"Preliminary Results of Experimental Work in the Radionuclide Migration Program - The Sorptive Character of Tuffaceous Rock in a 'Static' versus 'Dynamic' Mode of Testing," University of California Report, UCID 17064 (1976).

"Silresim: A Hazardous Waste Case Study," Proceedings of the American Defense Preparedness Association. 12th Annual Environmental Symposium, Langley Air Force Base, Volume V, pp. 43-46 (1982)